Block practice · Blood, heart and circulation

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

Identify blood components and match them to their features and jobs

Describe phagocytosis, antibody production and barriers to pathogens

Name heart chambers and major vessels from a diagram

Describe what red cells and plasma carry, and from where to where

Trace the route of blood through the heart, lungs and body

Describe how blood clots and explain why clotting matters

Explain how a blocked coronary artery damages the heart

Describe how heart disease is prevented and treated

Identify arteries, veins and capillaries and relate their structure to function

Explain red-cell adaptations and the effects of more or less oxygen carriage

Explain systolic, diastolic and changing pressure around the circulation

Use ABO antigens and antibodies to choose donors and explain clumping

Explain how tissue fluid forms, what it does and why it builds up

Practice workbook · Identify blood components and match them to their features and jobs

Recognising red cells, white cells, platelets and plasma in smears or drawings, and tick grids or naming prompts pairing each with what it contains or does.

14 question parts: 1 worked-example retry and 13 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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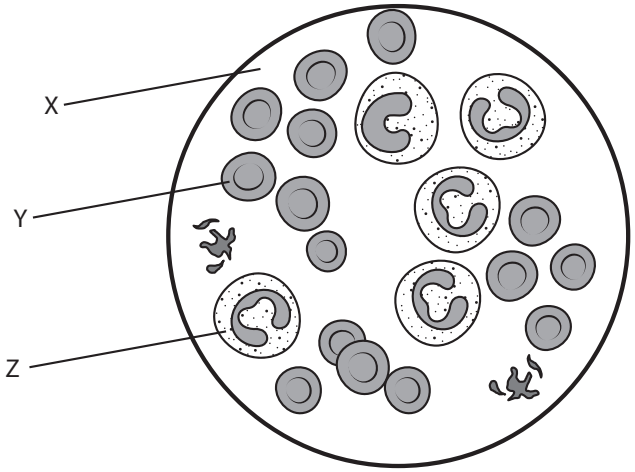
Type ID: HBI-BLOOD-COMPONENTS-IDENTIFY-AND-FUNCTION

1. Question 2 · part a.i · 3 marks

June 2024 Paper 1R

Worked-example retry · Attempt part a.i only.

2 (a) The diagram shows components, X, Y and Z, in a sample of human blood.



(i) Name components X, Y and Z.

(3)

X .....

Y .....

Z .....

(ii) Describe the function of component Y.

(2)

.....

.....

.....

.....

(iii) A sample of blood from a patient is analysed in a hospital.

A scientist finds that the number of component Z in the blood sample is higher than normal.

State a possible reason for the increase in the number of component Z.

(1)

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1. Question 2 · part a.i · 3 marks · Working and answer

Working and answer

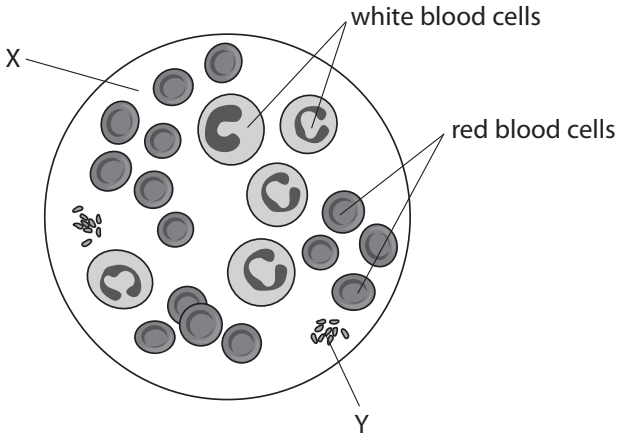
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2. Question 1 · part b · 2 marks

November 2020 Paper 1R

Further practice · Attempt part b only.

1 The diagram shows a sample of human blood.



(a) State the function of red blood cells.

(1)

.....

.....

(b) Give the name of X and Y.

(2)

X .....

Y .....

(c) (i) Calculate the ratio of red blood cells to white blood cells.

Write the ratio in its simplest form.

(2)

ratio = ...

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2. Question 1 · part b · 2 marks · Working and answer

Working and answer

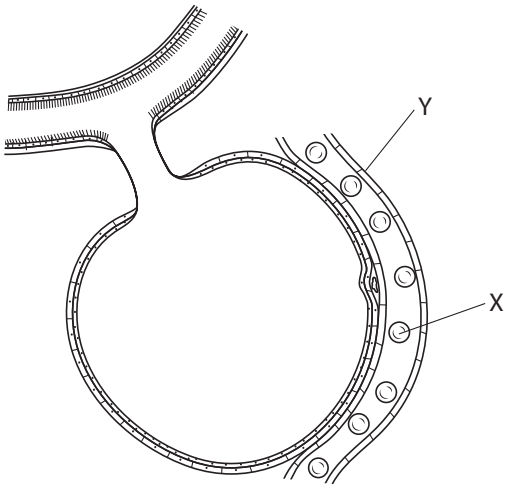
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3. Question 3 · part a.i · 1 mark

November 2020 Paper 2

Further practice · Attempt part a.i only.

3 The diagram shows an alveolus and its blood supply.



(a) (i) What is the name of blood component X? **(1)**

- ☒ **A** plasma
- ☒ **B** platelet
- ☒ **C** red blood cell
- ☒ **D** white blood cell

(ii) State three ways in which air in the alveolus differs from air in the atmosphere. **(3)**

1 .....

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2 .....

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3 .....

.....

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3. Question 3 · part a.i · 1 mark · Working and answer

Working and answer

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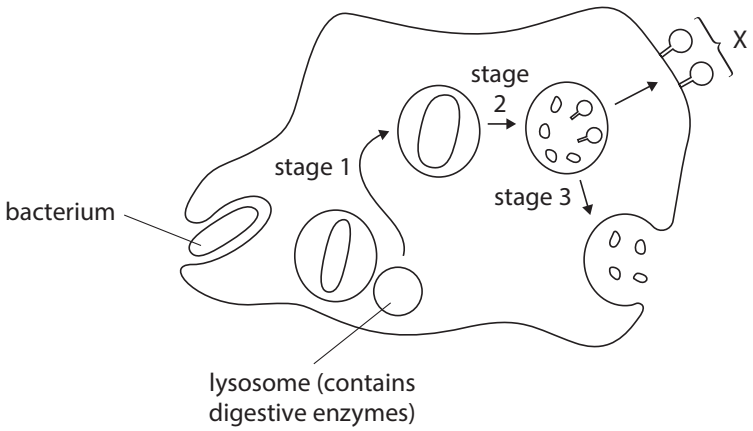
4. Question 6 · part b.ii · 4 marks

June 2023 Paper 1

Further practice · Attempt part b.ii only.

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6 The diagram shows a white blood cell engulfing a bacterium and the stages that follow.



(a) (i) State the name of the process used by white blood cells to engulf bacteria.

(1)

(ii) Describe what is happening at stages 1, 2 and 3 shown in the diagram.

(3)

stage 1

stage 2

stage 3

(iii) Suggest the function of the structures labelled X.

(2)

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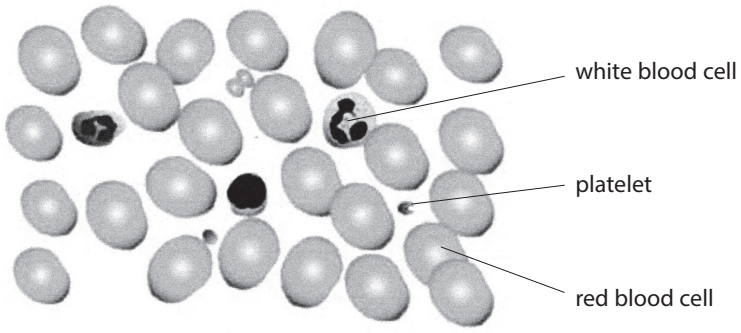
4. Question 6 · part b.ii · 4 marks

June 2023 Paper 1

Further practice · Attempt part b.ii only.

(b) A student observes a prepared slide of blood under a light microscope.

The student draws this diagram of what could be seen.



(i) Describe how the slide of human blood could be prepared.

(3)

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4. Question 6 · part b.ii · 4 marks

June 2023 Paper 1

Further practice · Attempt part b.ii only.

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(ii) Discuss the accuracy of the student's diagram.

(4)

(iii) Explain why the student would be advised to use a prepared slide rather than making one from their own blood.

(2)

(Total for Question 6 = 15 marks)

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4. Question 6 · part b.ii · 4 marks · Working and answer

Working and answer

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5. Question 7 · part a · 2 marks

November 2020 Paper 2

Further practice · Attempt part a only.

7 (a) Give two differences between whole blood and plasma.

(2)

1 .....

.....

2 .....

.....

(b) Ebola is an infectious disease. One symptom of Ebola is blood loss.

The World Health Organisation (WHO) has a procedure for treating Ebola.

This is the WHO's procedure.

- find a person who has recovered from Ebola
- observe this person to make sure that they are disease free for 28 days
- take blood from this person and test the blood for other diseases
- separate the plasma from the blood
- transfer this plasma to another person who has Ebola

Explain how the WHO procedure helps to treat a person who has Ebola.

(6)

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P 6 2 0 5 3 A 0 2 0 2 4

5. Question 7 · part a · 2 marks · Working and answer

Working and answer

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6. Question 2 · part a.iv · 2 marks

November 2020 Paper 2R

Further practice · Attempt part a.iv only.

2 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Treatment of Ebola

Ebola is caused by an RNA virus. There is no proven treatment available for Ebola. However, whole blood collected from someone who has recovered from infection has been used as a treatment. The results have been promising in a small group of Ebola patients.

Guidance for national health authorities and blood transfusion services describes the stages needed to collect whole blood (CWB) or plasma (CP) from Ebola-recovered patients. This can be used for transfusion to patients as a treatment for early Ebola.

The guidance states how to

- identify people who have recovered from Ebola as potential blood donors
- screen donors' blood for A, B, O and Rhesus groupings
- screen donors' blood for transfusion-transmissible infections
- collect blood and care for donors
- obtain agreement of Ebola patients for the treatment
- identify patient's blood grouping
- store and transport CWB and CP to the places where transfusions are to be given
- select Ebola patients for transfusion
- transfuse the blood into the patient

Donated CWB should never be frozen and should be stored between 2 °C and 6 °C.

CWB and CP donations should be transfused to the Ebola patients using standard transfusion procedures.

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6. Question 2 · part a.iv · 2 marks

November 2020 Paper 2R

Further practice · Attempt part a.iv only.

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(a) (i) Explain what is meant by an RNA virus (line 1).

(2)

(ii) The table lists five diseases. Put a tick (✓) next to the diseases that should be tested for in the donors' blood.

(2)

Disease	Blood tested (✓)
anaemia	
cystic fibrosis	
gonorrhoea	
HIV	
scurvy	

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(iii) Explain why it is necessary to identify the blood group of the blood collected from donors who have recovered from Ebola (line 9).

(3)

6. Question 2 · part a.iv · 2 marks

November 2020 Paper 2R

Further practice · Attempt part a.iv only.

(iv) Suggest why donated CWB should never be frozen (line 17).

(2)

(v) Explain how the blood of people who have recovered from Ebola is likely to be effective in treating the disease.

(2)

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(b) The nitrogenous base composition of another virus was found to be adenine 13%, guanine 26%, uracil 20% and cytosine 41%.

Explain how these figures prove that this is a virus with a single strand of RNA.

(3)

(Total for Question 2 = 14 marks)

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6. Question 2 · part a.iv · 2 marks · Working and answer

Working and answer

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7. Question 1 · part whole · 6 marks

June 2019 Paper 1R

Further practice · Attempt part whole only.

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1 The table lists blood components and their features.

Place ticks (✓) in boxes to show the features of each blood component.

(6)

	Feature					
Blood component	**has a nucleus**	**contains haemoglobin**	**carries hormones**	**engulfs pathogens**	**produces antibodies**	**helps blood clotting**
red blood cell						
phagocyte						
lymphocyte						
platelets						
plasma						

(Total for Question 1 = 6 marks)

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7. Question 1 · part whole · 6 marks · Working and answer

Working and answer

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8. Question 1 · part whole · 4 marks

June 2021 Paper 1

Further practice · Attempt part whole only.

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1 The table lists four features of blood. It also shows four main components of blood.

Put a tick (✓) in a box if the component shows the feature.

(4)

Feature	Components			
	Red blood cell	**Plasma**	**Platelet**	**Phagocyte**
transports oxygen				
has a nucleus				
consists of 90% water				
involved in blood clotting				

(Total for Question 1 = 4 marks)

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Working and answer

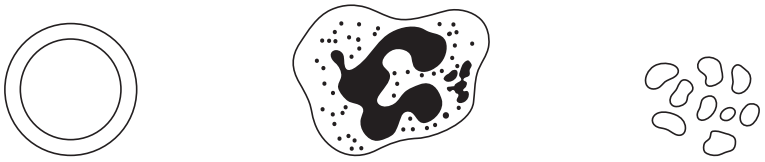
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9. Question 3 · part a · 4 marks

November 2021 Paper 1

Further practice · Attempt part a only.

3 The diagram shows some of the components of blood.



A B C

(a) Complete the table by giving the name of the component and its function.

Component A has been completed as an example.

(4)

Component	Name	Function
A	red blood cell	transports oxygen
B		
C		

(b) (i) Describe the process of blood clotting.

(5)

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9. Question 3 · part a · 4 marks · Working and answer

Working and answer

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10. Question 1 · part a · 5 marks

November 2023 Paper 1

Further practice · Attempt part a only.

1 (a) The table gives components of the blood and their features.

Complete the table by putting a tick (✓) in the box if the blood component shows the feature.

(5)

Feature	Blood component				
	red blood cell	phagocyte	lymphocyte	platelet	plasma
transports oxygen					
contains haemoglobin					
is a fluid					
produces antibodies					
engulfs pathogens					

(b) Explain why it is important that blood clots when the skin is cut open.

(4)

(Total for Question 1 = 9 marks)

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10. Question 1 · part a · 5 marks · Working and answer

Working and answer

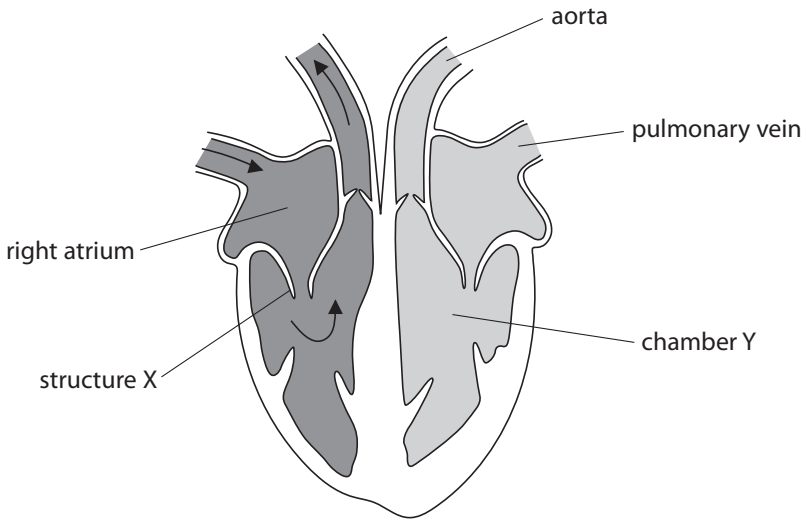
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11. Question 2 · part c.i · 1 mark

January 2023 Paper 2

Further practice · Attempt part c.i only.

2 (a) The diagram shows a section through the heart.



(i) Name chamber Y. **(1)**

(ii) Explain the function of structure X. **(2)**

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P 7 2 6 0 2 A 0 4 2 8

11. Question 2 · part c.i · 1 mark

January 2023 Paper 2

Further practice · Attempt part c.i only.

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(iii) The table lists some properties of blood.

Add ticks (✓) to those boxes that show the properties of blood in the aorta.

(2)

Properties of blood	Blood in the aorta
oxygenated	
deoxygenated	
high pressure	
medium pressure	
low pressure	

(iv) State where blood in the left side of the heart has come from.

(1)

(b) In one year in the United Kingdom, there were 173 heart transplants.

Heart transplant operations require hearts to be donated.

Some hearts cannot be used because the donor is too old.

Give two other reasons why a heart might not be suitable for a transplant.

(2)

1 .....

.....

2 .....

.....

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11. Question 2 · part c.i · 1 mark

January 2023 Paper 2

Further practice · Attempt part c.i only.

(c) (i) Which blood component causes blood clotting?

(1)

- ☒ **A** lymphocytes
- ☒ **B** phagocytes
- ☒ **C** platelets
- ☒ **D** red blood cells

(ii) The diagram summarises part of the process that causes a blood clot.



Anticoagulant drugs prevent a blood clot forming.

These drugs stop the enzyme thrombin being produced.

Describe how a lack of thrombin prevents a blood clot forming.

(2)

(Total for Question 2 = 11 marks)

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11. Question 2 · part c.i · 1 mark · Working and answer

Working and answer

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12. Question 5 · part c · 1 mark

June 2024 Paper 2R

Further practice · Attempt part c only.

5 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

The heart is a double-pump that is part of the human circulatory system. It helps ensure that the blood and its components reach all cells in the body.

Some components of the blood are made in the bone marrow. An average adult with a mass of 70 kg has 5.5 dm^3 of blood. About 45% of the blood is made up of different cells. There are several different types of cell in the blood but red blood cells are the most common. In males, 1.0 dm^3 of blood contains 5.9×10^{12} red blood cells. In females the same volume of blood contains 5.5×10^{12} red blood cells. Haemoglobin makes up a large part of red blood cells. This protein is vital to the function of red blood cells.

Red blood cells are transported around the body in a liquid known as plasma. Only 10% of blood plasma is made up of components that are used or produced by body cells. The rest of the plasma is made up of water. Some components of blood plasma are used by body cells in chemical reactions, some act as messengers or have a role in blood clotting and others are excreted from the body as waste.

(a) Name the main tissue that makes up the walls of the heart. **(1)**

(b) Explain why the heart is known as a double-pump. (Lines 1 to 2) **(3)**

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P 7 4 4 9 5 A 0 1 4 2 8

12. Question 5 · part c · 1 mark

June 2024 Paper 2R

Further practice · Attempt part c only.

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(c) Red blood cells are the most common type of cell found in the blood.

Which of these are also types of cell found in the blood?

(1)

- ☒ **A** phagocytes and platelets
- ☒ **B** antibodies and hormones
- ☒ **C** lymphocytes and phagocytes
- ☒ **D** platelets and lymphocytes

(d) (i) Name a component of the blood plasma that acts as a messenger. (Lines 12 to 14)

(1)

.....

(ii) Name a component of the blood that is involved in blood clotting.

(1)

.....

(e) Calculate the number of red blood cells found in the blood of an adult male with a mass of 95 kg. (Lines 3 to 7)

Give your answer in standard form.

(3)

number of red blood cells = ...

(Total for Question 5 = 10 marks)

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12. Question 5 · part c · 1 mark · Working and answer

Working and answer

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13. Question 5 · part d.ii · 1 mark

June 2024 Paper 2R

Further practice · Attempt part d.ii only.

5 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

The heart is a double-pump that is part of the human circulatory system. It helps ensure that the blood and its components reach all cells in the body.

Some components of the blood are made in the bone marrow. An average adult with a mass of 70 kg has 5.5 dm^3 of blood. About 45% of the blood is made up of different cells. There are several different types of cell in the blood but red blood cells are the most common. In males, 1.0 dm^3 of blood contains 5.9×10^{12} red blood cells. In females the same volume of blood contains 5.5×10^{12} red blood cells. Haemoglobin makes up a large part of red blood cells. This protein is vital to the function of red blood cells.

Red blood cells are transported around the body in a liquid known as plasma. Only 10% of blood plasma is made up of components that are used or produced by body cells. The rest of the plasma is made up of water. Some components of blood plasma are used by body cells in chemical reactions, some act as messengers or have a role in blood clotting and others are excreted from the body as waste.

(a) Name the main tissue that makes up the walls of the heart. **(1)**

(b) Explain why the heart is known as a double-pump. (Lines 1 to 2) **(3)**

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P 7 4 4 9 5 A 0 1 4 2 8

13. Question 5 · part d.ii · 1 mark

June 2024 Paper 2R

Further practice · Attempt part d.ii only.

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(c) Red blood cells are the most common type of cell found in the blood.

Which of these are also types of cell found in the blood?

(1)

- ☒ **A** phagocytes and platelets
- ☒ **B** antibodies and hormones
- ☒ **C** lymphocytes and phagocytes
- ☒ **D** platelets and lymphocytes

(d) (i) Name a component of the blood plasma that acts as a messenger. (Lines 12 to 14)

(1)

.....

(ii) Name a component of the blood that is involved in blood clotting.

(1)

.....

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(e) Calculate the number of red blood cells found in the blood of an adult male with a mass of 95 kg. (Lines 3 to 7)

Give your answer in standard form.

(3)

number of red blood cells = ...

(Total for Question 5 = 10 marks)

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13. Question 5 · part d.ii · 1 mark · Working and answer

Working and answer

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14. Question 5 · part d.i · 1 mark

June 2024 Paper 2R

Further practice · Attempt part d.i only.

5 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

The heart is a double-pump that is part of the human circulatory system. It helps ensure that the blood and its components reach all cells in the body.

Some components of the blood are made in the bone marrow. An average adult with a mass of 70 kg has 5.5 dm^3 of blood. About 45% of the blood is made up of different cells. There are several different types of cell in the blood but red blood cells are the most common. In males, 1.0 dm^3 of blood contains 5.9×10^{12} red blood cells. In females the same volume of blood contains 5.5×10^{12} red blood cells. Haemoglobin makes up a large part of red blood cells. This protein is vital to the function of red blood cells.

Red blood cells are transported around the body in a liquid known as plasma. Only 10% of blood plasma is made up of components that are used or produced by body cells. The rest of the plasma is made up of water. Some components of blood plasma are used by body cells in chemical reactions, some act as messengers or have a role in blood clotting and others are excreted from the body as waste.

(a) Name the main tissue that makes up the walls of the heart. **(1)**

(b) Explain why the heart is known as a double-pump. (Lines 1 to 2) **(3)**

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P 7 4 4 9 5 A 0 1 4 2 8

14. Question 5 · part d.i · 1 mark

June 2024 Paper 2R

Further practice · Attempt part d.i only.

DO NOT WRITE IN THIS AREA

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(c) Red blood cells are the most common type of cell found in the blood.

Which of these are also types of cell found in the blood?

(1)

- ☒ **A** phagocytes and platelets
- ☒ **B** antibodies and hormones
- ☒ **C** lymphocytes and phagocytes
- ☒ **D** platelets and lymphocytes

(d) (i) Name a component of the blood plasma that acts as a messenger. (Lines 12 to 14)

(1)

.....

(ii) Name a component of the blood that is involved in blood clotting.

(1)

.....

(e) Calculate the number of red blood cells found in the blood of an adult male with a mass of 95 kg. (Lines 3 to 7)

Give your answer in standard form.

(3)

number of red blood cells = ...

(Total for Question 5 = 10 marks)

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14. Question 5 · part d.i · 1 mark · Working and answer

Working and answer

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Practice workbook · Describe phagocytosis, antibody production and barriers to pathogens

White-cell defence (engulfing and digesting, antibodies) and the physical and chemical barriers that keep pathogens out.

9 question parts: 1 worked-example retry and 8 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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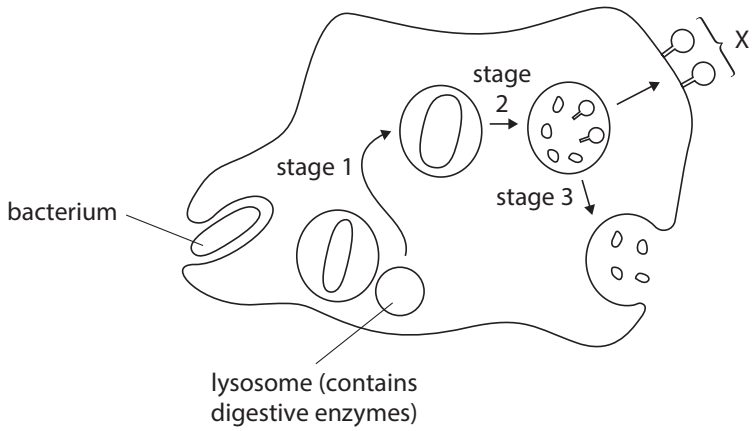
Type ID: HBI-DESCRIBE-PHAGOCYTOSIS-AND-BODY-DEFENCES

1. Question 6 · part a.ii · 3 marks

June 2023 Paper 1

Worked-example retry · Attempt part a.ii only.

6 The diagram shows a white blood cell engulfing a bacterium and the stages that follow.



(a) (i) State the name of the process used by white blood cells to engulf bacteria.

(1)

(ii) Describe what is happening at stages 1, 2 and 3 shown in the diagram.

(3)

stage 1

stage 2

stage 3

(iii) Suggest the function of the structures labelled X.

(2)

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1. Question 6 · part a.ii · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

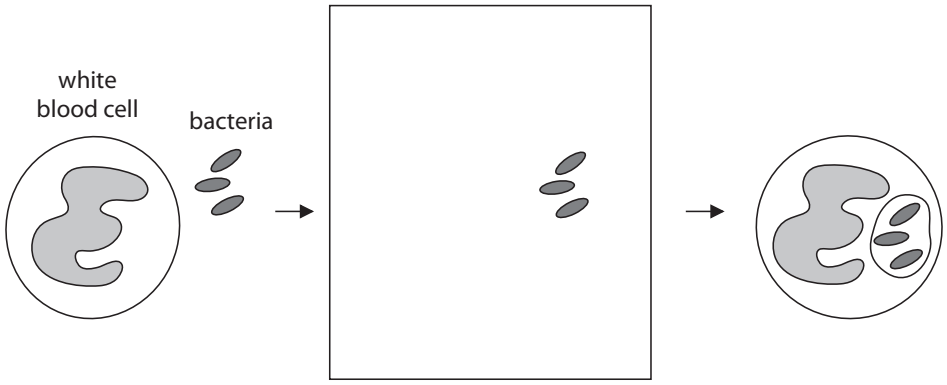
2. Question 2 · part a.i · 1 mark

June 2019 Paper 1

Further practice · Attempt part a.i only.

2 (a) Bacteria can cause disease in humans.

The diagram shows how one type of white blood cell helps to defend the body against disease.



(i) Complete the diagram by drawing the shape of the white blood cell at stage 2.

(1)

(ii) The box lists words associated with bacteria and disease.

acids	enzymes	erythrocytes	lymphocytes	phagocytes	toxins

Use words from the box to complete the sentences.

(2)

White blood cells called ..... engulf bacteria.

These white blood cells contain ..... to digest bacteria.

(iii) One way that white blood cells defend the body from disease is shown in the diagram.

State another way in which white blood cells defend the body against disease.

(1)

.....

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2. Question 2 · part a.i · 1 mark · Working and answer

Working and answer

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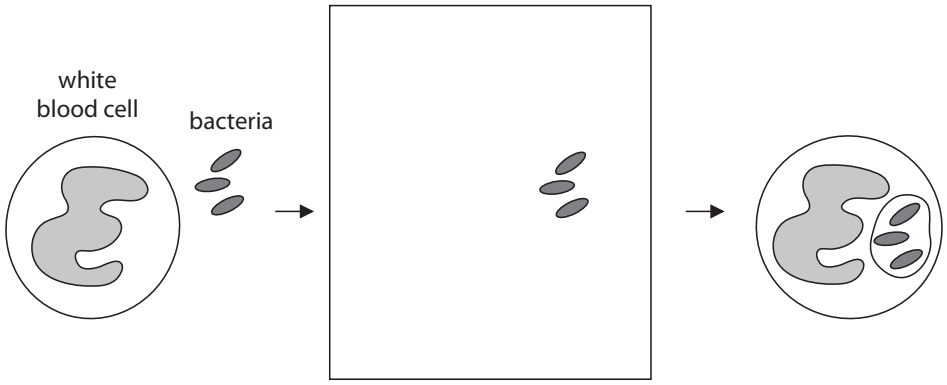
3. Question 2 · part a.ii · 2 marks

June 2019 Paper 1

Further practice · Attempt part a.ii only.

2 (a) Bacteria can cause disease in humans.

The diagram shows how one type of white blood cell helps to defend the body against disease.



(i) Complete the diagram by drawing the shape of the white blood cell at stage 2.

(1)

(ii) The box lists words associated with bacteria and disease.

acids	enzymes	erythrocytes	lymphocytes	phagocytes	toxins

Use words from the box to complete the sentences.

(2)

White blood cells called ..... engulf bacteria.

These white blood cells contain ..... to digest bacteria.

(iii) One way that white blood cells defend the body from disease is shown in the diagram.

State another way in which white blood cells defend the body against disease.

(1)

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3. Question 2 · part a.ii · 2 marks · Working and answer

Working and answer

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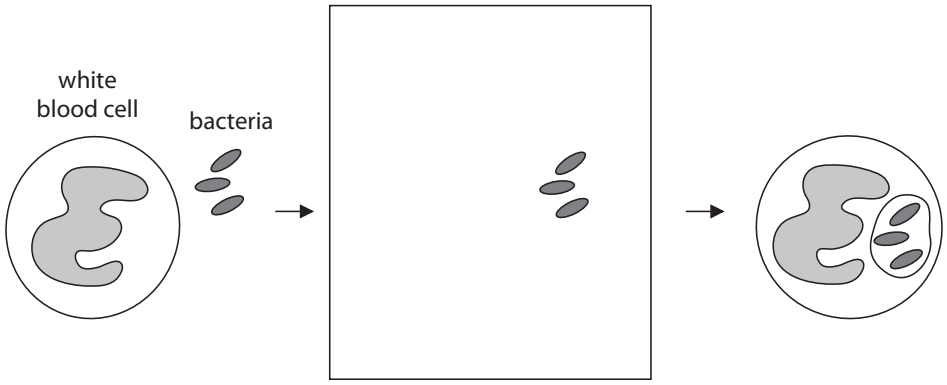
4. Question 2 · part a.iii · 1 mark

June 2019 Paper 1

Further practice · Attempt part a.iii only.

2 (a) Bacteria can cause disease in humans.

The diagram shows how one type of white blood cell helps to defend the body against disease.



(i) Complete the diagram by drawing the shape of the white blood cell at stage 2.

(1)

(ii) The box lists words associated with bacteria and disease.

acids	enzymes	erythrocytes	lymphocytes	phagocytes	toxins

Use words from the box to complete the sentences.

(2)

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These white blood cells contain ..... to digest bacteria.

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State another way in which white blood cells defend the body against disease.

(1)

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4. Question 2 · part a.iii · 1 mark

June 2019 Paper 1

Further practice · Attempt part a.iii only.

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(b) The table lists structures found in some cells.

Place ticks in boxes to show which structures are found in bacterial cells and which are found in human skin cells.

One row has been completed for you.

(3)

Structures	Bacterial cell	Human skin cell
nucleus		
DNA		
cytoplasm	✓	✓
cell wall		

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(c) Viruses can also cause diseases in humans.

Many viruses contain RNA as their genetic material.

Which statement describes the structure of RNA?

(1)

☒ **A** a double-stranded helix containing the bases ATGC

☒ **B** a double-stranded helix containing the bases AUGC

☒ **C** a single-stranded helix containing the bases ATGC

☒ **D** a single-stranded helix containing the bases AUGC

(Total for Question 2 = 8 marks)

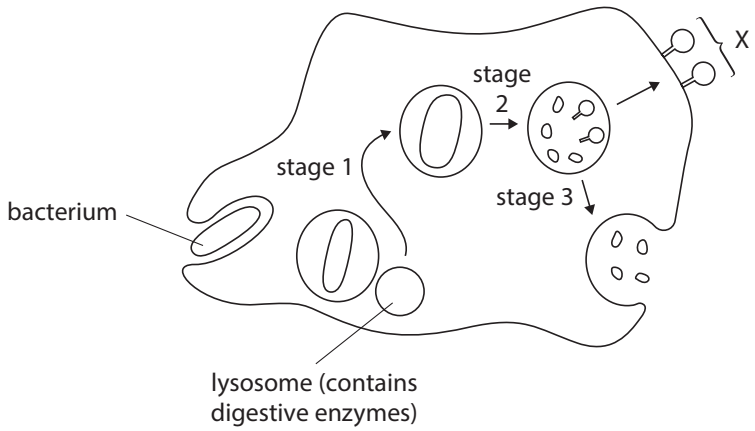
Working and answer

5. Question 6 · part a.i · 1 mark

June 2023 Paper 1

Further practice · Attempt part a.i only.

6 The diagram shows a white blood cell engulfing a bacterium and the stages that follow.



(a) (i) State the name of the process used by white blood cells to engulf bacteria.

(1)

(ii) Describe what is happening at stages 1, 2 and 3 shown in the diagram.

(3)

stage 1

stage 2

stage 3

(iii) Suggest the function of the structures labelled X.

(2)

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5. Question 6 · part a.i · 1 mark · Working and answer

Working and answer

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6. Question 8 · part a.i · 1 mark

June 2025 Paper 2R

Further practice · Attempt part a.i only.

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8 (a) Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

The human body has physical and chemical barriers that reduce the risk of pathogens entering the body. These barriers are known as the first line of defence. If the barriers are penetrated, then the second line of defence comes into action.

The cells involved in the second line of defence form part of the non-specific line of defence. They do not target specific pathogens but they play an important role in stimulating cells involved in the next stages of defence against disease.

The third line of defence involves specific immune responses by cells. These cells rely on the presence and detection of marker chemicals found on the surface of microscopic invaders. The cells of the specific immune system have receptor proteins that interact with the marker chemicals. The marker chemicals cause the cells to become activated and produce antibodies. The production of antibodies in this way is known as natural, active immunity and results in long-term immunity. However, there are other ways that antibodies can provide protection against disease. Natural passive immunity and artificial passive immunity are two similar processes that minimise the risk of too much harm being caused by pathogens in the body.

(i) Name a physical barrier that prevents the entry of pathogens into the body. (lines 1 to 3)

(1)

.....

(ii) Suggest how chemical barriers can reduce the harm caused to the body by pathogens. (lines 1 to 3)

(1)

.....

(iii) Phagocytes are one type of cell involved in the second line of defence.

Describe the process of phagocytosis. (lines 4 to 6)

(2)

.....

.....

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6. Question 8 · part a.i · 1 mark · Working and answer

Working and answer

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7. Question 8 · part a.ii · 1 mark

June 2025 Paper 2R

Further practice · Attempt part a.ii only.

DO NOT WRITE IN THIS AREA

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8 (a) Read the passage below.

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(1)

.....

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Describe the process of phagocytosis. (lines 4 to 6)

(2)

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7. Question 8 · part a.ii · 1 mark · Working and answer

Working and answer

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8. Question 8 · part a.iii · 2 marks

June 2025 Paper 2R

Further practice · Attempt part a.iii only.

DO NOT WRITE IN THIS AREA

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(i) Name a physical barrier that prevents the entry of pathogens into the body. (lines 1 to 3)

(1)

.....

(ii) Suggest how chemical barriers can reduce the harm caused to the body by pathogens. (lines 1 to 3)

(1)

.....

(iii) Phagocytes are one type of cell involved in the second line of defence.

Describe the process of phagocytosis. (lines 4 to 6)

(2)

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8. Question 8 · part a.iii · 2 marks

June 2025 Paper 2R

Further practice · Attempt part a.iii only.

(iv) State the scientific name for the marker chemicals found on the surface of microscopic invaders. (lines 7 to 11)

(1)

(v) Suggest what is meant by a specific immune response. (line 7)

(2)

(vi) Describe the differences between natural passive immunity and artificial passive immunity. (lines 14 to 16)

You should give examples in your answer.

(2)

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8. Question 8 · part a.iii · 2 marks · Working and answer

Working and answer

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9. Question 4 · part b.ii · 2 marks

November 2024 Paper 2

Further practice · Attempt part b.ii only.

4 Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

When looking at data, it is clear that people in lower income countries are more likely to die from infectious diseases (approximately 240 per 100 000) than non-infectious diseases (approximately 180 per 100 000). Infectious diseases such as diarrhoea and malaria can be easily eliminated or treated. It is also possible to vaccinate against diseases such as TB. These diseases are uncommon in higher income countries because of preventative medicine.

The only common diseases present in both higher and lower income countries are respiratory tract infections. These infections are usually influenza, pneumonia and bronchitis. Whilst influenza is a viral infection, the other infections are usually bacterial and can be treated. Many people at risk in the higher income countries are encouraged to get preventative treatment. These are often provided free of charge for the elderly who are most at risk and those who work in indoor spaces such as schools, offices and hospitals. The effect on mortality in lower income countries is more than two times as great as in higher income countries.

The greatest killers in higher income countries are heart disease and stroke. These two diseases kill approximately twice as many people per 100 000 as in lower income countries. These are often seen as lifestyle diseases. When we compare the number of deaths caused by non-infectious diseases, in higher income countries there are 445 deaths per 100 000 and in lower income countries there are 185 deaths per 100 000. There are also more cases of other diseases such as dementia and cancers in higher income countries.

(a) State one way the infectious diseases diarrhoea and malaria can be prevented (lines 3 and 4).

(2)

diarrhoea

malaria

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9. Question 4 · part b.ii · 2 marks

November 2024 Paper 2

Further practice · Attempt part b.ii only.

(b) (i) State how pneumonia and bronchitis can be treated (lines 8 to 10).

(1)

(ii) Give one reason why elderly people and people who work in indoor spaces are at risk from respiratory tract infections. (lines 12 and 13).

(2)

elderly people

indoor spaces

(c) (i) Give two factors that may contribute to lifestyle diseases (line 18).

(2)

1

2

(ii) In a higher income country of 5 000 000 people, calculate the number of people who will have non-infectious diseases (lines 20 and 21).

Give your answer to three significant figures.

(3)

number of people =

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9. Question 4 · part b.ii · 2 marks · Working and answer

Working and answer

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Practice workbook · Name heart chambers and major vessels from a diagram

Labelled heart or circulation diagrams asking for a chamber, a connected vessel, or properties of blood in it.

13 question parts: 1 worked-example retry and 12 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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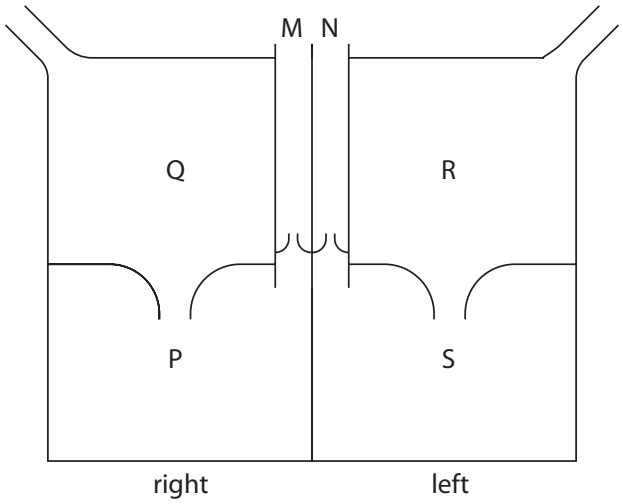
Type ID: HBI-NAME-HEART-CHAMBERS-AND-VESSELS

1. Question 1 · part a.i · 1 mark

January 2022 Paper 1

Worked-example retry · Attempt part a.i only.

1 The diagram shows a human heart in simple form.



(a) (i) Which of these is the name of blood vessel M?

(1)

- ☒ **A** aorta
- ☒ **B** pulmonary artery
- ☒ **C** pulmonary vein
- ☒ **D** vena cava

(ii) Which of these is the name of blood vessel N?

(1)

- ☒ **A** aorta
- ☒ **B** pulmonary artery
- ☒ **C** pulmonary vein
- ☒ **D** vena cava

(iii) Draw arrows on the diagram to show the flow of blood through the heart.

(2)

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1. Question 1 · part a.i · 1 mark · Working and answer

Working and answer

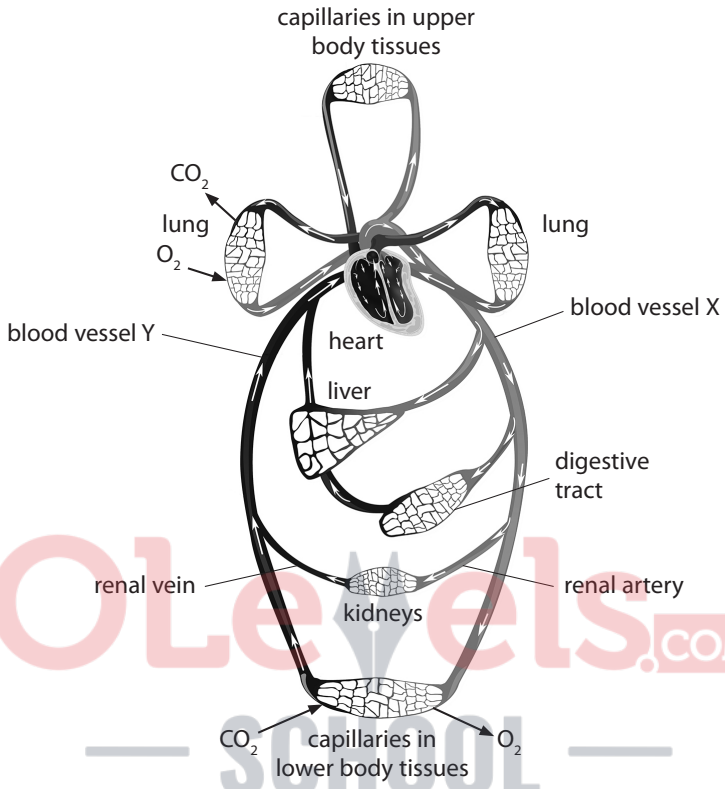
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2. Question 6 · part a.i · 1 mark

June 2019 Paper 2

Further practice · Attempt part a.i only.

6 The diagram shows the human circulatory system.



(Source: © Designua/Shutterstock)

(a) (i) What is the name of blood vessel X?

(1)

- ☒ **A** aorta
- ☒ **B** pulmonary artery
- ☒ **C** pulmonary vein
- ☒ **D** vena cava

(ii) What is the name of blood vessel Y?

(1)

- ☒ **A** aorta
- ☒ **B** pulmonary artery
- ☒ **C** pulmonary vein
- ☒ **D** vena cava

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2. Question 6 · part a.i · 1 mark · Working and answer

Working and answer

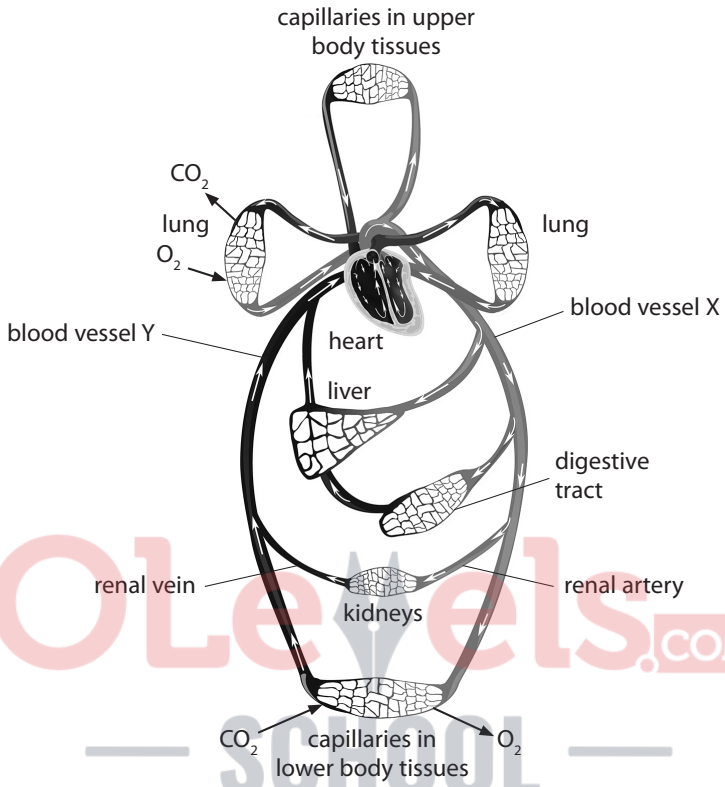
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3. Question 6 · part a.ii · 1 mark

June 2019 Paper 2

Further practice · Attempt part a.ii only.

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(Source: © Designua/Shutterstock)

(a) (i) What is the name of blood vessel X?

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3. Question 6 · part a.ii · 1 mark

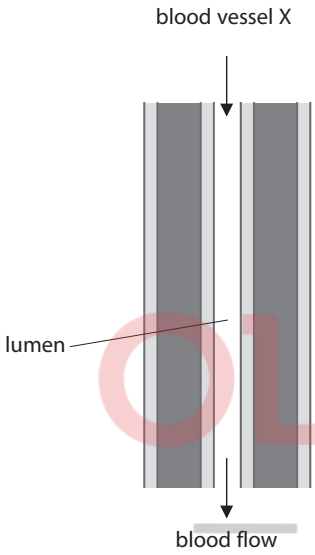
June 2019 Paper 2

Further practice · Attempt part a.ii only.

(iii) The diagram shows a longitudinal section through blood vessel X.

Draw a diagram to show a longitudinal section through blood vessel Y.

(3)



(iv) Explain the differences in the structures of blood vessel X and blood vessel Y.

(3)

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3. Question 6 · part a.ii · 1 mark · Working and answer

Working and answer

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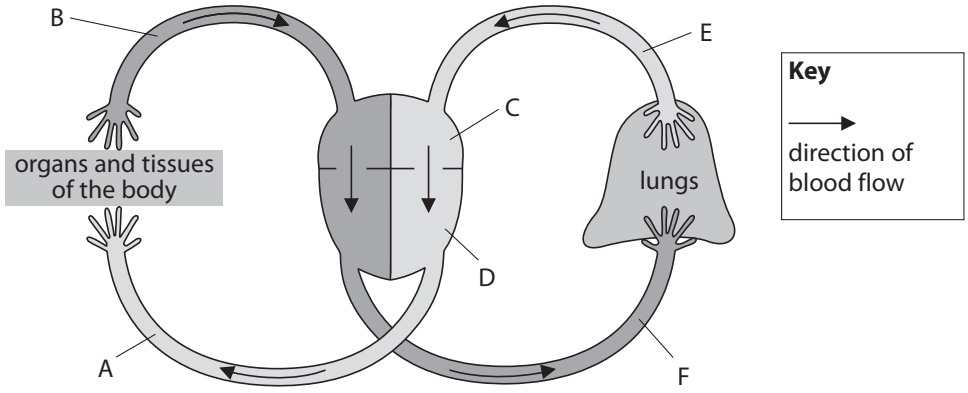
4. Question 4 · part a · 2 marks

January 2020 Paper 1

Further practice · Attempt part a only.

4 William Harvey was the first person to describe the circulation of blood around the body.

The diagram shows a model of the human circulatory system based on the research carried out by Harvey.



(a) Harvey suggested that each side of the heart was divided into two chambers.

State the names of chambers C and D.

(2)

chamber C .....

chamber D .....

(b) Harvey also suggested that blood is transported to and from the heart in different types of vessel.

Explain how the diagram supports Harvey's suggestion.

(4)

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4. Question 4 · part a · 2 marks · Working and answer

Working and answer

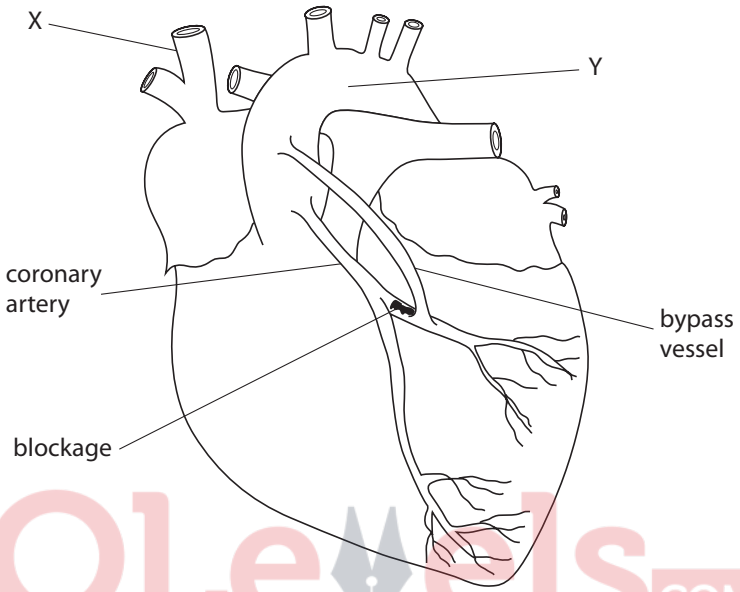
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5. Question 3 · part a.i · 1 mark

November 2020 Paper 2R

Further practice · Attempt part a.i only.

3 The diagram shows a person's heart. The person has had a heart bypass operation because a coronary artery had become partly blocked.



(a) (i) What is the name of blood vessel Y?

(1)

- ☒ **A** aorta
- ☒ **B** pulmonary artery
- ☒ **C** pulmonary vein
- ☒ **D** vena cava

(ii) To which part of the heart is blood vessel X connected?

(1)

- ☒ **A** left atrium
- ☒ **B** left ventricle
- ☒ **C** right atrium
- ☒ **D** right ventricle

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5. Question 3 · part a.i · 1 mark · Working and answer

Working and answer

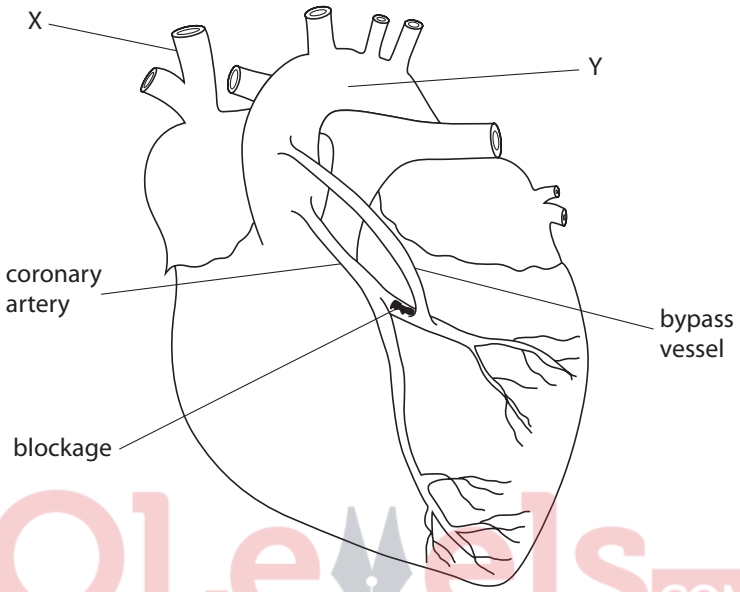
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6. Question 3 · part a.ii · 1 mark

November 2020 Paper 2R

Further practice · Attempt part a.ii only.

3 The diagram shows a person's heart. The person has had a heart bypass operation because a coronary artery had become partly blocked.



(a) (i) What is the name of blood vessel Y?

(1)

- ☒ **A** aorta
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- ☒ **C** pulmonary vein
- ☒ **D** vena cava

(ii) To which part of the heart is blood vessel X connected?

(1)

- ☒ **A** left atrium
- ☒ **B** left ventricle
- ☒ **C** right atrium
- ☒ **D** right ventricle

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6. Question 3 · part a.ii · 1 mark

November 2020 Paper 2R

Further practice · Attempt part a.ii only.

(b) (i) Before the bypass operation, the coronary artery was becoming blocked at the point shown on the diagram.

Draw a circle on the diagram to show the area of the heart that was affected by the blockage before the bypass operation.

(1)

(ii) Describe what effect this blockage would have on the heart tissue if the bypass vessel was not inserted.

(3)

(iii) Suggest how the bypass vessel allows the heart to work more effectively.

(2)

(Total for Question 3 = 8 marks)

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6. Question 3 · part a.ii · 1 mark · Working and answer

Working and answer

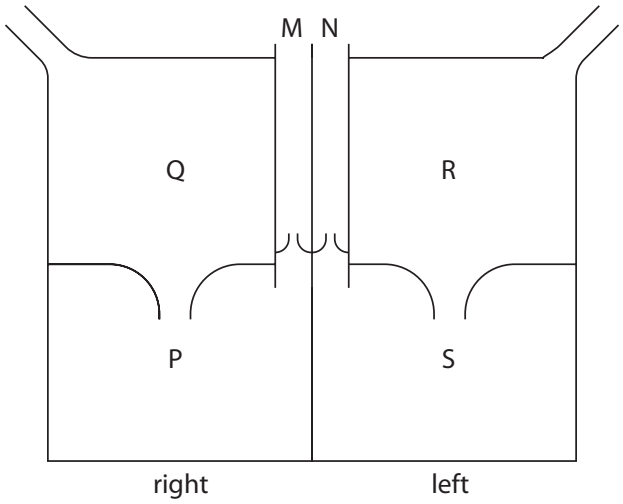
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7. Question 1 · part a.ii · 1 mark

January 2022 Paper 1

Further practice · Attempt part a.ii only.

1 The diagram shows a human heart in simple form.



(a) (i) Which of these is the name of blood vessel M?

(1)

- ☒ **A** aorta
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(1)

- ☒ **A** aorta
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- ☒ **C** pulmonary vein
- ☒ **D** vena cava

(iii) Draw arrows on the diagram to show the flow of blood through the heart.

(2)

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7. Question 1 · part a.ii · 1 mark · Working and answer

Working and answer

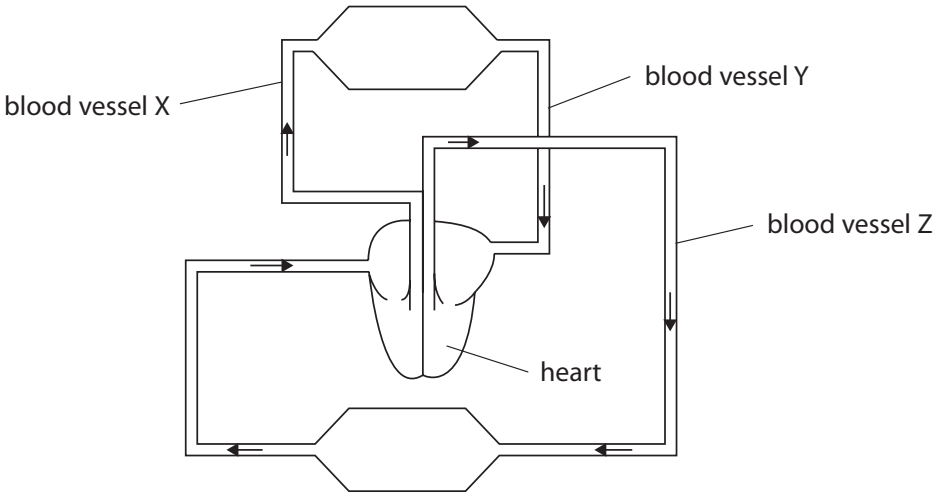
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8. Question 1 · part a · 1 mark

June 2022 Paper 2

Further practice · Attempt part a only.

1 The diagram shows a simple version of the human circulatory system.



(a) Name blood vessel X.

(1)

.....

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8. Question 1 · part a · 1 mark

June 2022 Paper 2

Further practice · Attempt part a only.

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(b) The box gives words about the circulatory system.

body	oxygenated	lungs	ventricle
	deoxygenated	atrium	

Complete the sentences using words from the box.

(5)

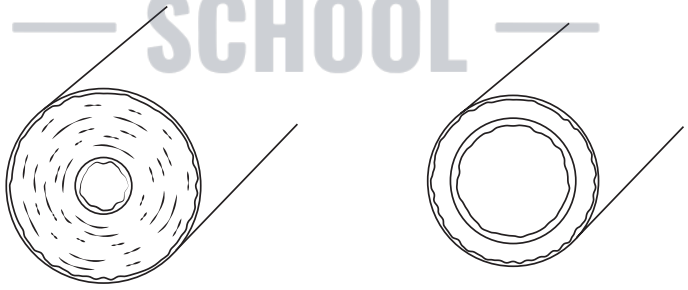
Blood vessel Y transports ..... blood from the

..... to the left ..... of the heart.

Blood vessel Z transports blood from the left ..... of the heart

to the .....

(c) The diagram shows the structure of blood vessel X and blood vessel Y.



blood vessel X blood vessel Y

Compare the structure of blood vessel X with the structure of blood vessel Y.

(3)

.....

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.....

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.....

(Total for Question 1 = 9 marks)

8. Question 1 · part a · 1 mark · Working and answer

Working and answer

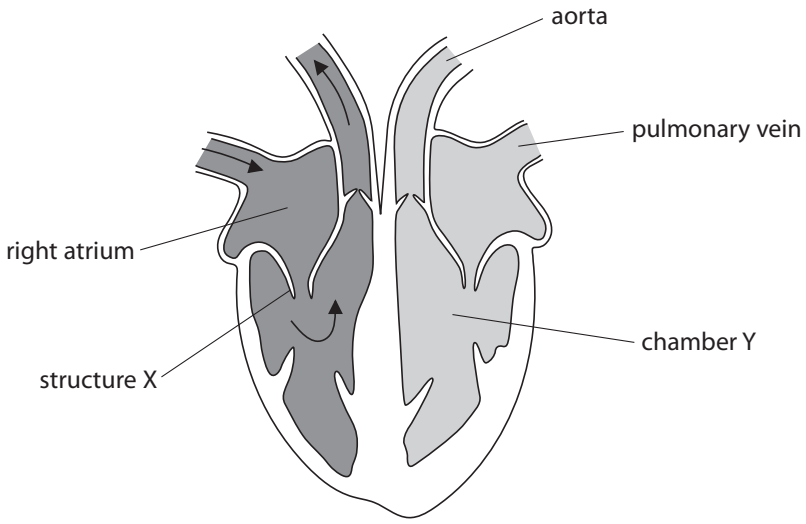
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9. Question 2 · part a.i · 1 mark

January 2023 Paper 2

Further practice · Attempt part a.i only.

2 (a) The diagram shows a section through the heart.



(i) Name chamber Y. **(1)**

(ii) Explain the function of structure X. **(2)**

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P 7 2 6 0 2 A 0 4 2 8

9. Question 2 · part a.i · 1 mark · Working and answer

Working and answer

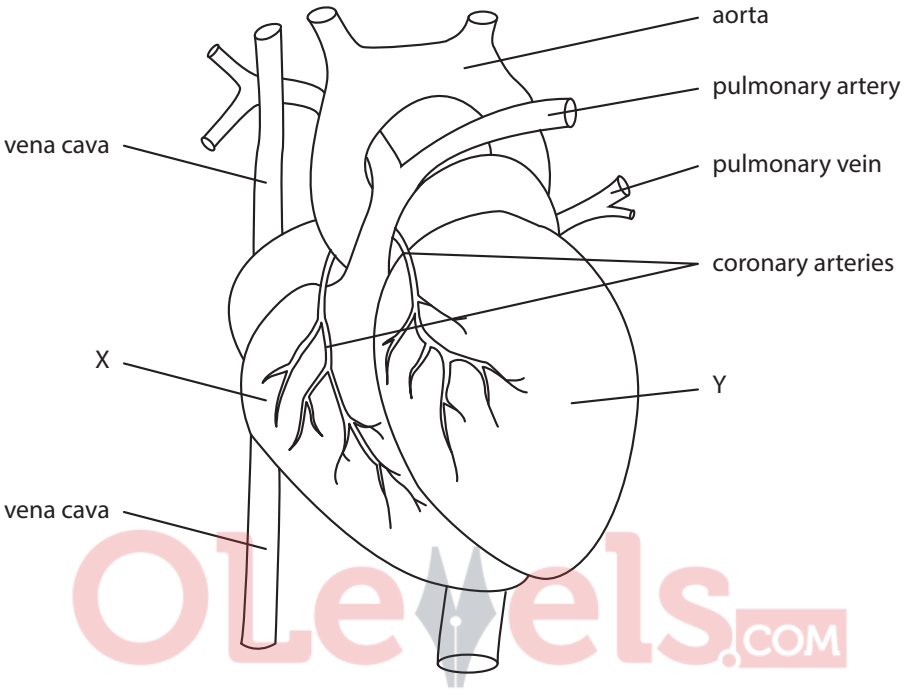
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10. Question 7 · part a · 1 mark

November 2023 Paper 1

Further practice · Attempt part a only.

7 The diagram shows the heart as seen from the front.



(a) The vena cava carries deoxygenated blood.

Name a different blood vessel, shown on the diagram, that also carries deoxygenated blood.

(1)

.....

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10. Question 7 · part a · 1 mark

November 2023 Paper 1

Further practice · Attempt part a only.

(b) Blood chambers X and Y have walls that are different in thickness.

Explain this difference.

(5)

(c) The coronary arteries can become blocked.

Explain the effect of this blockage on the functioning of the heart.

(3)

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10. Question 7 · part a · 1 mark · Working and answer

Working and answer

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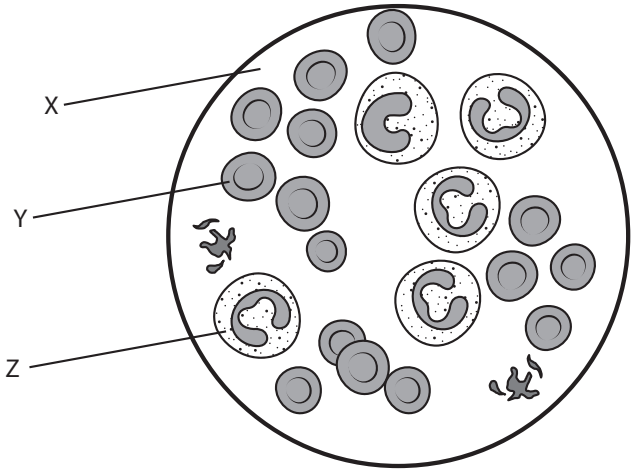
11. Question 2 · part c.i · 1 mark

June 2024 Paper 1R

Further practice · Attempt part c.i only.

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

2 (a) The diagram shows components, X, Y and Z, in a sample of human blood.



(i) Name components X, Y and Z.

(3)

X .....

Y .....

Z .....

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(ii) Describe the function of component Y.

(2)

.....

.....

.....

.....

(iii) A sample of blood from a patient is analysed in a hospital.

A scientist finds that the number of component Z in the blood sample is higher than normal.

State a possible reason for the increase in the number of component Z.

(1)

.....

.....

11. Question 2 · part c.i · 1 mark

June 2024 Paper 1R

Further practice · Attempt part c.i only.

(b) Blood is transported around the body inside blood vessels.

Draw **one** straight line from each blood vessel to a feature of the blood vessel.

(2)

Blood vessel	Feature of blood vessel
artery	contains valves to prevent backflow of blood
vein	has muscular walls to maintain high blood pressure
capillary	has thin walls for exchange of materials

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11. Question 2 · part c.i · 1 mark

June 2024 Paper 1R

Further practice · Attempt part c.i only.

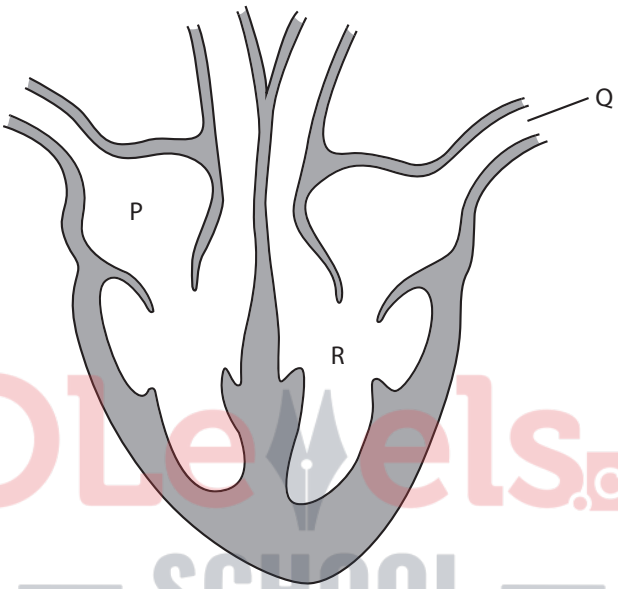
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(c) The heart pumps blood around the body.

The diagram shows a human heart.



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(i) The heart is divided into two halves.

Each half is divided into two chambers.

Name chamber R.

(1)

.....

(ii) Describe the route that the blood travels from part P to part Q.

(4)

.....

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(Total for Question 2 = 13 marks)

11. Question 2 · part c.i · 1 mark · Working and answer

Working and answer

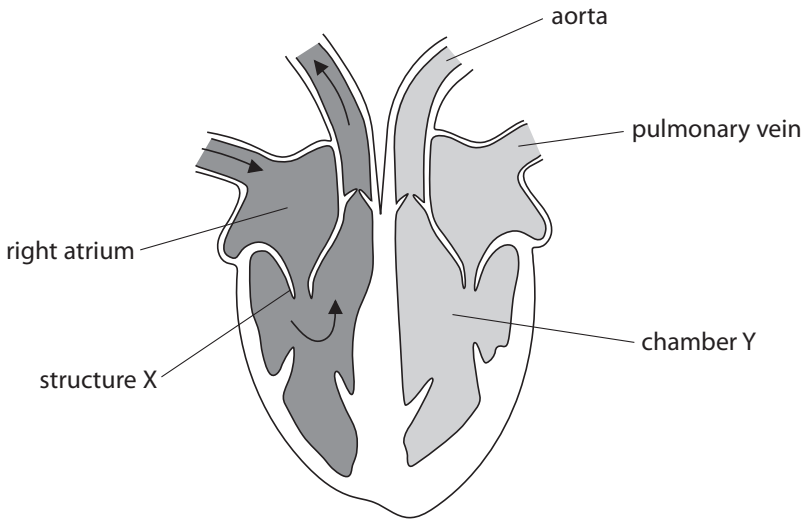
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12. Question 2 · part a.iv · 1 mark

January 2023 Paper 2

Further practice · Attempt part a.iv only.

2 (a) The diagram shows a section through the heart.



(i) Name chamber Y. **(1)**

(ii) Explain the function of structure X. **(2)**

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P 7 2 6 0 2 A 0 4 2 8

12. Question 2 · part a.iv · 1 mark

January 2023 Paper 2

Further practice · Attempt part a.iv only.

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(iii) The table lists some properties of blood.

Add ticks (✓) to those boxes that show the properties of blood in the aorta.

(2)

Properties of blood	Blood in the aorta
oxygenated	
deoxygenated	
high pressure	
medium pressure	
low pressure	

(iv) State where blood in the left side of the heart has come from.

(1)

.....

(b) In one year in the United Kingdom, there were 173 heart transplants.

Heart transplant operations require hearts to be donated.

Some hearts cannot be used because the donor is too old.

Give two other reasons why a heart might not be suitable for a transplant.

(2)

1 .....

.....

2 .....

.....

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12. Question 2 · part a.iv · 1 mark · Working and answer

Working and answer

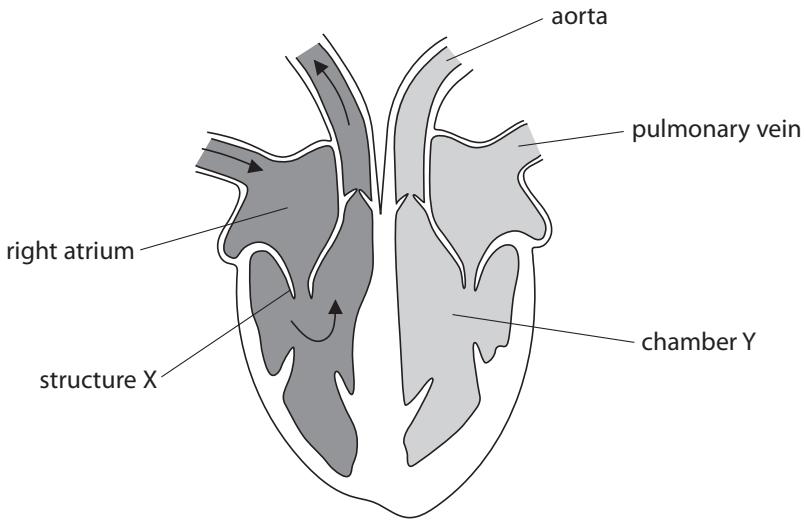
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13. Question 2 · part a.iii · 2 marks

January 2023 Paper 2

Further practice · Attempt part a.iii only.

2 (a) The diagram shows a section through the heart.



(i) Name chamber Y. **(1)**

(ii) Explain the function of structure X. **(2)**

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P 7 2 6 0 2 A 0 4 2 8

13. Question 2 · part a.iii · 2 marks

January 2023 Paper 2

Further practice · Attempt part a.iii only.

DO NOT WRITE IN THIS AREA

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DO NOT WRITE IN THIS AREA

(iii) The table lists some properties of blood.

Add ticks (✓) to those boxes that show the properties of blood in the aorta.

(2)

Properties of blood	Blood in the aorta
oxygenated	
deoxygenated	
high pressure	
medium pressure	
low pressure	

(iv) State where blood in the left side of the heart has come from.

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Heart transplant operations require hearts to be donated.

Some hearts cannot be used because the donor is too old.

Give two other reasons why a heart might not be suitable for a transplant.

(2)

1 .....

.....

2 .....

.....

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13. Question 2 · part a.iii · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

Practice workbook · Describe what red cells and plasma carry, and from where to where

Function of red blood cells and plasma in carrying oxygen, nutrients, wastes, hormones, heat and drugs around the body.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-DESCRIBE-BLOOD-TRANSPORT-ROLES

1. Question 6 · part b · 5 marks

June 2024 Paper 1

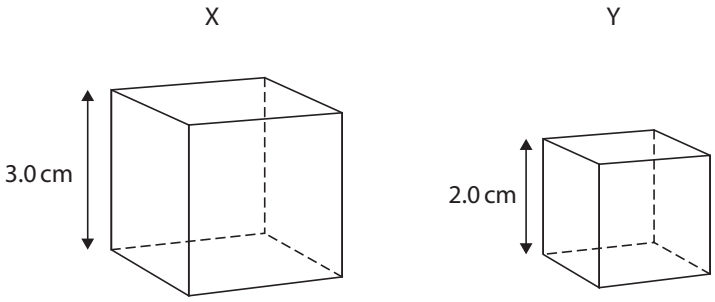
Worked-example retry · Attempt part b only.

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6 (a) Humans have a small surface area to volume ratio (SA : V) compared with some other organisms.

Cube X represents a human.

Cube Y represents an organism with a larger SA : V.



The table gives the SA : V of cube Y.

Cube	Surface area in cm^2	Volume in cm^3	SA : V
X			
Y	24.0	8.0	3:1

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(i) Calculate the surface area of cube X.

(2)

surface area = ... cm^2

(ii) Calculate the volume of cube X.

(2)

volume = ... cm^3

(iii) Calculate the SA : V of cube X.

(2)

SA : V = ... : 1

1. Question 6 · part b · 5 marks

June 2024 Paper 1

Worked-example retry · Attempt part b only.

(b) Multicellular organisms, with a small SA:V ratio, such as humans, need an efficient transport system to deliver substances to body cells.

Describe the role of the human blood transport system in delivering substances to body cells.

(5)

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1. Question 6 · part b · 5 marks

June 2024 Paper 1

Worked-example retry · Attempt part b only.

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(c) Bacteria are able to exchange materials directly with their environment as they have a very high SA : V ratio.

They do not have an internal transport system.

Describe how bacteria exchange materials with their environment.

(2)

(Total for Question 6 = 13 marks)

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1. Question 6 · part b · 5 marks · Working and answer

Working and answer

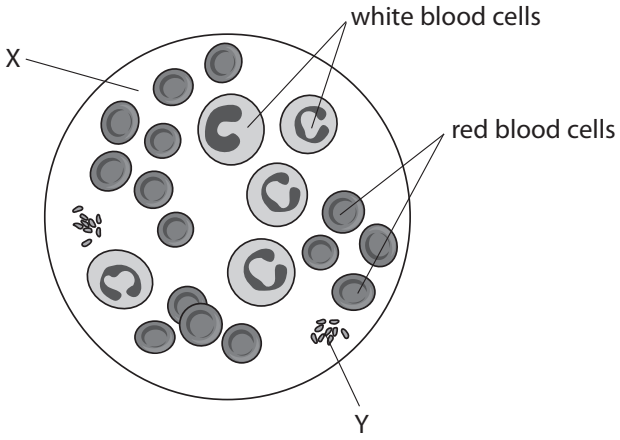
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2. Question 1 · part a · 1 mark

November 2020 Paper 1R

Further practice · Attempt part a only.

1 The diagram shows a sample of human blood.



(a) State the function of red blood cells.

(1)

.....

.....

(b) Give the name of X and Y.

(2)

X .....

Y .....

(c) (i) Calculate the ratio of red blood cells to white blood cells.

Write the ratio in its simplest form.

(2)

ratio = ...

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2. Question 1 · part a · 1 mark · Working and answer

Working and answer

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3. Question 1 · part a.i · 3 marks

November 2021 Paper 2

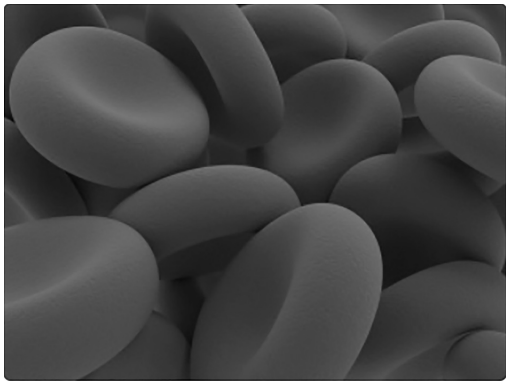
Further practice · Attempt part a.i only.

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1 (a) The photograph shows a group of red blood cells.



(Source: RomanenkoAlexey/Shutterstock)

(i) Describe the function of red blood cells.

(3)

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(ii) Explain how the structure of the red blood cell is adapted to its function.

(4)

3. Question 1 · part a.i · 3 marks · Working and answer

Working and answer

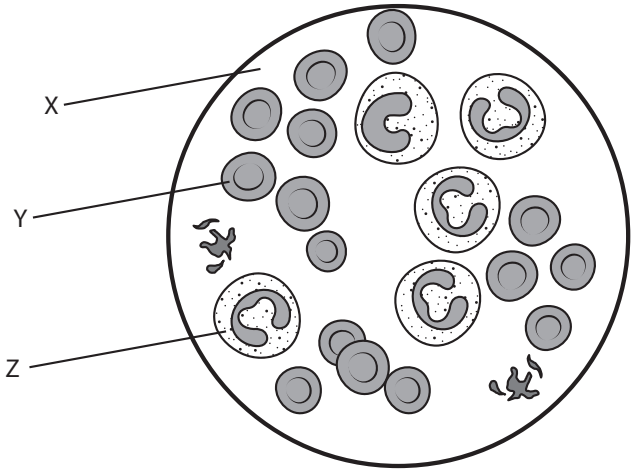
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4. Question 2 · part a.ii · 2 marks

June 2024 Paper 1R

Further practice · Attempt part a.ii only.

2 (a) The diagram shows components, X, Y and Z, in a sample of human blood.



(i) Name components X, Y and Z.

(3)

X .....

Y .....

Z .....

(ii) Describe the function of component Y.

(2)

.....

.....

.....

.....

(iii) A sample of blood from a patient is analysed in a hospital.

A scientist finds that the number of component Z in the blood sample is higher than normal.

State a possible reason for the increase in the number of component Z.

(1)

.....

.....

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4. Question 2 · part a.ii · 2 marks · Working and answer

Working and answer

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5. Question 5 · part b.ii · 2 marks

November 2020 Paper 2R

Further practice · Attempt part b.ii only.

5 A scientist investigates how the drug, atropine, affects the heart rate in humans.

Atropine paralyses the vagus nerve which carries impulses from the brain to the heart.

The heart rates of ten people are measured at intervals of two minutes for eight minutes, after which, atropine is injected.

The scientist continues to measure the heart rates.

The scientist's results are shown in the table.

Time in minutes	Heart rate in beats per minute
0	90
2	81
4	78
6	80
8	82
atropine injected → 9	100
10	120
11	148
12	160
13	138
15	128
17	120
19	116
21	109
23	100

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5. Question 5 · part b.ii · 2 marks

November 2020 Paper 2R

Further practice · Attempt part b.ii only.

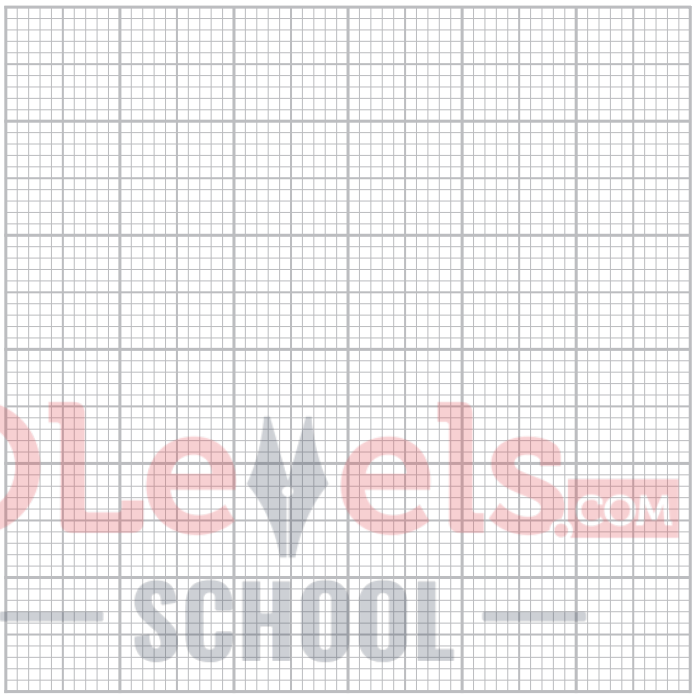
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(a) (i) Plot the scientist's results on the grid. Draw a curve of best fit.

(6)



(ii) Explain why ten people are injected with atropine rather than just one person.

(2)

(b) (i) Calculate the percentage change in the heart rate at 12 minutes compared to the heart rate at the start of the investigation.

(2)

percentage change = ...%

5. Question 5 · part b.ii · 2 marks

November 2020 Paper 2R

Further practice · Attempt part b.ii only.

(ii) Explain why this percentage change is not achieved as soon as atropine is injected.

(2)

(c) Suggest the function of the vagus nerve.

(2)

(Total for Question 5 = 14 marks)

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Working and answer

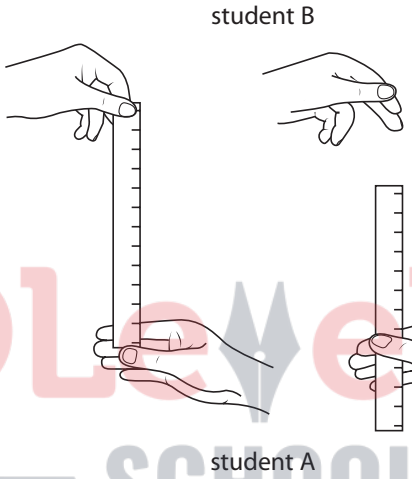
6. Question 5 · part c · 3 marks

November 2021 Paper 2

Further practice · Attempt part c only.

5 Two students use this method to investigate if caffeine affects the speed of a reflex action.

- student A sits in a chair holding out a hand with their thumb and forefinger open
- student B holds a metre rule with the 0 mm mark just above the open thumb and forefinger of student A
- student B releases the ruler
- student A catches the ruler as quickly as possible by closing their thumb and forefinger



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The method is repeated 10 times.

The student catching the ruler now drinks a cup of coffee and waits for 30 minutes.

The investigation is then repeated.

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6. Question 5 · part c · 3 marks

November 2021 Paper 2

Further practice · Attempt part c only.

The table shows the student's results.

Conditions \ Trial	Distance travelled by ruler in mm										
	1	2	3	4	5	6	7	8	9	10	mean
without coffee	210	190	200	205	207	195	198	195	199	201	200
with coffee	165	160	163	167	165	170	175	170	180	175	

(a) Complete the table by calculating the missing mean value.

(2)

(b) Coffee contains caffeine.

State the effect of caffeine on the reflex action.

(1)

(c) Explain why the student waited 30 minutes after drinking the coffee before continuing with the investigation.

(3)

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6. Question 5 · part c · 3 marks

November 2021 Paper 2

Further practice · Attempt part c only.

(d) The student who catches the ruler does not eat or drink anything for several hours before the investigation.

Suggest why this improves the validity of the investigation.

(2)

.....

.....

.....

.....

(e) A cup of coffee contains 100 mg of caffeine. A can of cola contains 50 mg of caffeine.

If the student catching the ruler had drunk a can of cola instead of coffee, estimate the mean distance travelled by the ruler.

(1)

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mean distance = ... mm

(Total for Question 5 = 9 marks)

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6. Question 5 · part c · 3 marks · Working and answer

Working and answer

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Practice workbook · Trace the route of blood through the heart, lungs and body

Ordering chambers, vessels and organs blood passes through, including the double-pump idea and arrows on diagrams.

9 question parts: 1 worked-example retry and 8 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-TRACE-DOUBLE-CIRCULATION-ROUTE

1. Question 5 · part a · 5 marks

June 2023 Paper 2

Worked-example retry · Attempt part a only.

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5 (a) A red blood cell travelling from the liver to the kidney passes through six different arteries and veins.

Complete the flow chart by naming these blood vessels in order. The first one has been done for you.

(5)

LIVER

↓

hepatic vein

↓

.....

↓

.....

↓

.....

↓

.....

↓

.....

↓

KIDNEY

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P 7 2 5 8 7 A 0 1 3 2 4

Turn over ▶

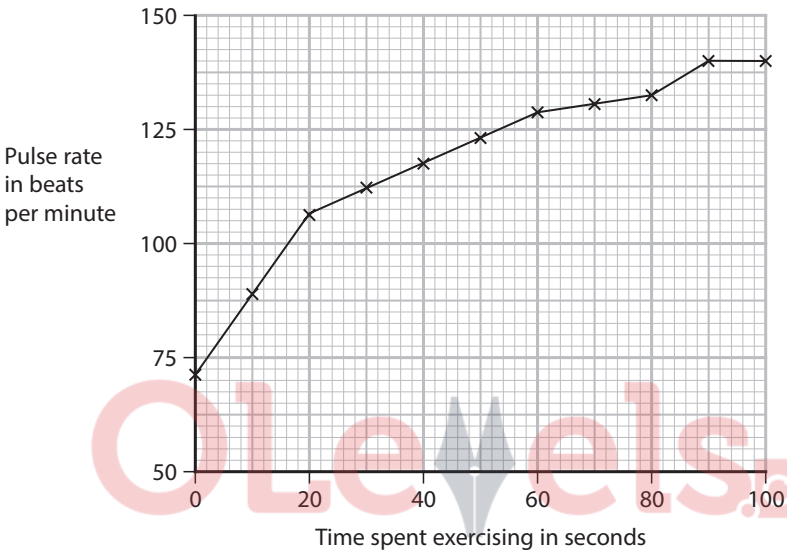
1. Question 5 · part a · 5 marks

June 2023 Paper 2

Worked-example retry · Attempt part a only.

(b) A student investigates the effect of time spent exercising on pulse rate.

The graph shows the student's results.



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1. Question 5 · part a · 5 marks · Working and answer

Working and answer

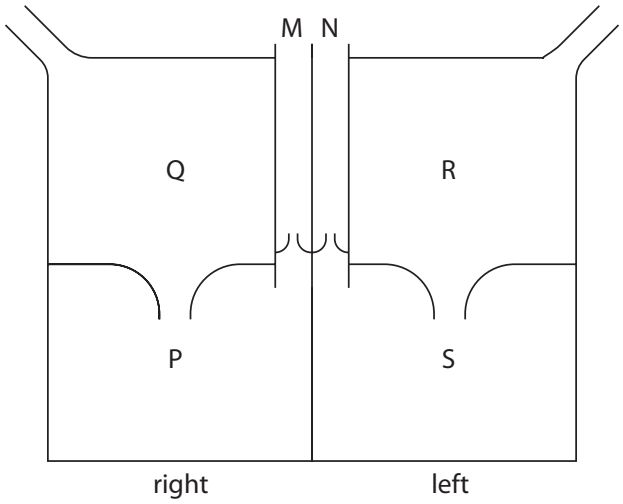
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2. Question 1 · part a.iii · 2 marks

January 2022 Paper 1

Further practice · Attempt part a.iii only.

1 The diagram shows a human heart in simple form.



(a) (i) Which of these is the name of blood vessel M?

(1)

- ☒ **A** aorta
- ☒ **B** pulmonary artery
- ☒ **C** pulmonary vein
- ☒ **D** vena cava

(ii) Which of these is the name of blood vessel N?

(1)

- ☒ **A** aorta
- ☒ **B** pulmonary artery
- ☒ **C** pulmonary vein
- ☒ **D** vena cava

(iii) Draw arrows on the diagram to show the flow of blood through the heart.

(2)

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2. Question 1 · part a.iii · 2 marks

January 2022 Paper 1

Further practice · Attempt part a.iii only.

(iv) Explain why the blood can only flow in the direction of the arrows.

(2)

(b) The width of the walls of each of the four chambers is measured.

These are the results.

0.2 cm 1.0 cm 1.5 cm 0.2 cm

(i) Complete the table by putting the correct width next to the letter of each chamber.

(3)

Chamber	Width in cm
P	
Q	
R	
S	

(ii) Explain the differences in the widths.

(3)

(Total for Question 1 = 12 marks)

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2. Question 1 · part a.iii · 2 marks · Working and answer

Working and answer

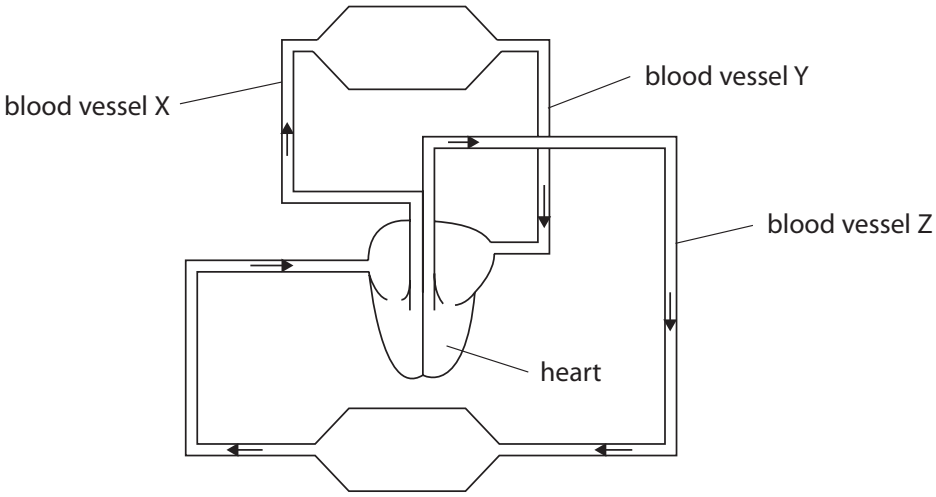
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3. Question 1 · part b · 5 marks

June 2022 Paper 2

Further practice · Attempt part b only.

1 The diagram shows a simple version of the human circulatory system.



(a) Name blood vessel X. **(1)**

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3. Question 1 · part b · 5 marks

June 2022 Paper 2

Further practice · Attempt part b only.

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(b) The box gives words about the circulatory system.

body	oxygenated	lungs	ventricle
	deoxygenated	atrium	

Complete the sentences using words from the box.

(5)

Blood vessel Y transports ..... blood from the

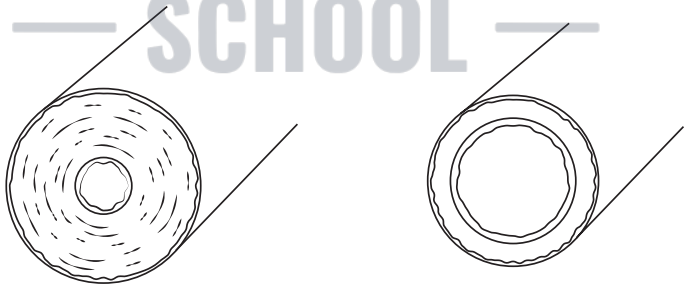
..... to the left ..... of the heart.

Blood vessel Z transports blood from the left ..... of the heart

to the .....

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(c) The diagram shows the structure of blood vessel X and blood vessel Y.



blood vessel X blood vessel Y

Compare the structure of blood vessel X with the structure of blood vessel Y.

(3)

.....

.....

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.....

.....

.....

(Total for Question 1 = 9 marks)

3. Question 1 · part b · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

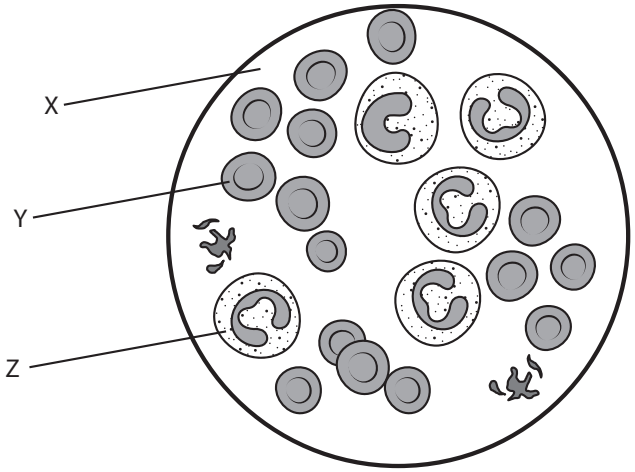
4. Question 2 · part c.ii · 4 marks

June 2024 Paper 1R

Further practice · Attempt part c.ii only.

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2 (a) The diagram shows components, X, Y and Z, in a sample of human blood.



(i) Name components X, Y and Z.

(3)

X .....

Y .....

Z .....

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(ii) Describe the function of component Y.

(2)

.....

.....

.....

.....

(iii) A sample of blood from a patient is analysed in a hospital.

A scientist finds that the number of component Z in the blood sample is higher than normal.

State a possible reason for the increase in the number of component Z.

(1)

.....

.....

4. Question 2 · part c.ii · 4 marks

June 2024 Paper 1R

Further practice · Attempt part c.ii only.

(b) Blood is transported around the body inside blood vessels.

Draw **one** straight line from each blood vessel to a feature of the blood vessel.

(2)

Blood vessel	Feature of blood vessel
artery	contains valves to prevent backflow of blood
vein	has muscular walls to maintain high blood pressure
capillary	has thin walls for exchange of materials

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4. Question 2 · part c.ii · 4 marks

June 2024 Paper 1R

Further practice · Attempt part c.ii only.

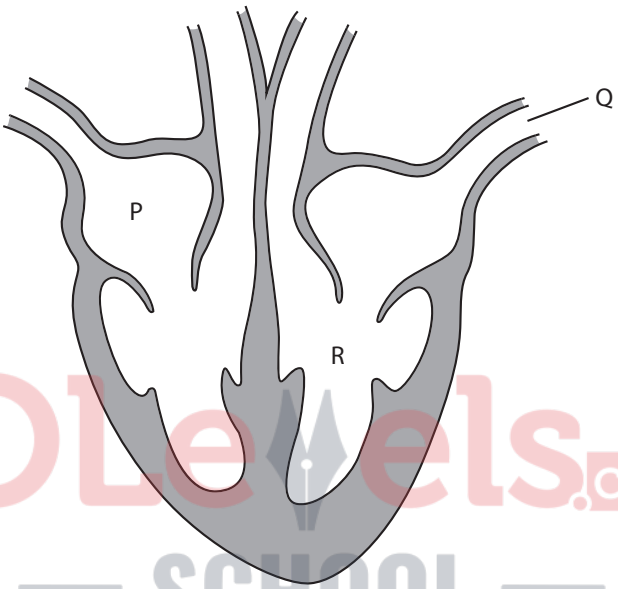
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(c) The heart pumps blood around the body.

The diagram shows a human heart.



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(i) The heart is divided into two halves.

Each half is divided into two chambers.

Name chamber R.

(1)

(ii) Describe the route that the blood travels from part P to part Q.

(4)

(Total for Question 2 = 13 marks)

4. Question 2 · part c.ii · 4 marks · Working and answer

Working and answer

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5. Question 5 · part b · 3 marks

June 2024 Paper 2R

Further practice · Attempt part b only.

5 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

The heart is a double-pump that is part of the human circulatory system. It helps ensure that the blood and its components reach all cells in the body.

Some components of the blood are made in the bone marrow. An average adult with a mass of 70 kg has 5.5 dm^3 of blood. About 45% of the blood is made up of different cells. There are several different types of cell in the blood but red blood cells are the most common. In males, 1.0 dm^3 of blood contains 5.9×10^{12} red blood cells. In females the same volume of blood contains 5.5×10^{12} red blood cells. Haemoglobin makes up a large part of red blood cells. This protein is vital to the function of red blood cells.

Red blood cells are transported around the body in a liquid known as plasma. Only 10% of blood plasma is made up of components that are used or produced by body cells. The rest of the plasma is made up of water. Some components of blood plasma are used by body cells in chemical reactions, some act as messengers or have a role in blood clotting and others are excreted from the body as waste.

(a) Name the main tissue that makes up the walls of the heart. **(1)**

(b) Explain why the heart is known as a double-pump. (Lines 1 to 2) **(3)**

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5. Question 5 · part b · 3 marks

June 2024 Paper 2R

Further practice · Attempt part b only.

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(c) Red blood cells are the most common type of cell found in the blood.

Which of these are also types of cell found in the blood?

(1)

- ☒ **A** phagocytes and platelets
- ☒ **B** antibodies and hormones
- ☒ **C** lymphocytes and phagocytes
- ☒ **D** platelets and lymphocytes

(d) (i) Name a component of the blood plasma that acts as a messenger. (Lines 12 to 14)

(1)

.....

(ii) Name a component of the blood that is involved in blood clotting.

(1)

.....

(e) Calculate the number of red blood cells found in the blood of an adult male with a mass of 95 kg. (Lines 3 to 7)

Give your answer in standard form.

(3)

number of red blood cells = ...

(Total for Question 5 = 10 marks)

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5. Question 5 · part b · 3 marks · Working and answer

Working and answer

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6. Question 1 · part a · 8 marks

June 2025 Paper 1R

Further practice · Attempt part a only.

1 (a) The sentences are about the heart.

Complete the sentences by using words or numbers from the box.

(8)

80	100	120	atria	body	diastolic	elastic tissue
lungs	muscle	systolic	tendon	vena cava	ventricles	

The heart consists of four chambers.

The two upper chambers are called ..... .

The left upper chamber receives blood from the ..... whilst the right upper chamber receives blood from the body through the ..... .

The two lower chambers are called ..... .

The left lower chamber pumps blood to the ..... .

This left lower chamber has a very thick wall made of ..... .

When the chamber contracts, it generates a pressure known as the ..... pressure.

The average value for this pressure is about ..... mmHg.

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P 7 4 6 2 8 A 0 2 2 0

6. Question 1 · part a · 8 marks

June 2025 Paper 1R

Further practice · Attempt part a only.

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(b) Cardiac output is the volume of blood pushed out of the heart in one minute.

Cardiac output is calculated using this formula.

cardiac output = heart rate × stroke volume

The stroke volume is the volume, in dm^3, of blood pushed out during one heartbeat. The heart rate is the number of beats per minute.

Calculate the stroke volume for a person with a heart rate of 68 beats per minute and a cardiac output of 6.1 dm^3.
Give your answer in cm^3.

(3)

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stroke volume = ... cm^3

(Total for Question 1 = 11 marks)

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6. Question 1 · part a · 8 marks · Working and answer

Working and answer

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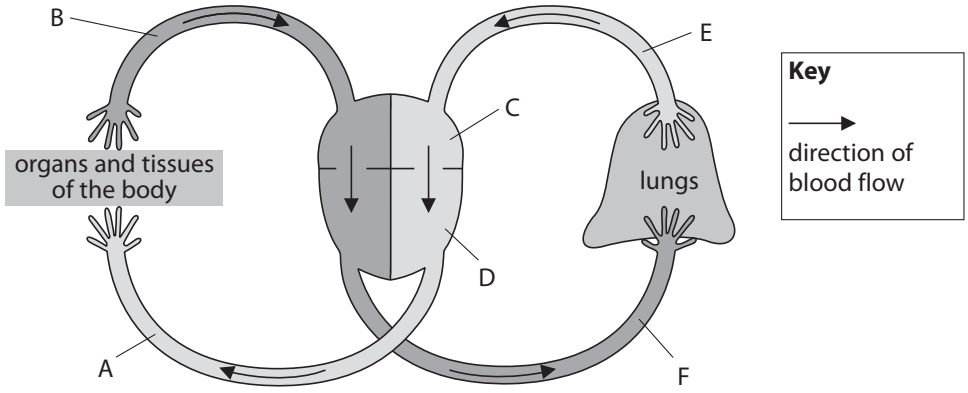
7. Question 4 · part b · 4 marks

January 2020 Paper 1

Further practice · Attempt part b only.

4 William Harvey was the first person to describe the circulation of blood around the body.

The diagram shows a model of the human circulatory system based on the research carried out by Harvey.



(a) Harvey suggested that each side of the heart was divided into two chambers.

State the names of chambers C and D.

(2)

chamber C .....

chamber D .....

(b) Harvey also suggested that blood is transported to and from the heart in different types of vessel.

Explain how the diagram supports Harvey's suggestion.

(4)

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P 5 9 9 0 9 A 0 1 0 2 8

7. Question 4 · part b · 4 marks

January 2020 Paper 1

Further practice · Attempt part b only.

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(c) Before Harvey's discoveries some scientists thought that the blood was circulated around the body by the liver.

State why this idea is not correct.

(1)

(d) Harvey's research did not include capillaries.

Describe the function of capillaries.

(2)

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(Total for Question 4 = 9 marks)

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7. Question 4 · part b · 4 marks · Working and answer

Working and answer

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8. Question 3 · part a.iii · 3 marks

November 2023 Paper 1

Further practice · Attempt part a.iii only.

3 The table shows the flow of blood to different organs in a person at rest and during exercise.

	Blood flow in cm^3 per minute	
	at rest	during exercise
Heart	250	750
Skeletal muscles	1000	12 500
Skin	400	1900
Gut	1400	600
Brain	750	750
Total flow	5600	17 500

(a) (i) Give the ratio of the blood flow to the heart when at rest and during exercise.

Give your answer in the form 1 : n

(2)

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Ratio = ... : .....

(ii) Calculate the percentage increase in the total flow of blood between rest and exercise.

Give your answer to two significant figures.

(3)

percentage increase = ... %

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8. Question 3 · part a.iii · 3 marks

November 2023 Paper 1

Further practice · Attempt part a.iii only.

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(iii) Explain why the total blood flow at rest is greater than the flow to the heart, skeletal muscle, skin, gut and brain added together.

(3)

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(b) (i) Describe the main changes between the blood flow values at rest and during exercise.

(3)

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(ii) Explain the change in blood flow rate to the skin when at rest and during exercise.

(4)

(Total for Question 3 = 15 marks)

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8. Question 3 · part a.iii · 3 marks · Working and answer

Working and answer

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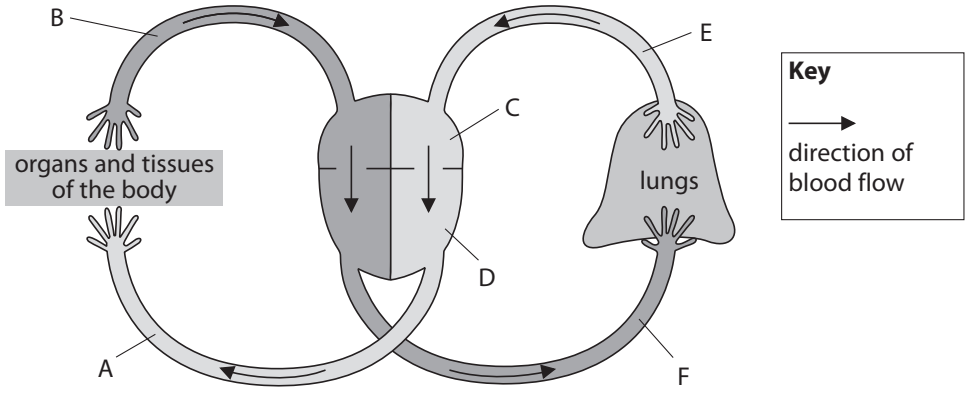
9. Question 4 · part c · 1 mark

January 2020 Paper 1

Further practice · Attempt part c only.

4 William Harvey was the first person to describe the circulation of blood around the body.

The diagram shows a model of the human circulatory system based on the research carried out by Harvey.



(a) Harvey suggested that each side of the heart was divided into two chambers.

State the names of chambers C and D.

(2)

chamber C .....

chamber D .....

(b) Harvey also suggested that blood is transported to and from the heart in different types of vessel.

Explain how the diagram supports Harvey's suggestion.

(4)

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9. Question 4 · part c · 1 mark

January 2020 Paper 1

Further practice · Attempt part c only.

DO NOT WRITE IN THIS AREA

(c) Before Harvey's discoveries some scientists thought that the blood was circulated around the body by the liver.

State why this idea is not correct.

(1)

.....

.....

(d) Harvey's research did not include capillaries.

Describe the function of capillaries.

(2)

.....

.....

.....

.....

(Total for Question 4 = 9 marks)

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Working and answer

Practice workbook · Describe how blood clots and explain why clotting matters

The sequence from damaged vessel and platelets to a fibrin mesh, labelled flow charts, the benefits of clotting at a wound and when clotting must be blocked.

10 question parts: 1 worked-example retry and 9 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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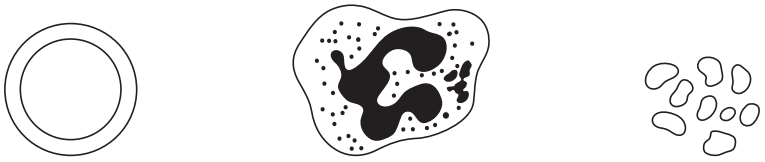
Type ID: HBI-BLOOD-CLOTTING-PROCESS-AND-IMPORTANCE

1. Question 3 · part b.i · 5 marks

November 2021 Paper 1

Worked-example retry · Attempt part b.i only.

3 The diagram shows some of the components of blood.



A B C

(a) Complete the table by giving the name of the component and its function.

Component A has been completed as an example.

(4)

Component	Name	Function
A	red blood cell	transports oxygen
B		
C		

(b) (i) Describe the process of blood clotting.

(5)

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1. Question 3 · part b.i · 5 marks

November 2021 Paper 1

Worked-example retry · Attempt part b.i only.

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(ii) Explain why it is important that blood clots.

(3)

(Total for Question 3 = 12 marks)

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1. Question 3 · part b.i · 5 marks · Working and answer

Working and answer

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2. Question 9 · part a.i · 3 marks

June 2019 Paper 2

Further practice · Attempt part a.i only.

9 Aspirin is a painkiller, but can also be prescribed to people who are at risk of having a stroke or a heart attack.

Aspirin dissolves blood clots and is also an enzyme inhibitor, which reduces the risk of more blood clots forming in blood vessels.

(a) (i) Describe how a blood clot forms in a blood vessel.

(3)

(ii) Explain how a blood clot in the coronary artery increases the risk of a heart attack.

(3)

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2. Question 9 · part a.i · 3 marks · Working and answer

Working and answer

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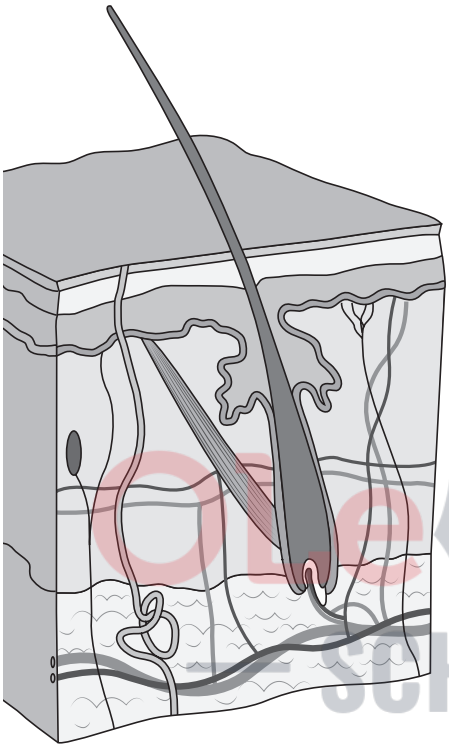
3. Question 5 · part d · 4 marks

January 2020 Paper 2

Further practice · Attempt part d only.

5 Burns can be first degree, second degree or third degree depending on how many layers of the skin are damaged.

The diagram shows the structure of the skin and the layers damaged by each type of burn.



layer 1	first degree burn
layers 1 and 2	second degree burn
layers 1, 2 and 3	third degree burn

(a) Name the three layers of skin that are damaged by a third degree burn.

(3)

layer 1 .....

layer 2 .....

layer 3 .....

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3. Question 5 · part d · 4 marks

January 2020 Paper 2

Further practice · Attempt part d only.

(b) Which two skin structures are damaged by a second degree burn?

(1)

- ☒ **A** blood vessels and sebaceous glands
- ☒ **B** fat tissue and blood vessels
- ☒ **C** hair follicles and sweat glands
- ☒ **D** sebaceous glands and fat tissue

(c) Explain why a person with a second or third degree burn has a high risk of infection.

(2)

(d) Blood clotting helps to prevent infection.

Describe the process of blood clotting.

(4)

(Total for Question 5 = 10 marks)

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3. Question 5 · part d · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 6 · part a.ii · 4 marks

June 2022 Paper 2R

Further practice · Attempt part a.ii only.

6 Blood will clot when there is damage to the skin, such as a cut.

(a) (i) State two reasons why it is important for the blood to clot.

(2)

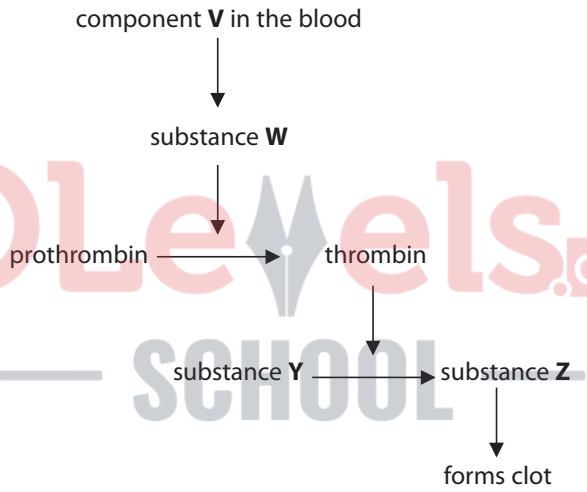
1 .....

.....

2 .....

.....

(ii) The diagram shows some of the stages in the process of blood clotting.



Use words from the box to identify V, W, Y and Z.

thrombokinase enzyme		white blood cells		platelets
haemoglobin	fibrinogen	fibrin	antigens	antibody

(4)

V .....

W .....

Y .....

Z .....

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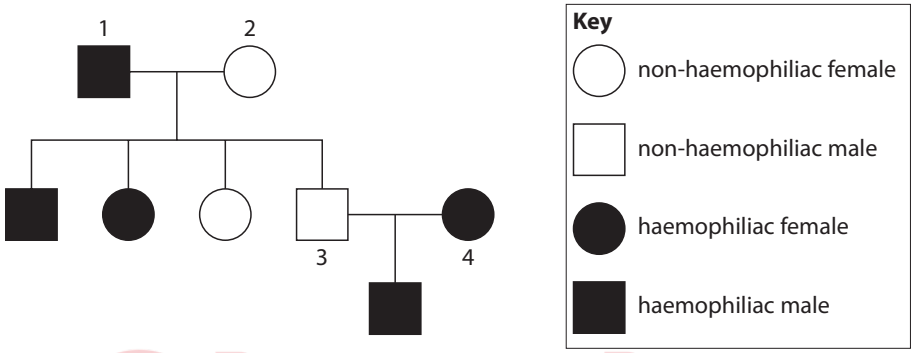
4. Question 6 · part a.ii · 4 marks

June 2022 Paper 2R

Further practice · Attempt part a.ii only.

(b) Haemophilia is an inherited condition where the blood of a person does not clot.

The diagram shows a family pedigree. Some members of the family have the condition.



(i) Deduce the genotypes of family members 1 and 2 showing how you obtained your answer.

(5)

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4. Question 6 · part a.ii · 4 marks · Working and answer

Working and answer

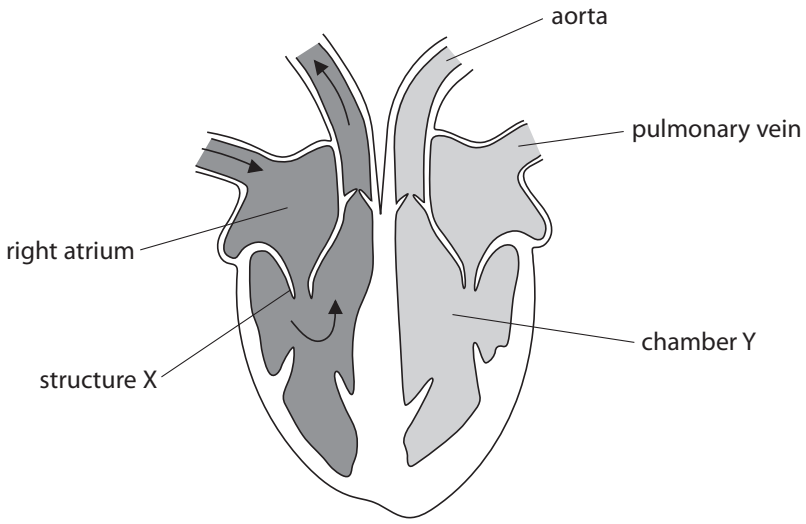
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5. Question 2 · part c.ii · 2 marks

January 2023 Paper 2

Further practice · Attempt part c.ii only.

2 (a) The diagram shows a section through the heart.



(i) Name chamber Y. **(1)**

(ii) Explain the function of structure X. **(2)**

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P 7 2 6 0 2 A 0 4 2 8

5. Question 2 · part c.ii · 2 marks

January 2023 Paper 2

Further practice · Attempt part c.ii only.

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(iii) The table lists some properties of blood.

Add ticks (✓) to those boxes that show the properties of blood in the aorta.

(2)

Properties of blood	Blood in the aorta
oxygenated	
deoxygenated	
high pressure	
medium pressure	
low pressure	

(iv) State where blood in the left side of the heart has come from.

(1)

(b) In one year in the United Kingdom, there were 173 heart transplants.

Heart transplant operations require hearts to be donated.

Some hearts cannot be used because the donor is too old.

Give two other reasons why a heart might not be suitable for a transplant.

(2)

1 .....

.....

2 .....

.....

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5. Question 2 · part c.ii · 2 marks

January 2023 Paper 2

Further practice · Attempt part c.ii only.

(c) (i) Which blood component causes blood clotting?

(1)

- ☒ **A** lymphocytes
- ☒ **B** phagocytes
- ☒ **C** platelets
- ☒ **D** red blood cells

(ii) The diagram summarises part of the process that causes a blood clot.



Anticoagulant drugs prevent a blood clot forming.

These drugs stop the enzyme thrombin being produced.

Describe how a lack of thrombin prevents a blood clot forming.

(2)

(Total for Question 2 = 11 marks)

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5. Question 2 · part c.ii · 2 marks · Working and answer

Working and answer

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6. Question 6 · part a.i · 3 marks

November 2020 Paper 2R

Further practice · Attempt part a.i only.

6 Haemophilia is a condition in which blood does not clot. It is caused by a sex-linked allele.

(a) (i) Explain why the process of blood clotting is important.

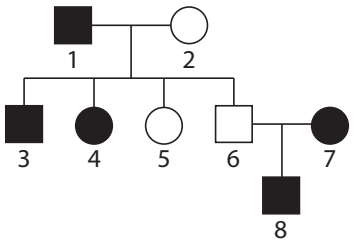
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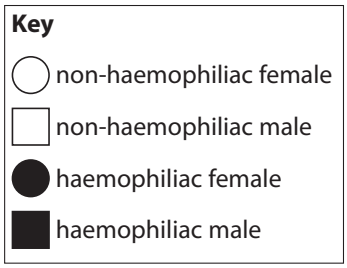
(ii) State what is meant by the term **sex-linked**.

(2)

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(b) The pedigree shows the pattern of inheritance of haemophilia in a family.





X^h is the allele for haemophilia and X^H is the allele for clotting.

(i) What is the genotype of person 3?

(1)

- ☒ **A** $X^h X^h$
- ☒ **B** $X^H X^h$
- ☒ **C** $X^H Y$
- ☒ **D** $X^h Y$

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6. Question 6 · part a.i · 3 marks · Working and answer

Working and answer

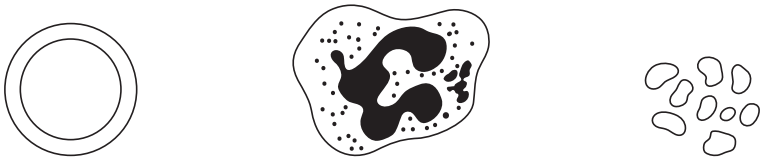
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7. Question 3 · part b.ii · 3 marks

November 2021 Paper 1

Further practice · Attempt part b.ii only.

3 The diagram shows some of the components of blood.



A **B** **C**

(a) Complete the table by giving the name of the component and its function.

Component A has been completed as an example.

(4)

Component	Name	Function
A	red blood cell	transports oxygen
B		
C		

(b) (i) Describe the process of blood clotting.

(5)

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7. Question 3 · part b.ii · 3 marks

November 2021 Paper 1

Further practice · Attempt part b.ii only.

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(ii) Explain why it is important that blood clots.

(3)

(Total for Question 3 = 12 marks)

Working and answer

8. Question 6 · part a.i · 2 marks

June 2022 Paper 2R

Further practice · Attempt part a.i only.

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6 Blood will clot when there is damage to the skin, such as a cut.

(a) (i) State two reasons why it is important for the blood to clot.

(2)

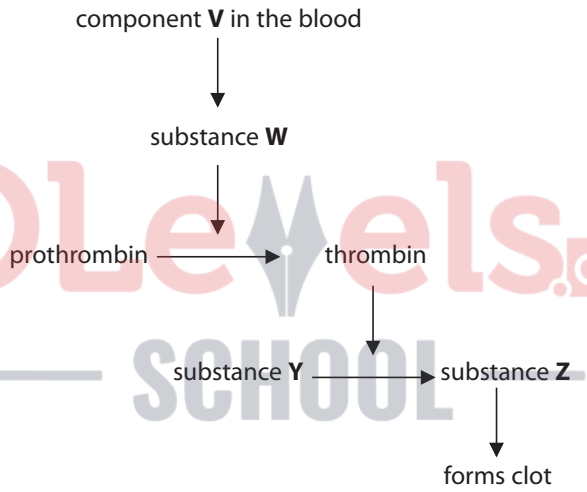
1 .....

.....

2 .....

.....

(ii) The diagram shows some of the stages in the process of blood clotting.



Use words from the box to identify V, W, Y and Z.

thrombokinase enzyme		white blood cells		platelets
haemoglobin	fibrinogen	fibrin	antigens	antibody

(4)

V .....

W .....

Y .....

Z .....

8. Question 6 · part a.i · 2 marks · Working and answer

Working and answer

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9. Question 1 · part b · 4 marks

November 2023 Paper 1

Further practice · Attempt part b only.

1 (a) The table gives components of the blood and their features.

Complete the table by putting a tick (✓) in the box if the blood component shows the feature.

(5)

Feature	Blood component				
	red blood cell	phagocyte	lymphocyte	platelet	plasma
transports oxygen					
contains haemoglobin					
is a fluid					
produces antibodies					
engulfs pathogens					

(b) Explain why it is important that blood clots when the skin is cut open.

(4)

(Total for Question 1 = 9 marks)

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9. Question 1 · part b · 4 marks · Working and answer

Working and answer

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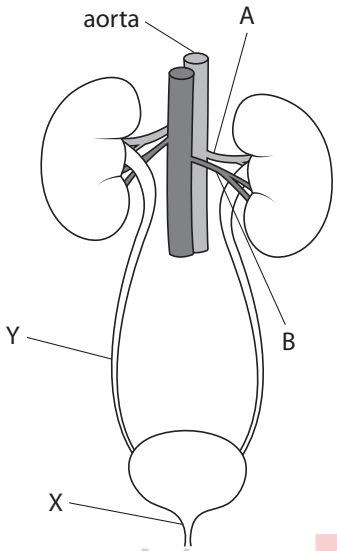
10. Question 8 · part b.ii · 3 marks

January 2020 Paper 1

Further practice · Attempt part b.ii only.

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8 The diagram shows part of the human renal system.



(a) (i) Give the names of structures X and Y.

(2)

structure X .....

structure Y .....

(ii) Describe the function of the aorta.

(2)

.....

.....

.....

.....

(iii) Compare the composition of the blood in vessels A and B.

(2)

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10. Question 8 · part b.ii · 3 marks

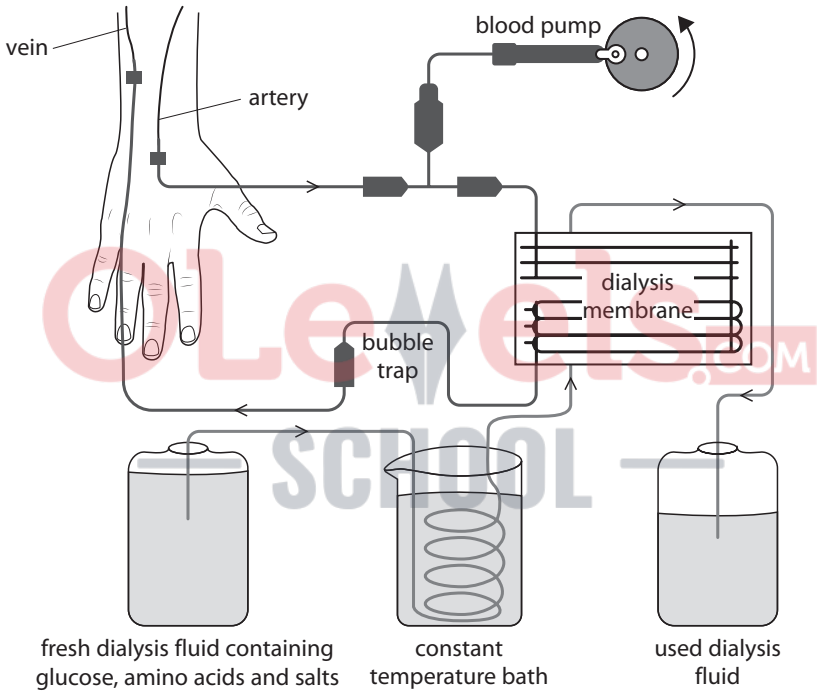
January 2020 Paper 1

Further practice · Attempt part b.ii only.

(b) Dialysis is a treatment used for people with kidney failure.

Blood containing waste substances is pumped through a dialysis machine. The dialysis machine filters the waste from the blood. The cleaned blood is then returned to the patient.

The diagram shows how a dialysis machine filters the blood of a person with kidney failure.



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10. Question 8 · part b.ii · 3 marks

January 2020 Paper 1

Further practice · Attempt part b.ii only.

(i) Explain how diffusion removes waste substances from the patient's blood inside the dialysis machine.

(4)

(ii) A chemical is added to the patient's blood as it enters the dialysis machine. This chemical prevents the blood from clotting.

Explain why it is important to add this chemical as the blood enters the machine, rather than just before the blood returns to the body.

(3)

(Total for Question 8 = 13 marks)

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10. Question 8 · part b.ii · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

Practice workbook · Explain how a blocked coronary artery damages the heart

Causes and consequences of narrowing or blocking arteries supplying heart muscle or other tissue.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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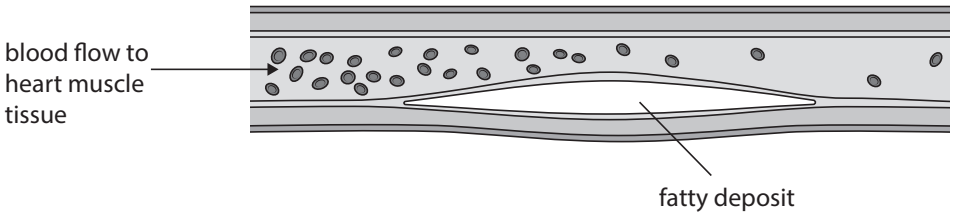
Type ID: HBI-EXPLAIN-CORONARY-BLOCKAGE-EFFECTS

1. Question 7 · part a · 4 marks

June 2024 Paper 2

Worked-example retry · Attempt part a only.

7 (a) The diagram shows a fatty deposit in an artery carrying blood to the heart muscle.



(Source: adapted from https://www.heartandstroke.ca/heart-disease/conditions/atherosclerosis)

Explain how a build-up of a fatty deposit in an artery carrying blood to the heart muscle can cause heart disease.

(4)

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1. Question 7 · part a · 4 marks

June 2024 Paper 2

Worked-example retry · Attempt part a only.

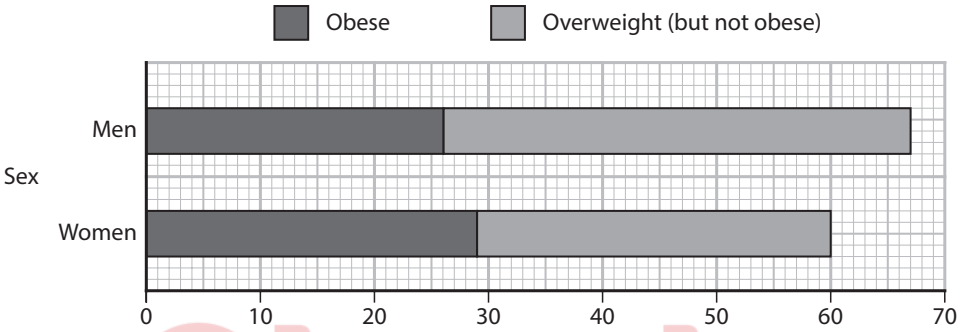
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(b) People with obesity are more at risk of heart disease than those who are overweight.

The chart shows the percentage of men and women in the United Kingdom who were either overweight or obese in 2018.



(Source: https://digital.nhs.uk/data-and-information/publications/statistical/health-survey-for-england/2018/summary)

(i) In 2018 there were 66.5 million people living in the UK. Of these people, 33 million were men.

Calculate how many more women than men were classified as obese.

(3)

extra number of women = ...

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P 7 4 6 3 0 A 0 1 9 2 4

Turn over ▶

1. Question 7 · part a · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 9 · part a.ii · 3 marks

June 2019 Paper 2

Further practice · Attempt part a.ii only.

9 Aspirin is a painkiller, but can also be prescribed to people who are at risk of having a stroke or a heart attack.

Aspirin dissolves blood clots and is also an enzyme inhibitor, which reduces the risk of more blood clots forming in blood vessels.

(a) (i) Describe how a blood clot forms in a blood vessel.

(3)

(ii) Explain how a blood clot in the coronary artery increases the risk of a heart attack.

(3)

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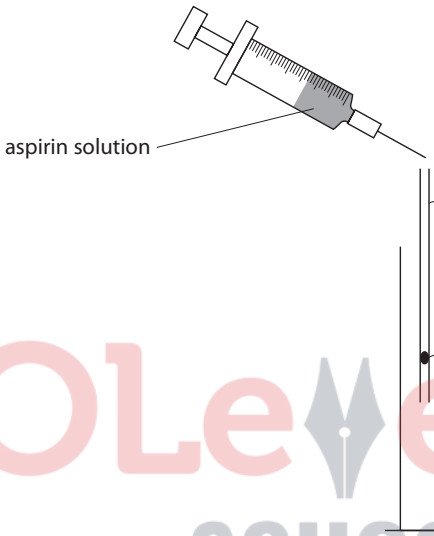
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2. Question 9 · part a.ii · 3 marks

June 2019 Paper 2

Further practice · Attempt part a.ii only.

(b) A student uses this apparatus to investigate the effect of aspirin on the rate of blood flow through blood vessels.



The student uses this method.

- dissolve one aspirin tablet in water
- use a syringe to pass the aspirin solution through a straw containing a blood clot
- record the total volume of aspirin solution in the measuring cylinder every five minutes

The student repeats the method with solutions formed from two aspirin tablets and then from three aspirin tablets.

(i) Give a control variable for this investigation.

(1)

(ii) Describe a suitable control test for this investigation.

(1)

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2. Question 9 · part a.ii · 3 marks · Working and answer

Working and answer

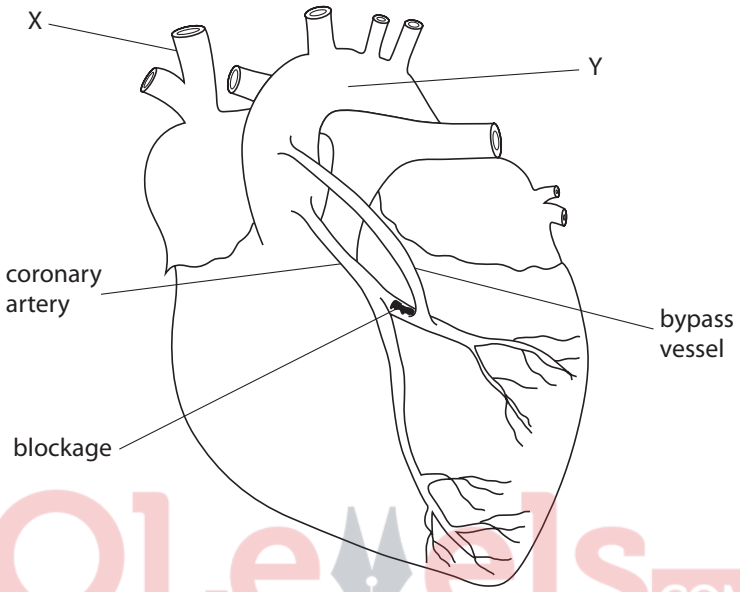
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3. Question 3 · part b.i · 1 mark

November 2020 Paper 2R

Further practice · Attempt part b.i only.

3 The diagram shows a person's heart. The person has had a heart bypass operation because a coronary artery had become partly blocked.



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(a) (i) What is the name of blood vessel Y?

(1)

- ☒ **A** aorta
- ☒ **B** pulmonary artery
- ☒ **C** pulmonary vein
- ☒ **D** vena cava

(ii) To which part of the heart is blood vessel X connected?

(1)

- ☒ **A** left atrium
- ☒ **B** left ventricle
- ☒ **C** right atrium
- ☒ **D** right ventricle

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3. Question 3 · part b.i · 1 mark

November 2020 Paper 2R

Further practice · Attempt part b.i only.

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(b) (i) Before the bypass operation, the coronary artery was becoming blocked at the point shown on the diagram.

Draw a circle on the diagram to show the area of the heart that was affected by the blockage before the bypass operation.

(1)

(ii) Describe what effect this blockage would have on the heart tissue if the bypass vessel was not inserted.

(3)

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(iii) Suggest how the bypass vessel allows the heart to work more effectively.

(2)

(Total for Question 3 = 8 marks)

3. Question 3 · part b.i · 1 mark · Working and answer

Working and answer

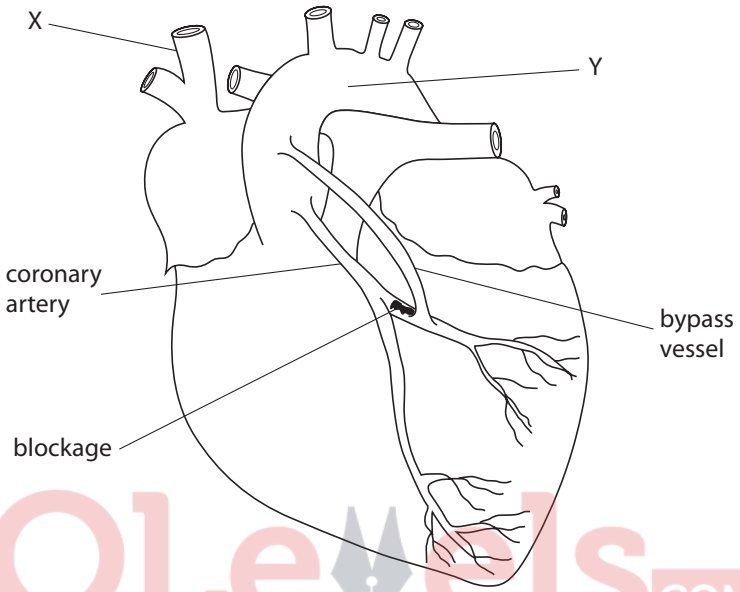
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4. Question 3 · part b.ii · 3 marks

November 2020 Paper 2R

Further practice · Attempt part b.ii only.

3 The diagram shows a person's heart. The person has had a heart bypass operation because a coronary artery had become partly blocked.



(a) (i) What is the name of blood vessel Y?

(1)

- ☒ **A** aorta
- ☒ **B** pulmonary artery
- ☒ **C** pulmonary vein
- ☒ **D** vena cava

(ii) To which part of the heart is blood vessel X connected?

(1)

- ☒ **A** left atrium
- ☒ **B** left ventricle
- ☒ **C** right atrium
- ☒ **D** right ventricle

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4. Question 3 · part b.ii · 3 marks

November 2020 Paper 2R

Further practice · Attempt part b.ii only.

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(b) (i) Before the bypass operation, the coronary artery was becoming blocked at the point shown on the diagram.

Draw a circle on the diagram to show the area of the heart that was affected by the blockage before the bypass operation.

(1)

(ii) Describe what effect this blockage would have on the heart tissue if the bypass vessel was not inserted.

(3)

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(iii) Suggest how the bypass vessel allows the heart to work more effectively.

(2)

(Total for Question 3 = 8 marks)

4. Question 3 · part b.ii · 3 marks · Working and answer

Working and answer

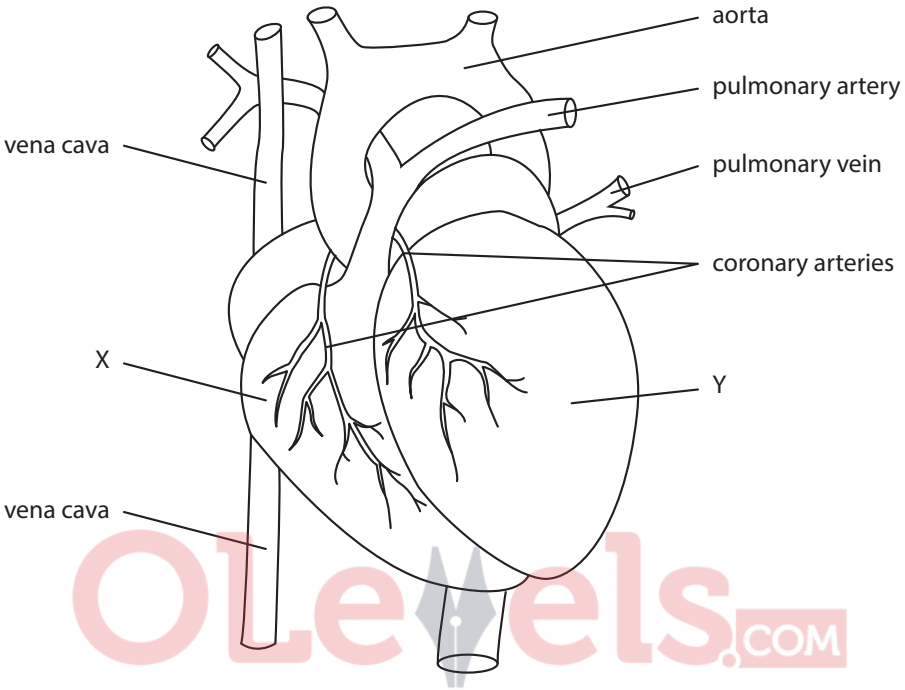
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5. Question 7 · part c · 3 marks

November 2023 Paper 1

Further practice · Attempt part c only.

7 The diagram shows the heart as seen from the front.



(a) The vena cava carries deoxygenated blood.

Name a different blood vessel, shown on the diagram, that also carries deoxygenated blood.

(1)

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5. Question 7 · part c · 3 marks

November 2023 Paper 1

Further practice · Attempt part c only.

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(b) Blood chambers X and Y have walls that are different in thickness.

Explain this difference.

(5)

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(c) The coronary arteries can become blocked.

Explain the effect of this blockage on the functioning of the heart.

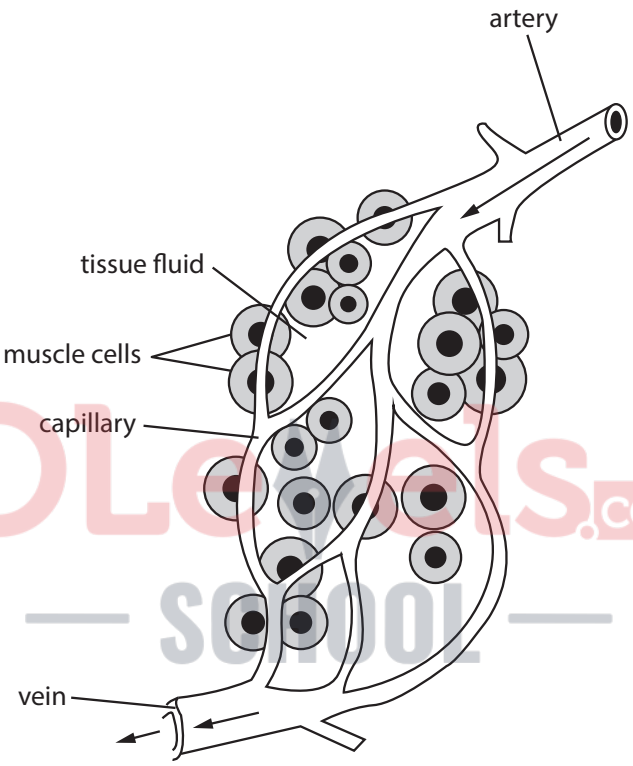
(3)

5. Question 7 · part c · 3 marks

November 2023 Paper 1

Further practice · Attempt part c only.

(d) The diagram shows a capillary bed.



Use the diagram to explain the function of the tissue fluid.

(3)

(Total for Question 7 = 12 marks)

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5. Question 7 · part c · 3 marks · Working and answer

Working and answer

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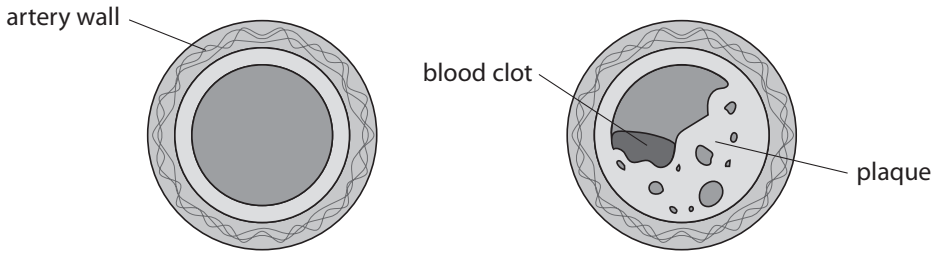
6. Question 6 · part a · 4 marks

June 2024 Paper 2R

Further practice · Attempt part a only.

6 (a) Cholesterol is a fatty substance that can build up inside arteries forming waxy deposits called plaques.

The diagram shows a healthy artery and an artery with a plaque.



healthy artery artery with a plaque

(Source: adapted from https://www.dreamstime.com/artery-plaque-blood-clot-medical-illustration-diseased-compared-to-healthy-cross-section-diagram-shows-image120895536)

Explain how plaques might affect the heart of an individual.

(4)

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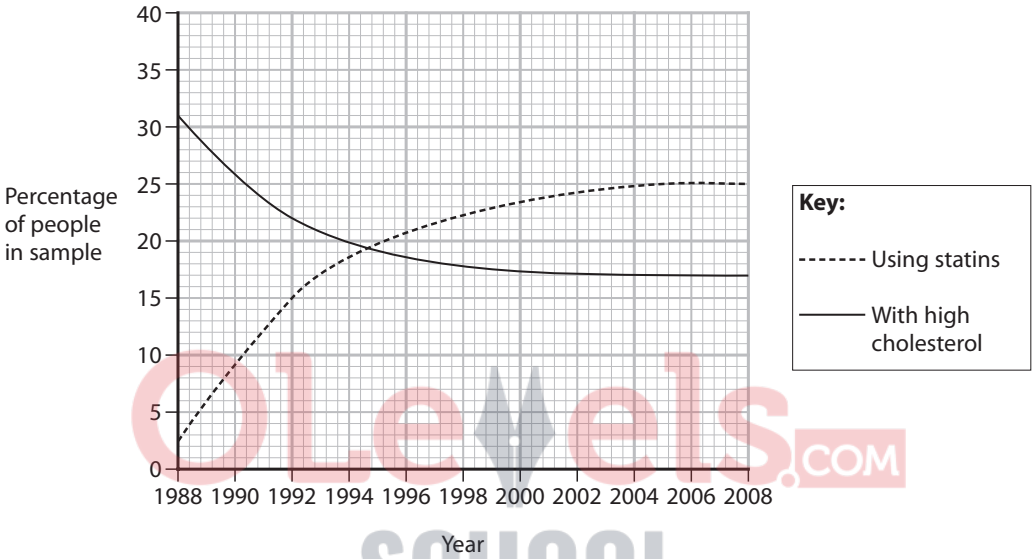
6. Question 6 · part a · 4 marks

June 2024 Paper 2R

Further practice · Attempt part a only.

(b) Scientists investigated the effect of statins on cholesterol levels in a sample of people.

The graph shows the results of their investigation.



(Source: https://www.health.harvard.edu/heart-health/trends-in-high-cholesterol-and-statin-use)

(i) Give the relationship between the use of statins and the percentage of people with high cholesterol levels.

(1)

.....

.....

(ii) Calculate the ratio of the percentage of people with high cholesterol levels in 1988 compared with the percentage of people with high cholesterol levels in 2005.

Give your answer in the form of 1 : n

(3)

ratio = ...

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6. Question 6 · part a · 4 marks · Working and answer

Working and answer

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7. Question 4 · part c · 2 marks

November 2024 Paper 1

Further practice · Attempt part c only.

4 Carbohydrates are large molecules found in many body cells.

(a) Glycogen is a carbohydrate.

(i) Describe how the liver produces glycogen.

(2)

(ii) Name the hormone that stimulates the liver to produce glycogen.

(1)

(iii) Name a hormone that stimulates the liver to increase blood sugar levels.

(1)

(b) Consuming a diet high in carbohydrates can increase the BMI of a person.

The table gives information on the BMI categories of individuals.

BMI	Category
less than 18.5	underweight
18.5 to 24.9	healthy weight
25 to 29.9	overweight
30 to 34.9	obese
35 to 39.9	severely obese
more than 39.9	morbidly obese

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7. Question 4 · part c · 2 marks

November 2024 Paper 1

Further practice · Attempt part c only.

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(i) A person has a height of 1.69 m and a mass of 67.0 kg.

Calculate the BMI of this person.

(2)

BMI .....

(ii) Give the BMI category of this person.

(1)

.....

(c) Obesity is a risk factor for heart disease.

Explain how obesity can cause heart disease.

(2)

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(Total for Question 4 = 9 marks)

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Working and answer

8. Question 4 · part d · 3 marks

June 2021 Paper 2

Further practice · Attempt part d only.

4 Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

Obesity and diabetes have been linked to a 20% increase in the number of leg and foot amputations.

Data from Diabetes UK shows that there were 26 378 lower limb amputations related to diabetes from the period 2014 to 2017. This is an increase from the period 2010 to 2013 when there were 22 092 amputations related to diabetes.

Minor lower limb amputations are below the ankle and major lower limb amputations are below the knee. Minor lower limb amputations increased by 26.5% and major lower limb amputations increased by 4.0%.

In 2019 in the United Kingdom 3.8 million people had been diagnosed with diabetes and 90% of these people had Type 2 diabetes.

(a) (i) Describe the causes of obesity.

(3)

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(ii) Give two health risks, other than diabetes, associated with obesity.

(2)

1 .....

2 .....

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8. Question 4 · part d · 3 marks

June 2021 Paper 2

Further practice · Attempt part d only.

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(b) Diabetes is caused as a result of insulin not being secreted or not working correctly.

(i) Which organ secretes insulin?

(1)

- ☒ **A** kidney
- ☒ **B** liver
- ☒ **C** pancreas
- ☒ **D** thyroid

(ii) Insulin stimulates the conversion of glucose into glycogen.

In which organ does this occur?

(1)

- ☒ **A** kidney
- ☒ **B** liver
- ☒ **C** pancreas
- ☒ **D** thyroid

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(c) (i) Calculate the percentage increase in the number of lower limb amputations between the period 2010 to 2013 and the period 2014 to 2017.

(3)

percentage increase = ... %

(ii) Calculate the number of people in 2019 who had been diagnosed with Type 2 diabetes in the United Kingdom.

(1)

number of people = ... million

8. Question 4 · part d · 3 marks

June 2021 Paper 2

Further practice · Attempt part d only.

(d) Diabetes causes a narrowing of the arteries supplying blood to the lower limbs.

Suggest why this can lead to the need to amputate the lower limbs.

(3)

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(Total for Question 4 = 14 marks)

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Working and answer

Practice workbook · Describe how heart disease is prevented and treated

Lifestyle and diet changes, statins, stents, bypass surgery and transplant problems, including judging data from a statin study.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-HEART-DISEASE-PREVENTION-AND-TREATMENT

1. Question 4 · part a · 2 marks

November 2023 Paper 2

Worked-example retry · Attempt part a only.

4 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Atorvastatin belongs to a group of medicines known as statins. Statins are medicines that regulate lipid levels in the blood. Statins are used when changes of diet and lifestyle have failed to lower blood cholesterol levels.

Statins can also be used to reduce the risk of heart disease when cholesterol levels are normal. People with liver disease or kidney problems should avoid taking statins. Women who are pregnant or are breastfeeding should also avoid taking statins.

In the United Kingdom approximately 7.5 million people take statins. This is 12% of the total population. However, up to 20% of these people stop taking statins because of adverse side effects such as muscle aches, fatigue, feeling sick and joint pains.

A recent study looked at patients who had stopped taking statins because of side effect symptoms. Patients were asked to take statins for four months, then placebo tablets (tablets that look like statins but contain no active substances) for four months followed by no tablets for four months. Every day the patients were asked how bad their symptoms were using a score from 0 to 100. The table shows the results.

Four-month treatment	Mean score out of 100
statins	16.3
placebo	15.4
no tablets	8.0

(a) Describe the changes in diet and lifestyle that could lower cholesterol. (Lines 2 and 3)

(2)

.....

.....

.....

.....

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1. Question 4 · part a · 2 marks

November 2023 Paper 2

Worked-example retry · Attempt part a only.

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(b) (i) Explain why people with liver disease or kidney problems should avoid taking statins. (Lines 5 and 6)

(4)

(ii) Explain why women who are pregnant or breastfeeding should avoid taking statins. (Lines 6 and 7)

(3)

OLevels.COM SCHOOL

(c) (i) Using the results of the study, calculate how many times greater the score for symptoms when taking statins is compared with the score when not taking tablets.

(1)

times greater = ...

1. Question 4 · part a · 2 marks · Working and answer

Working and answer

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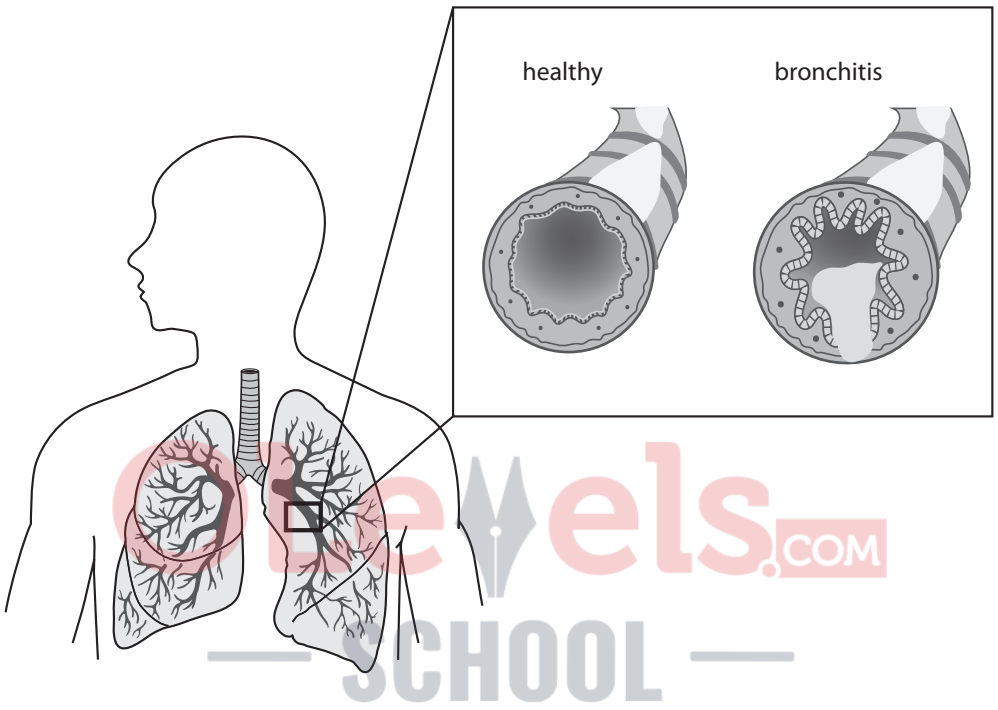
2. Question 6 · part c.iii · 1 mark

January 2020 Paper 1

Further practice · Attempt part c.iii only.

6 Chronic obstructive pulmonary disease (COPD) is a group of lung diseases that includes bronchitis and emphysema.

The diagram shows how bronchitis affects the lungs.



(a) Name the part of the lungs affected by bronchitis.

(1)

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2. Question 6 · part c.iii · 1 mark

January 2020 Paper 1

Further practice · Attempt part c.iii only.

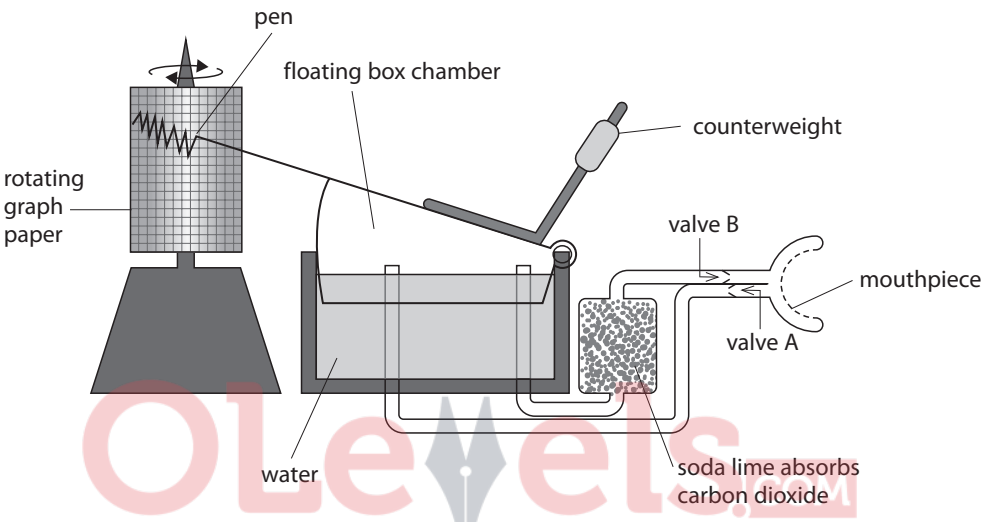
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(b) The diagram shows a spirometer.

A spirometer measures how quickly a person breathes in and out.



(i) Describe how a spirometer is used to obtain a trace of a person's breathing pattern.

(4)

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2. Question 6 · part c.iii · 1 mark

January 2020 Paper 1

Further practice · Attempt part c.iii only.

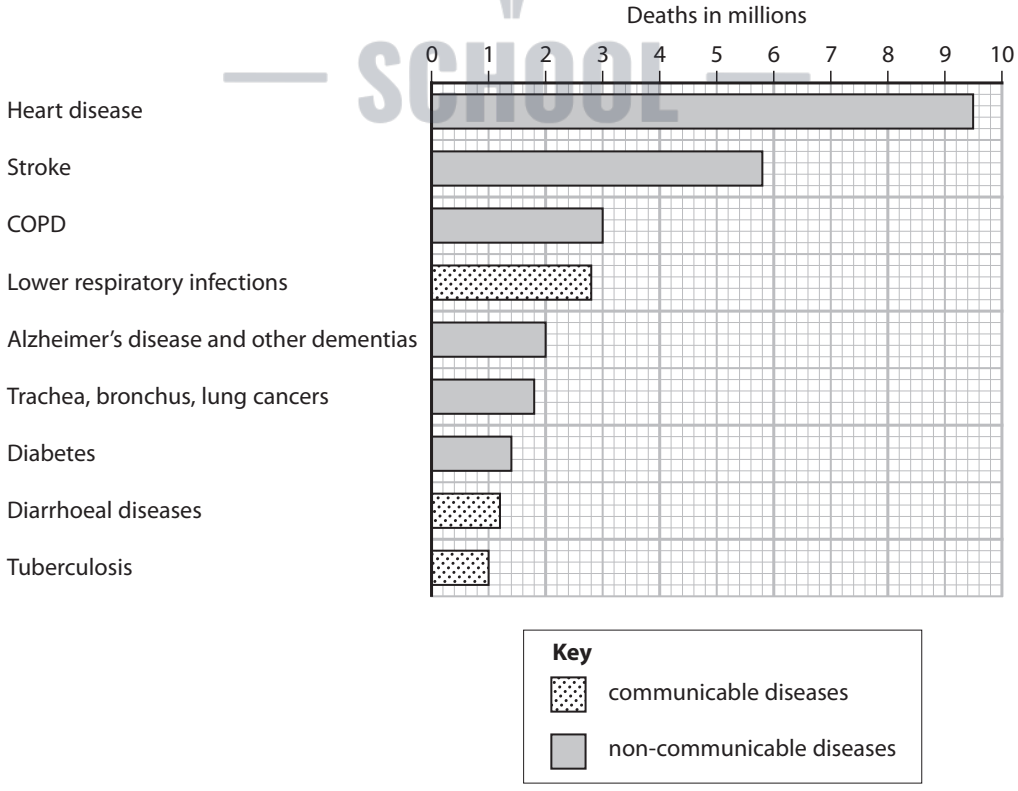
(ii) State one way to reduce risk to health when using a spirometer.

(1)

(iii) Explain why a person with bronchitis needs to take a longer breath to obtain a normal tidal volume.

(2)

(c) The bar chart shows the number of deaths in millions from different health conditions worldwide in 2016.



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2. Question 6 · part c.iii · 1 mark

January 2020 Paper 1

Further practice · Attempt part c.iii only.

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(i) More deaths are caused by non-communicable diseases than by communicable diseases.

Suggest what is meant by the term **non-communicable disease**.

(1)

.....

.....

.....

(ii) In 2000, the number of deaths from heart disease was 7.0 million.

Calculate the percentage change in the number of deaths from heart disease from 2000 to 2016.

(3)

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percentage change = ...%

(iii) Suggest a reason for the change in the number of deaths from heart disease from 2000 to 2016.

(1)

.....

.....

.....

(Total for Question 6 = 13 marks)

2. Question 6 · part c.iii · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 4 · part c.i · 2 marks

November 2024 Paper 2

Further practice · Attempt part c.i only.

4 Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

When looking at data, it is clear that people in lower income countries are more likely to die from infectious diseases (approximately 240 per 100 000) than non-infectious diseases (approximately 180 per 100 000). Infectious diseases such as diarrhoea and malaria can be easily eliminated or treated. It is also possible to vaccinate against diseases such as TB. These diseases are uncommon in higher income countries because of preventative medicine.

The only common diseases present in both higher and lower income countries are respiratory tract infections. These infections are usually influenza, pneumonia and bronchitis. Whilst influenza is a viral infection, the other infections are usually bacterial and can be treated. Many people at risk in the higher income countries are encouraged to get preventative treatment. These are often provided free of charge for the elderly who are most at risk and those who work in indoor spaces such as schools, offices and hospitals. The effect on mortality in lower income countries is more than two times as great as in higher income countries.

The greatest killers in higher income countries are heart disease and stroke. These two diseases kill approximately twice as many people per 100 000 as in lower income countries. These are often seen as lifestyle diseases. When we compare the number of deaths caused by non-infectious diseases, in higher income countries there are 445 deaths per 100 000 and in lower income countries there are 185 deaths per 100 000. There are also more cases of other diseases such as dementia and cancers in higher income countries.

(a) State one way the infectious diseases diarrhoea and malaria can be prevented (lines 3 and 4).

(2)

diarrhoea

malaria

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3. Question 4 · part c.i · 2 marks

November 2024 Paper 2

Further practice · Attempt part c.i only.

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(b) (i) State how pneumonia and bronchitis can be treated (lines 8 to 10).

(1)

(ii) Give one reason why elderly people and people who work in indoor spaces are at risk from respiratory tract infections. (lines 12 and 13).

(2)

elderly people

indoor spaces

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(c) (i) Give two factors that may contribute to lifestyle diseases (line 18).

(2)

1 .....

2 .....

(ii) In a higher income country of 5 000 000 people, calculate the number of people who will have non-infectious diseases (lines 20 and 21).

Give your answer to three significant figures.

(3)

number of people = ...

3. Question 4 · part c.i · 2 marks · Working and answer

Working and answer

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4. Question 4 · part c.ii · 4 marks

November 2023 Paper 2

Further practice · Attempt part c.ii only.

4 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Atorvastatin belongs to a group of medicines known as statins. Statins are medicines that regulate lipid levels in the blood. Statins are used when changes of diet and lifestyle have failed to lower blood cholesterol levels.

Statins can also be used to reduce the risk of heart disease when cholesterol levels are normal. People with liver disease or kidney problems should avoid taking statins. Women who are pregnant or are breastfeeding should also avoid taking statins.

In the United Kingdom approximately 7.5 million people take statins. This is 12% of the total population. However, up to 20% of these people stop taking statins because of adverse side effects such as muscle aches, fatigue, feeling sick and joint pains.

A recent study looked at patients who had stopped taking statins because of side effect symptoms. Patients were asked to take statins for four months, then placebo tablets (tablets that look like statins but contain no active substances) for four months followed by no tablets for four months. Every day the patients were asked how bad their symptoms were using a score from 0 to 100. The table shows the results.

Four-month treatment	Mean score out of 100
statins	16.3
placebo	15.4
no tablets	8.0

(a) Describe the changes in diet and lifestyle that could lower cholesterol. (Lines 2 and 3)

(2)

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4. Question 4 · part c.ii · 4 marks

November 2023 Paper 2

Further practice · Attempt part c.ii only.

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(b) (i) Explain why people with liver disease or kidney problems should avoid taking statins. (Lines 5 and 6)

(4)

(ii) Explain why women who are pregnant or breastfeeding should avoid taking statins. (Lines 6 and 7)

(3)

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(c) (i) Using the results of the study, calculate how many times greater the score for symptoms when taking statins is compared with the score when not taking tablets.

(1)

times greater = ...

4. Question 4 · part c.ii · 4 marks

November 2023 Paper 2

Further practice · Attempt part c.ii only.

(ii) Discuss the results of the study using information from the table.

(4)

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(Total for Question 4 = 14 marks)

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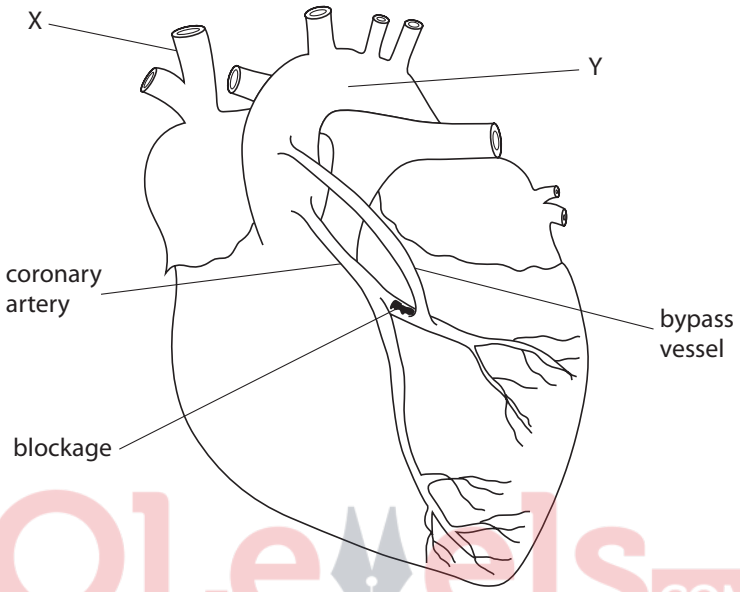
Working and answer

5. Question 3 · part b.iii · 2 marks

November 2020 Paper 2R

Further practice · Attempt part b.iii only.

3 The diagram shows a person's heart. The person has had a heart bypass operation because a coronary artery had become partly blocked.



(a) (i) What is the name of blood vessel Y?

(1)

- ☒ **A** aorta
- ☒ **B** pulmonary artery
- ☒ **C** pulmonary vein
- ☒ **D** vena cava

(ii) To which part of the heart is blood vessel X connected?

(1)

- ☒ **A** left atrium
- ☒ **B** left ventricle
- ☒ **C** right atrium
- ☒ **D** right ventricle

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P 6 2 0 5 4 A 0 8 2 4

5. Question 3 · part b.iii · 2 marks

November 2020 Paper 2R

Further practice · Attempt part b.iii only.

(b) (i) Before the bypass operation, the coronary artery was becoming blocked at the point shown on the diagram.

Draw a circle on the diagram to show the area of the heart that was affected by the blockage before the bypass operation.

(1)

(ii) Describe what effect this blockage would have on the heart tissue if the bypass vessel was not inserted.

(3)

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(iii) Suggest how the bypass vessel allows the heart to work more effectively.

(2)

(Total for Question 3 = 8 marks)

5. Question 3 · part b.iii · 2 marks · Working and answer

Working and answer

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6. Question 9 · part b · 4 marks

June 2022 Paper 1R

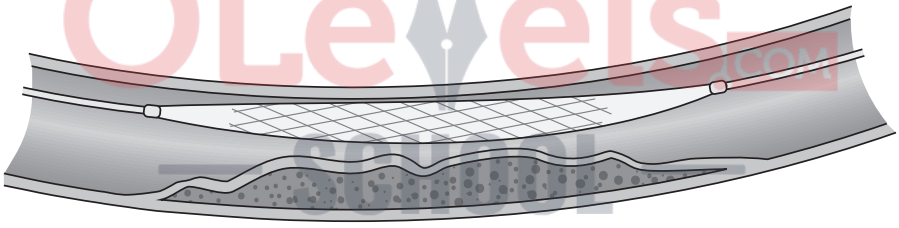
Further practice · Attempt part b only.

9 (a) Statins can be used to treat people with circulatory disorders.

Design a method that can be used to test the effectiveness of statins in treating circulatory disorders.

(3)

(b) The diagram shows a stent being used in the treatment of heart disease.



(Source: © DesignPrax/Shutterstock)

Explain how a stent can be used to treat heart disease.

(4)

(Total for Question 9 = 7 marks)

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6. Question 9 · part b · 4 marks · Working and answer

Working and answer

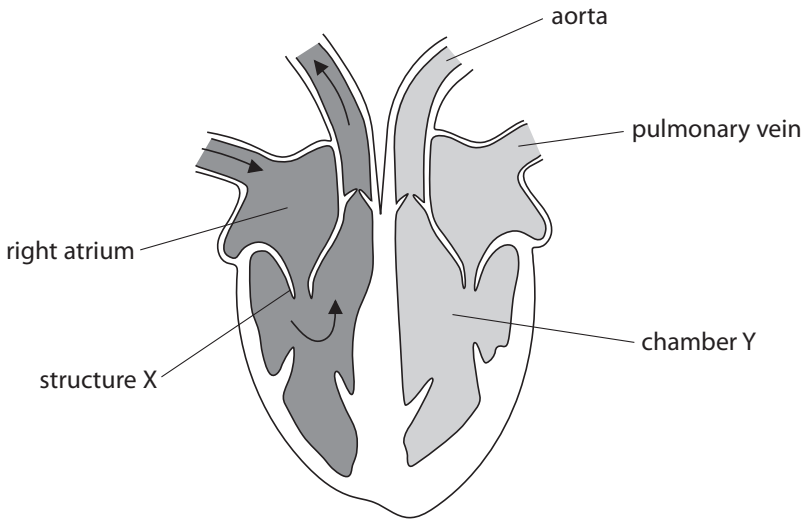
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7. Question 2 · part b · 2 marks

January 2023 Paper 2

Further practice · Attempt part b only.

2 (a) The diagram shows a section through the heart.



(i) Name chamber Y. **(1)**

(ii) Explain the function of structure X. **(2)**

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P 7 2 6 0 2 A 0 4 2 8

7. Question 2 · part b · 2 marks

January 2023 Paper 2

Further practice · Attempt part b only.

(iii) The table lists some properties of blood.

Add ticks (✓) to those boxes that show the properties of blood in the aorta.

(2)

Properties of blood	Blood in the aorta
oxygenated	
deoxygenated	
high pressure	
medium pressure	
low pressure	

(iv) State where blood in the left side of the heart has come from.

(1)

.....

(b) In one year in the United Kingdom, there were 173 heart transplants.

Heart transplant operations require hearts to be donated.

Some hearts cannot be used because the donor is too old.

Give two other reasons why a heart might not be suitable for a transplant.

(2)

1 .....

.....

2 .....

.....

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7. Question 2 · part b · 2 marks

January 2023 Paper 2

Further practice · Attempt part b only.

(c) (i) Which blood component causes blood clotting?

(1)

- ☒ **A** lymphocytes
- ☒ **B** phagocytes
- ☒ **C** platelets
- ☒ **D** red blood cells

(ii) The diagram summarises part of the process that causes a blood clot.



Anticoagulant drugs prevent a blood clot forming.

These drugs stop the enzyme thrombin being produced.

Describe how a lack of thrombin prevents a blood clot forming.

(2)

(Total for Question 2 = 11 marks)

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7. Question 2 · part b · 2 marks · Working and answer

Working and answer

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Practice workbook · Identify arteries, veins and capillaries and relate their structure to function

Comparing and explaining vessel structures, capillary adaptations for exchange, and the jobs of particular vessels.

17 question parts: 1 worked-example retry and 16 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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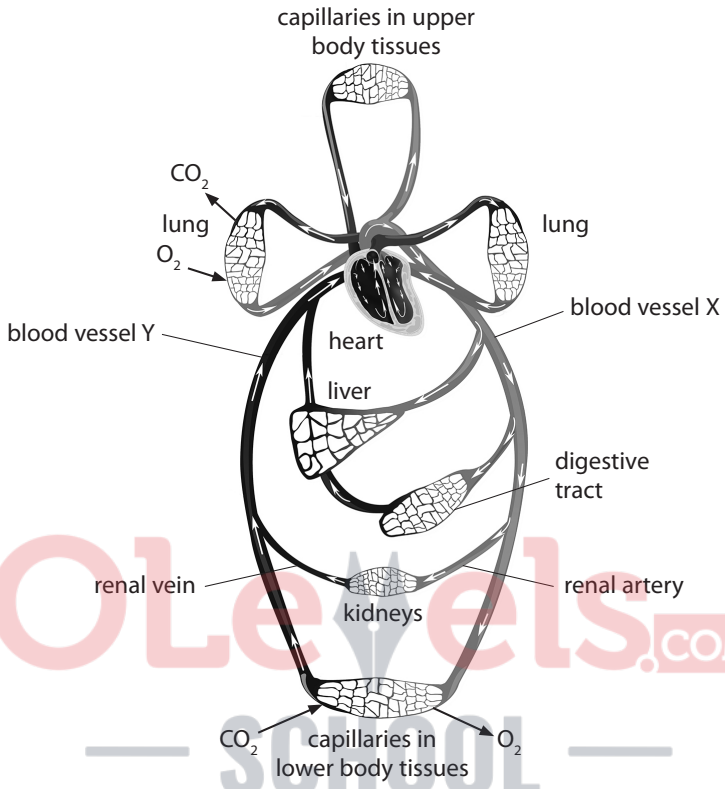
Type ID: HBI-RELATE-VESSEL-STRUCTURE-TO-FUNCTION

1. Question 6 · part a.iv · 3 marks

June 2019 Paper 2

Worked-example retry · Attempt part a.iv only.

6 The diagram shows the human circulatory system.



(Source: © Designua/Shutterstock)

(a) (i) What is the name of blood vessel X?

(1)

- ☒ **A** aorta
- ☒ **B** pulmonary artery
- ☒ **C** pulmonary vein
- ☒ **D** vena cava

(ii) What is the name of blood vessel Y?

(1)

- ☒ **A** aorta
- ☒ **B** pulmonary artery
- ☒ **C** pulmonary vein
- ☒ **D** vena cava

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1. Question 6 · part a.iv · 3 marks

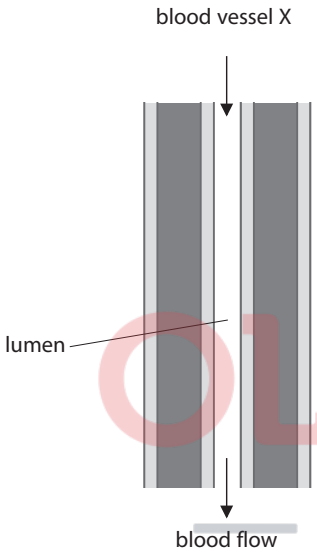
June 2019 Paper 2

Worked-example retry · Attempt part a.iv only.

(iii) The diagram shows a longitudinal section through blood vessel X.

Draw a diagram to show a longitudinal section through blood vessel Y.

(3)



(iv) Explain the differences in the structures of blood vessel X and blood vessel Y.

(3)

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1. Question 6 · part a.iv · 3 marks

June 2019 Paper 2

Worked-example retry · Attempt part a.iv only.

(b) Explain two adaptations of capillaries that increase the rate of diffusion of substances into body tissues.

(4)

1 .....

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2 .....

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(Total for Question 6 = 12 marks)

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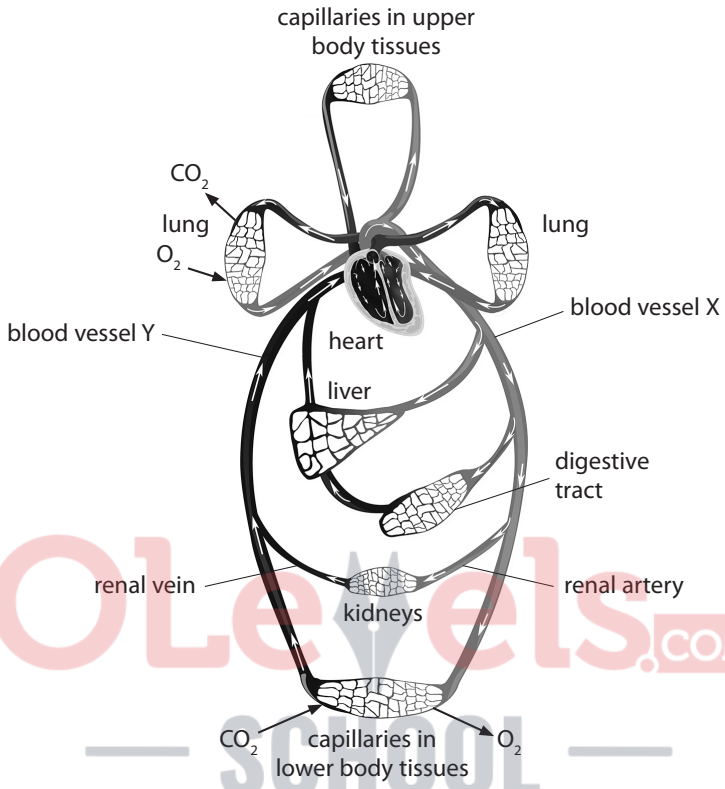
Working and answer

2. Question 6 · part a.iii · 3 marks

June 2019 Paper 2

Further practice · Attempt part a.iii only.

6 The diagram shows the human circulatory system.



(Source: © Designua/Shutterstock)

(a) (i) What is the name of blood vessel X?

(1)

- ☒ **A** aorta
- ☒ **B** pulmonary artery
- ☒ **C** pulmonary vein
- ☒ **D** vena cava

(ii) What is the name of blood vessel Y?

(1)

- ☒ **A** aorta
- ☒ **B** pulmonary artery
- ☒ **C** pulmonary vein
- ☒ **D** vena cava

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2. Question 6 · part a.iii · 3 marks

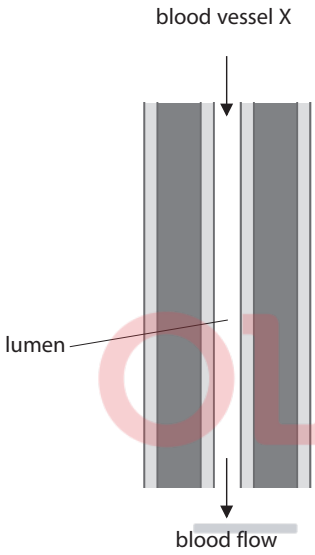
June 2019 Paper 2

Further practice · Attempt part a.iii only.

(iii) The diagram shows a longitudinal section through blood vessel X.

Draw a diagram to show a longitudinal section through blood vessel Y.

(3)



(iv) Explain the differences in the structures of blood vessel X and blood vessel Y.

(3)

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2. Question 6 · part a.iii · 3 marks · Working and answer

Working and answer

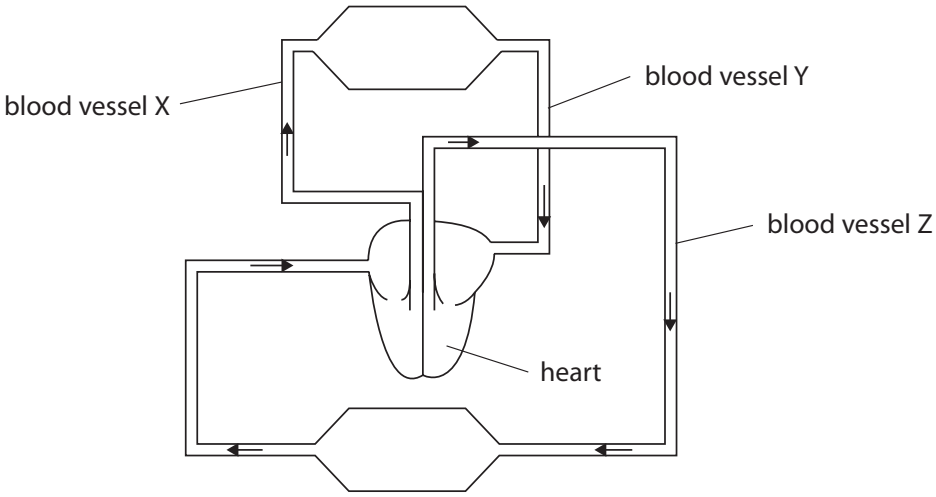
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3. Question 1 · part c · 3 marks

June 2022 Paper 2

Further practice · Attempt part c only.

1 The diagram shows a simple version of the human circulatory system.



(a) Name blood vessel X. **(1)**

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3. Question 1 · part c · 3 marks

June 2022 Paper 2

Further practice · Attempt part c only.

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(b) The box gives words about the circulatory system.

body	oxygenated	lungs	ventricle
	deoxygenated	atrium	

Complete the sentences using words from the box.

(5)

Blood vessel Y transports ..... blood from the

..... to the left ..... of the heart.

Blood vessel Z transports blood from the left ..... of the heart

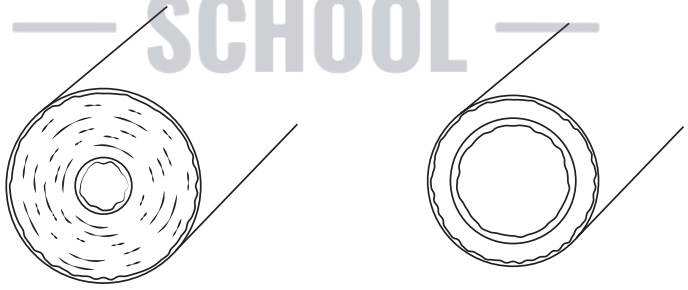
to the .....

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(c) The diagram shows the structure of blood vessel X and blood vessel Y.



blood vessel X blood vessel Y

Compare the structure of blood vessel X with the structure of blood vessel Y.

(3)

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.....

(Total for Question 1 = 9 marks)

3. Question 1 · part c · 3 marks · Working and answer

Working and answer

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4. Question 2 · part b · 2 marks

June 2024 Paper 1R

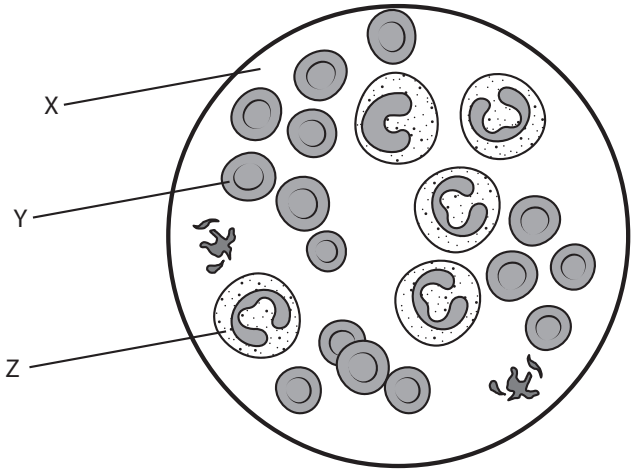
Further practice · Attempt part b only.

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2 (a) The diagram shows components, X, Y and Z, in a sample of human blood.



(i) Name components X, Y and Z.

(3)

X .....

Y .....

Z .....

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(ii) Describe the function of component Y.

(2)

.....

.....

.....

.....

(iii) A sample of blood from a patient is analysed in a hospital.

A scientist finds that the number of component Z in the blood sample is higher than normal.

State a possible reason for the increase in the number of component Z.

(1)

.....

.....

4. Question 2 · part b · 2 marks

June 2024 Paper 1R

Further practice · Attempt part b only.

(b) Blood is transported around the body inside blood vessels.

Draw **one** straight line from each blood vessel to a feature of the blood vessel.

(2)

Blood vessel	Feature of blood vessel
artery	contains valves to prevent backflow of blood
vein	has muscular walls to maintain high blood pressure
capillary	has thin walls for exchange of materials

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4. Question 2 · part b · 2 marks

June 2024 Paper 1R

Further practice · Attempt part b only.

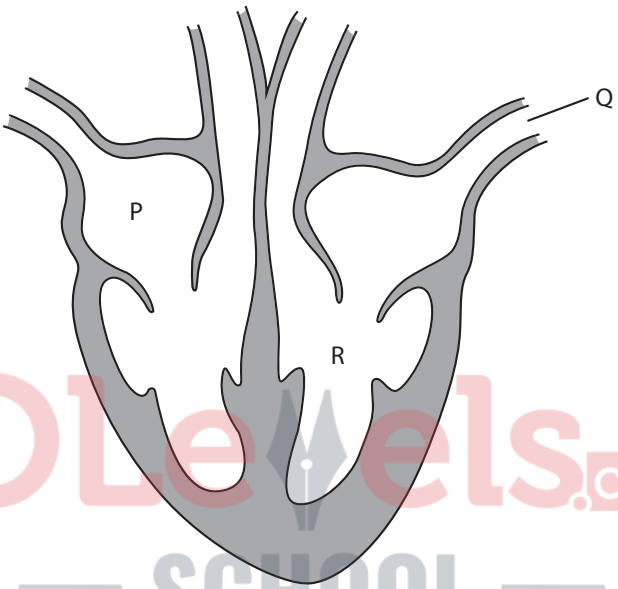
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(c) The heart pumps blood around the body.

The diagram shows a human heart.



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(i) The heart is divided into two halves.

Each half is divided into two chambers.

Name chamber R.

(1)

(ii) Describe the route that the blood travels from part P to part Q.

(4)

(Total for Question 2 = 13 marks)

4. Question 2 · part b · 2 marks · Working and answer

Working and answer

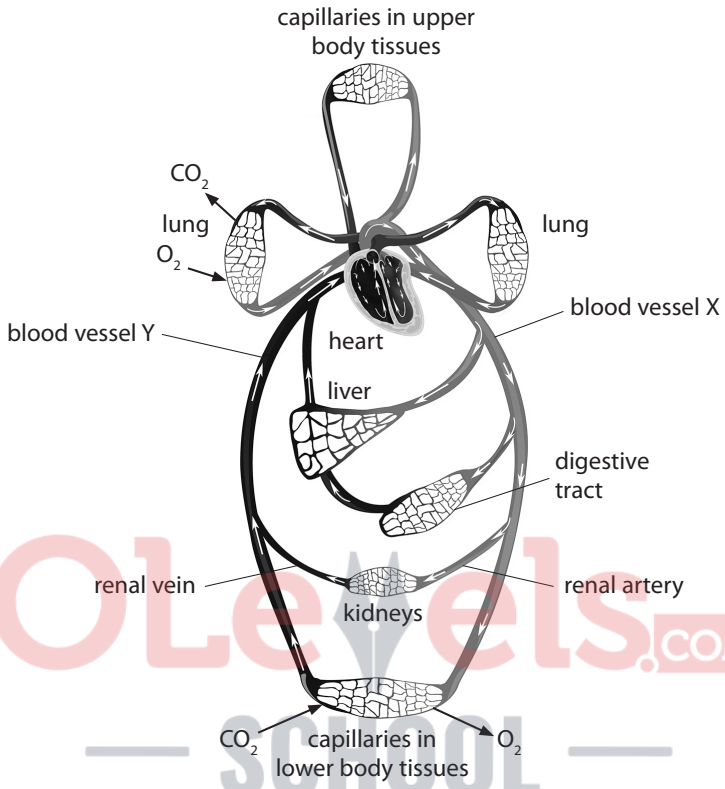
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5. Question 6 · part b · 4 marks

June 2019 Paper 2

Further practice · Attempt part b only.

6 The diagram shows the human circulatory system.



(Source: © Designua/Shutterstock)

(a) (i) What is the name of blood vessel X?

(1)

- ☒ **A** aorta
- ☒ **B** pulmonary artery
- ☒ **C** pulmonary vein
- ☒ **D** vena cava

(ii) What is the name of blood vessel Y?

(1)

- ☒ **A** aorta
- ☒ **B** pulmonary artery
- ☒ **C** pulmonary vein
- ☒ **D** vena cava

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5. Question 6 · part b · 4 marks

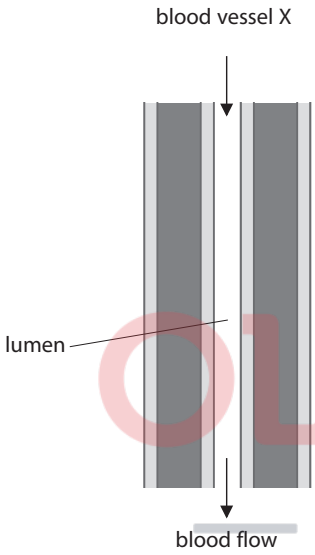
June 2019 Paper 2

Further practice · Attempt part b only.

(iii) The diagram shows a longitudinal section through blood vessel X.

Draw a diagram to show a longitudinal section through blood vessel Y.

(3)



(iv) Explain the differences in the structures of blood vessel X and blood vessel Y.

(3)

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5. Question 6 · part b · 4 marks

June 2019 Paper 2

Further practice · Attempt part b only.

(b) Explain two adaptations of capillaries that increase the rate of diffusion of substances into body tissues.

(4)

1 .....

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2 .....

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(Total for Question 6 = 12 marks)

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5. Question 6 · part b · 4 marks · Working and answer

Working and answer

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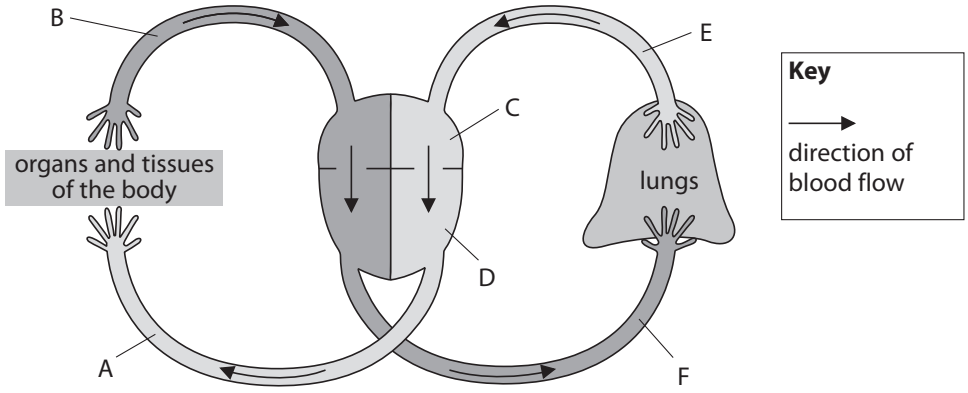
6. Question 4 · part d · 2 marks

January 2020 Paper 1

Further practice · Attempt part d only.

4 William Harvey was the first person to describe the circulation of blood around the body.

The diagram shows a model of the human circulatory system based on the research carried out by Harvey.



(a) Harvey suggested that each side of the heart was divided into two chambers.

State the names of chambers C and D.

(2)

chamber C .....

chamber D .....

(b) Harvey also suggested that blood is transported to and from the heart in different types of vessel.

Explain how the diagram supports Harvey's suggestion.

(4)

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6. Question 4 · part d · 2 marks

January 2020 Paper 1

Further practice · Attempt part d only.

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(c) Before Harvey's discoveries some scientists thought that the blood was circulated around the body by the liver.

State why this idea is not correct.

(1)

.....

.....

(d) Harvey's research did not include capillaries.

Describe the function of capillaries.

(2)

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(Total for Question 4 = 9 marks)

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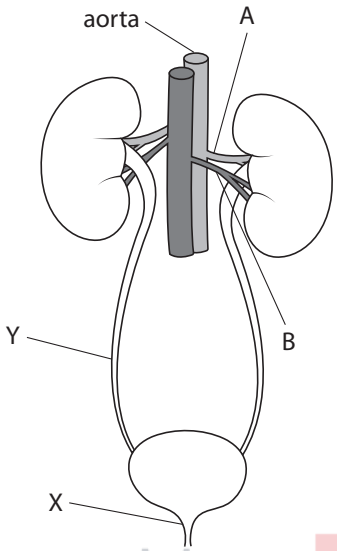
Working and answer

7. Question 8 · part a.ii · 2 marks

January 2020 Paper 1

Further practice · Attempt part a.ii only.

8 The diagram shows part of the human renal system.



(a) (i) Give the names of structures X and Y. **(2)**

structure X .....

structure Y .....

(ii) Describe the function of the aorta. **(2)**

.....

.....

.....

.....

(iii) Compare the composition of the blood in vessels A and B. **(2)**

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.....

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7. Question 8 · part a.ii · 2 marks · Working and answer

Working and answer

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8. Question 5 · part b · 4 marks

June 2025 Paper 2

Further practice · Attempt part b only.

5 Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

Losartan belongs to a group of drugs known as angiotensin-II receptor antagonists. The body produces angiotensin-II, which is a substance that binds to receptors in blood vessels, causing them to narrow. This causes an increase in blood pressure. Losartan prevents angiotensin-II binding to these receptors, lowering blood pressure.

If high blood pressure is left untreated, a decrease in kidney function can result in proteinuria. This is a condition that causes high levels of protein to be found in a person's urine. People with high blood pressure and a resultant thickening of the left ventricle can be treated with Losartan to reduce the risk of stroke.

People with high blood pressure should not take Losartan if their liver is not functioning properly. Women who are pregnant or who are breast feeding should not take Losartan.

(a) Describe how the action of Losartan differs from the action of an ACE inhibitor. (lines 4 and 5)

(4)

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8. Question 5 · part b · 4 marks

June 2025 Paper 2

Further practice · Attempt part b only.

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(b) Explain how the structure of a named type of blood vessel allows angiotensin-II to increase blood pressure. (lines 2 to 5)

(4)

(c) Explain why proteinuria is a sign of problems with kidney function. (lines 6 to 8)

(2)

(d) Suggest why high blood pressure causes the left ventricle to become thickened. (lines 8 to 9)

(2)

(e) Explain why people who have high blood pressure are advised not to take Losartan if their liver is not functioning properly. (lines 10 to 11)

(2)

(Total for Question 5 = 14 marks)

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8. Question 5 · part b · 4 marks · Working and answer

Working and answer

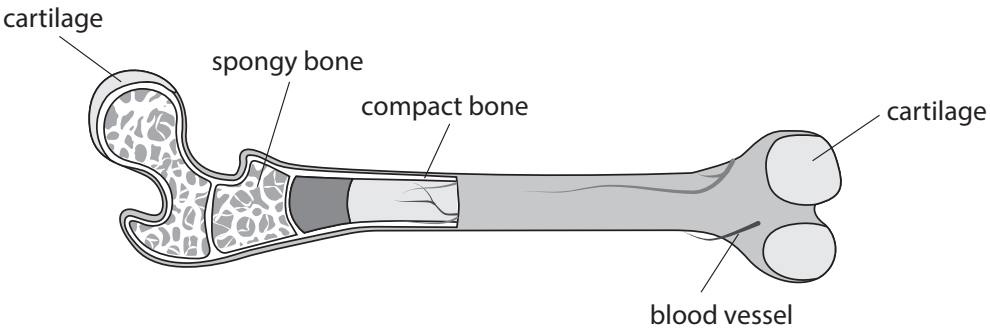
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9. Question 1 · part a.iii · 2 marks

June 2025 Paper 2R

Further practice · Attempt part a.iii only.

1 (a) The diagram shows a bone found in the human body.



(Source: adapted from: © Designua / Shutterstock)

(i) State the name of the type of bone shown in the diagram.

(1)

.....

(ii) Place one tick (✓) in each row to give the function of each part of the bone.

(3)

Part	Function		
	reduces friction	where red blood cells are produced	provides strength
compact bone			
spongy bone			
cartilage			

(iii) Describe the function of the blood vessel in the bone.

(2)

.....

.....

.....

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9. Question 1 · part a.iii · 2 marks

June 2025 Paper 2R

Further practice · Attempt part a.iii only.

(b) Bones are moved by muscles.

(i) Use the words in the box to complete the passage about bones and muscles.

(4)

cartilage	contracts	ligaments	lowers	pulls
pushes	raises	relaxes	tendons	

Muscles are attached to bones by ..... .

To raise the lower arm, the biceps ..... and the triceps ..... .

The biceps ..... on the radius, which raises the lower arm.

(ii) Energy is transferred by the biceps muscle when raising the lower arm.

The biceps muscle applies a force of 125 N to raise the lower arm a distance of 0.50 m.

Use this formula to calculate the energy transferred by the biceps muscle when raising the lower arm 0.50 m.

energy transferred (J) = force (N) × distance (m)

(2)

energy transferred = ... J

(Total for Question 1 = 12 marks)

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9. Question 1 · part a.iii · 2 marks · Working and answer

Working and answer

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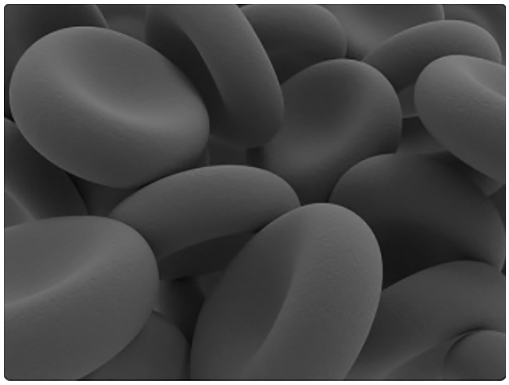
10. Question 1 · part b.i · 1 mark

November 2021 Paper 2

Further practice · Attempt part b.i only.

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1 (a) The photograph shows a group of red blood cells.



(Source: RomanenkoAlexey/Shutterstock)

(i) Describe the function of red blood cells.

(3)

OLevels.com SCHOOL

(ii) Explain how the structure of the red blood cell is adapted to its function.

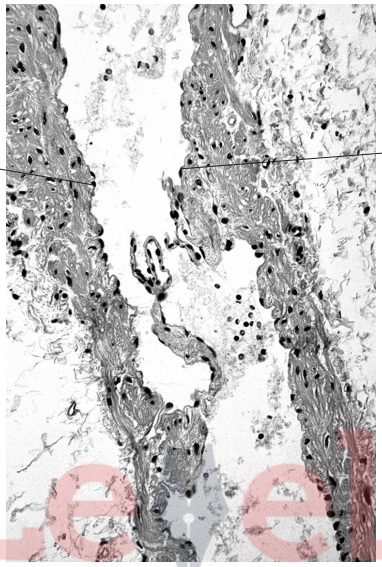
(4)

10. Question 1 · part b.i · 1 mark

November 2021 Paper 2

Further practice · Attempt part b.i only.

(b) The photograph shows a section through a blood vessel in a leg.



magnification ×10

(Source: © DR. GLADDEN WILLIS, VISUALS UNLIMITED/SCIENCE PHOTO LIBRARY)

(i) State the type of blood vessel shown in the photograph.

(1)

.....

(ii) Explain your choice of blood vessel.

(2)

.....

.....

.....

.....

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10. Question 1 · part b.i · 1 mark · Working and answer

Working and answer

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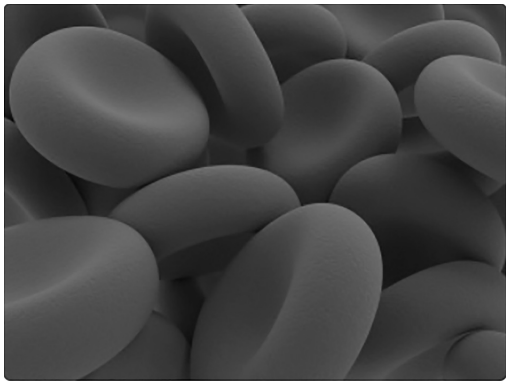
11. Question 1 · part b.ii · 2 marks

November 2021 Paper 2

Further practice · Attempt part b.ii only.

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1 (a) The photograph shows a group of red blood cells.



(Source: RomanenkoAlexey/Shutterstock)

(i) Describe the function of red blood cells.

(3)

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(ii) Explain how the structure of the red blood cell is adapted to its function.

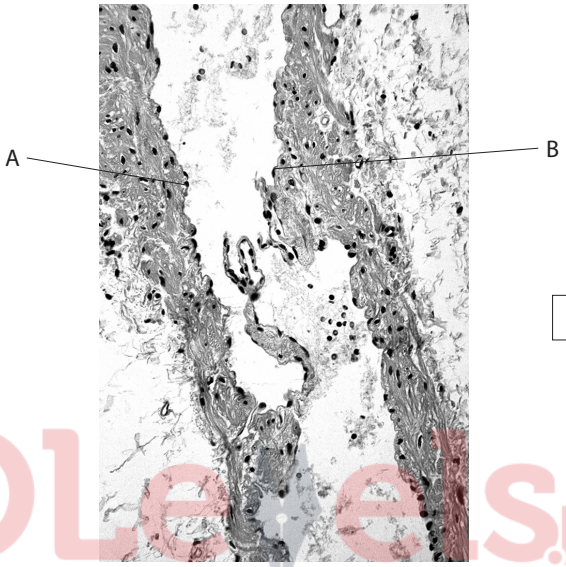
(4)

11. Question 1 · part b.ii · 2 marks

November 2021 Paper 2

Further practice · Attempt part b.ii only.

(b) The photograph shows a section through a blood vessel in a leg.



magnification ×10

(Source: © DR. GLADDEN WILLIS, VISUALS UNLIMITED/SCIENCE PHOTO LIBRARY)

(i) State the type of blood vessel shown in the photograph.

(1)

(ii) Explain your choice of blood vessel.

(2)

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P 6 7 0 6 2 R A 0 4 2 8

11. Question 1 · part b.ii · 2 marks

November 2021 Paper 2

Further practice · Attempt part b.ii only.

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(iii) Calculate the actual distance between points A and B.

(2)

distance = ... mm

(Total for Question 1 = 12 marks)

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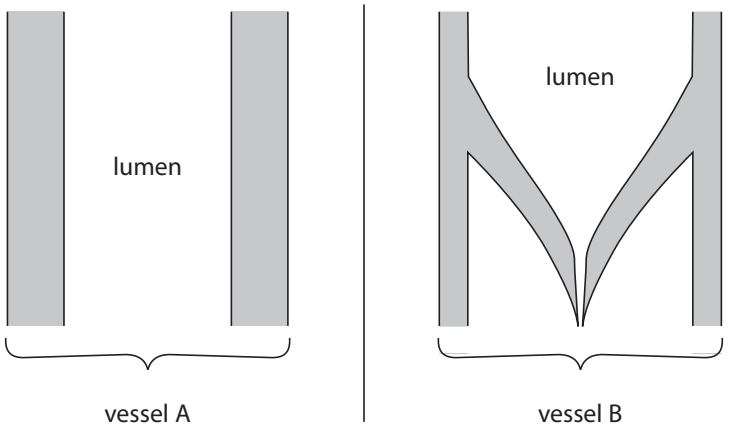
Working and answer

12. Question 3 · part a.ii · 4 marks

June 2022 Paper 2R

Further practice · Attempt part a.ii only.

3 (a) The diagram shows two types of blood vessel, A and B.



(i) Draw an arrow on the diagram to show the direction of blood flow in vessel B.

(1)

(ii) Identify each type of blood vessel, giving explanations for why there are differences in their structures.

(4)

Vessel A

Vessel B

(iii) State the name of one other type of blood vessel.

(1)

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12. Question 3 · part a.ii · 4 marks · Working and answer

Working and answer

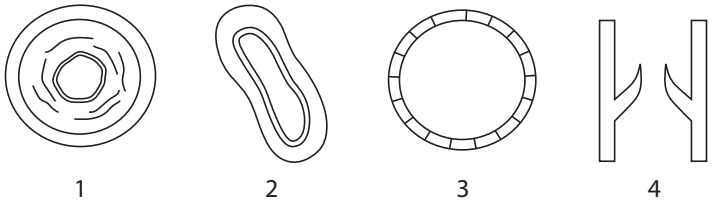
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13. Question 6 · part a.i · 4 marks

November 2024 Paper 2

Further practice · Attempt part a.i only.

6 (a) The diagram shows different sections through various blood vessels.



Not to scale

(i) Identify each type of blood vessel.

(4)

blood vessel 1

blood vessel 2

blood vessel 3

blood vessel 4

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13. Question 6 · part a.i · 4 marks

November 2024 Paper 2

Further practice · Attempt part a.i only.

(ii) State one reason for your choice for each blood vessel.

(4)

blood vessel 1

blood vessel 2

blood vessel 3

blood vessel 4

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13. Question 6 · part a.i · 4 marks · Working and answer

Working and answer

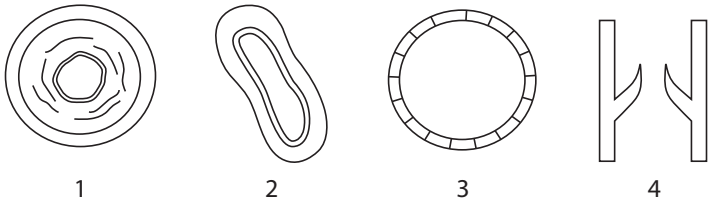
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14. Question 6 · part a.ii · 4 marks

November 2024 Paper 2

Further practice · Attempt part a.ii only.

6 (a) The diagram shows different sections through various blood vessels.



Not to scale

(i) Identify each type of blood vessel.

(4)

blood vessel 1

blood vessel 2

blood vessel 3

blood vessel 4

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14. Question 6 · part a.ii · 4 marks

November 2024 Paper 2

Further practice · Attempt part a.ii only.

(ii) State one reason for your choice for each blood vessel.

(4)

blood vessel 1

blood vessel 2

blood vessel 3

blood vessel 4

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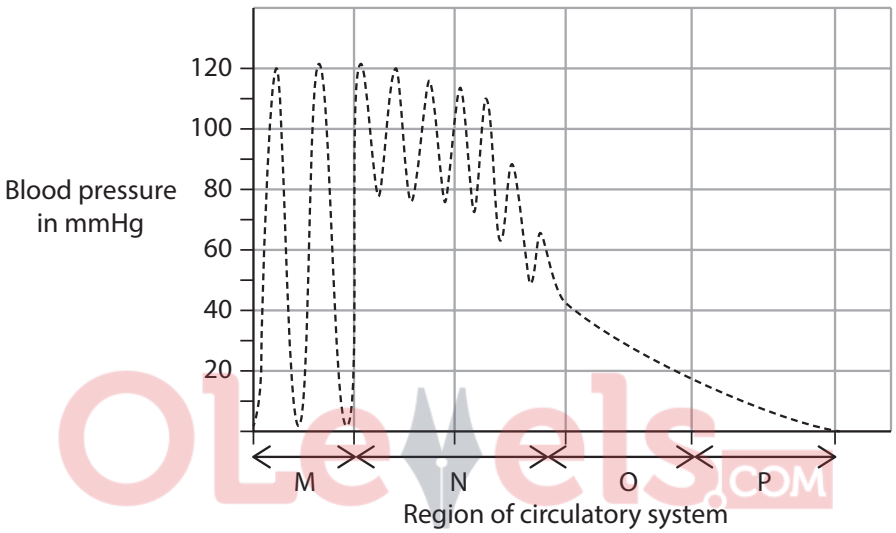
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14. Question 6 · part a.ii · 4 marks

November 2024 Paper 2

Further practice · Attempt part a.ii only.

(b) The graph shows changes in blood pressure in four regions of the circulatory system labelled M, N, O and P.



Identify each region, M, N, O and P, of the circulatory system.

(4)

M .....

N .....

O .....

P .....

(Total for Question 6 = 12 marks)

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14. Question 6 · part a.ii · 4 marks · Working and answer

Working and answer

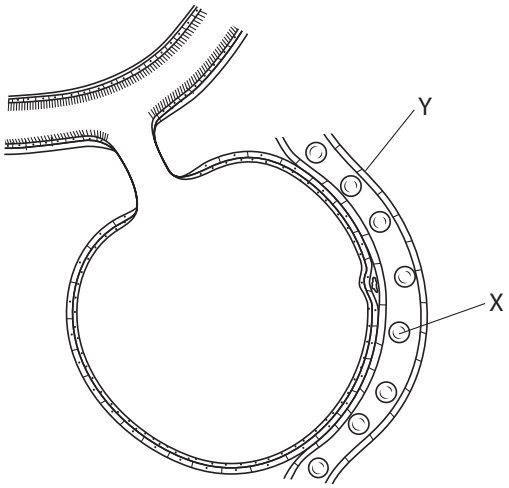
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15. Question 3 · part b · 2 marks

November 2020 Paper 2

Further practice · Attempt part b only.

3 The diagram shows an alveolus and its blood supply.



(a) (i) What is the name of blood component X? **(1)**

- ☒ **A** plasma
- ☒ **B** platelet
- ☒ **C** red blood cell
- ☒ **D** white blood cell

(ii) State three ways in which air in the alveolus differs from air in the atmosphere. **(3)**

1 .....

.....

2 .....

.....

3 .....

.....

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15. Question 3 · part b · 2 marks

November 2020 Paper 2

Further practice · Attempt part b only.

(iii) Give three features of alveoli that allow efficient gas exchange.

(3)

1 .....

.....

2 .....

.....

3 .....

.....

(b) Structure Y is a capillary.

Give two features in the diagram which show that structure Y is a capillary.

(2)

1 .....

.....

2 .....

.....

(c) Some people have a condition known as emphysema. One symptom of emphysema is the breakdown of elastic fibres in the lung tissue.

Suggest the effects that loss of elastic fibres have on a person's ability to breathe.

(2)

.....

.....

.....

.....

(Total for Question 3 = 11 marks)

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15. Question 3 · part b · 2 marks · Working and answer

Working and answer

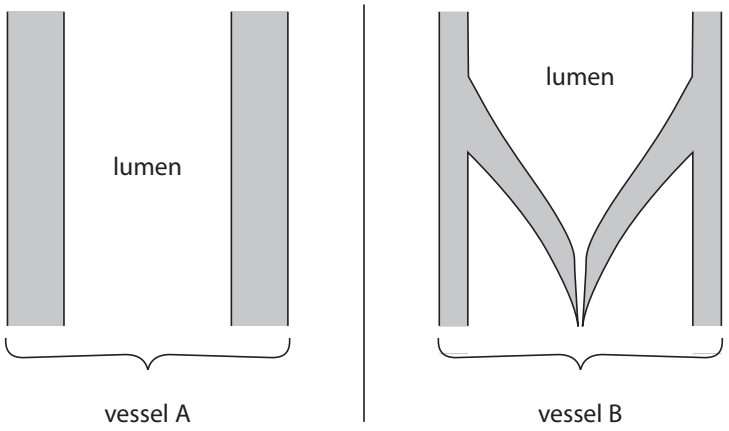
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16. Question 3 · part a.i · 1 mark

June 2022 Paper 2R

Further practice · Attempt part a.i only.

3 (a) The diagram shows two types of blood vessel, A and B.



(i) Draw an arrow on the diagram to show the direction of blood flow in vessel B.

(1)

(ii) Identify each type of blood vessel, giving explanations for why there are differences in their structures.

(4)

Vessel A

Vessel B

(iii) State the name of one other type of blood vessel.

(1)

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16. Question 3 · part a.i · 1 mark · Working and answer

Working and answer

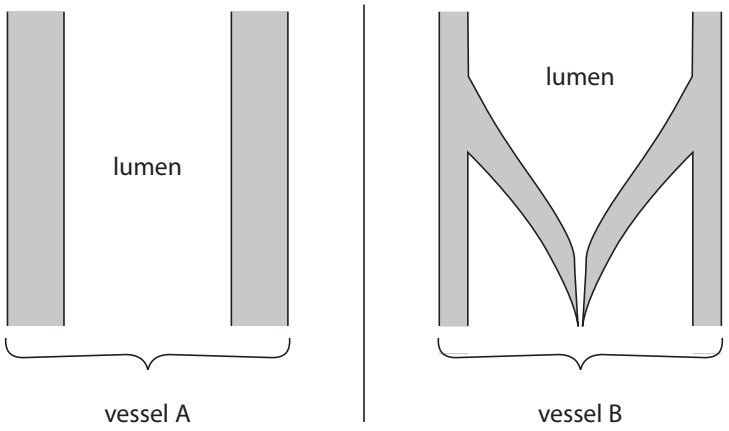
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17. Question 3 · part a.iii · 1 mark

June 2022 Paper 2R

Further practice · Attempt part a.iii only.

3 (a) The diagram shows two types of blood vessel, A and B.



(i) Draw an arrow on the diagram to show the direction of blood flow in vessel B.

(1)

(ii) Identify each type of blood vessel, giving explanations for why there are differences in their structures.

(4)

Vessel A

Vessel B

(iii) State the name of one other type of blood vessel.

(1)

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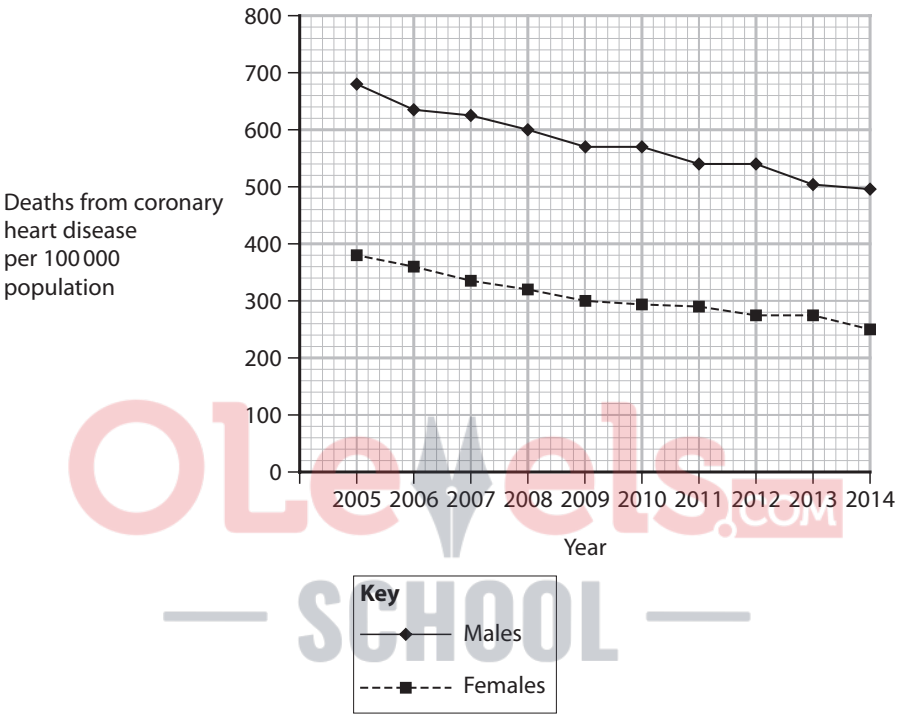
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17. Question 3 · part a.iii · 1 mark

June 2022 Paper 2R

Further practice · Attempt part a.iii only.

(b) Coronary heart disease kills many people each year. The graph shows the number of deaths per 100 000 of the whole population in a part of the world.



(i) Describe the trends shown by the graph.

(3)

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17. Question 3 · part a.iii · 1 mark · Working and answer

Working and answer

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Practice workbook · Explain red-cell adaptations and the effects of more or less oxygen carriage

How red-cell structure suits oxygen transport, and what follows when haemoglobin, red-cell number or oxygenation changes.

5 question parts: 1 worked-example retry and 4 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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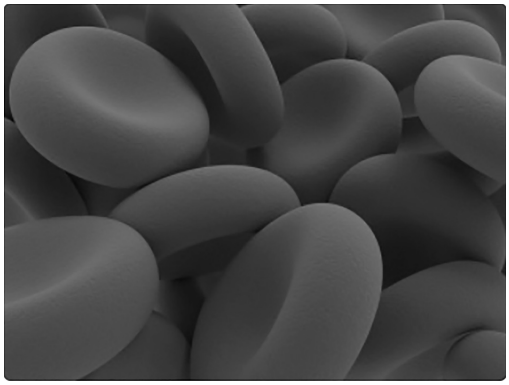
Type ID: HBI-EXPLAIN-OXYGEN-CARRIAGE-AND-ITS-CHANGES

1. Question 1 · part a.ii · 4 marks

November 2021 Paper 2

Worked-example retry · Attempt part a.ii only.

1 (a) The photograph shows a group of red blood cells.



(Source: RomanenkoAlexey/Shutterstock)

(i) Describe the function of red blood cells.

(3)

(ii) Explain how the structure of the red blood cell is adapted to its function.

(4)

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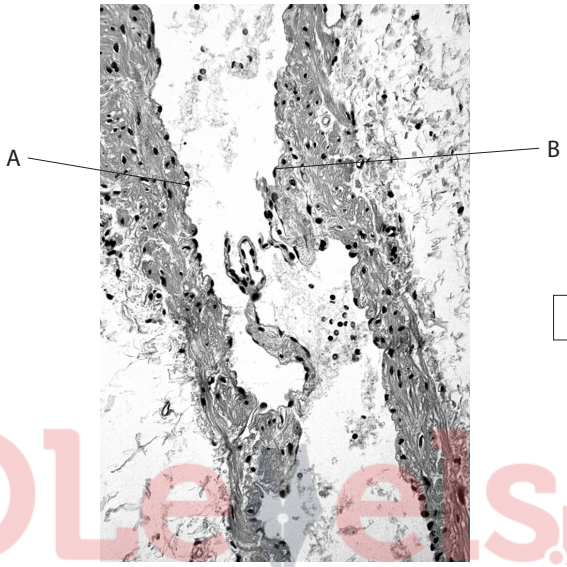
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1. Question 1 · part a.ii · 4 marks

November 2021 Paper 2

Worked-example retry · Attempt part a.ii only.

(b) The photograph shows a section through a blood vessel in a leg.



magnification ×10

(Source: © DR. GLADDEN WILLIS, VISUALS UNLIMITED/SCIENCE PHOTO LIBRARY)

(i) State the type of blood vessel shown in the photograph.

(1)

(ii) Explain your choice of blood vessel.

(2)

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1. Question 1 · part a.ii · 4 marks · Working and answer

Working and answer

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2. Question 5 · part b.iii · 5 marks

January 2023 Paper 2

Further practice · Attempt part b.iii only.

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5 Some athletes train at high altitudes to improve their performance.

Performance improves if more energy is released from aerobic respiration.

(a) (i) Complete the balanced symbol equation for aerobic respiration.

(2)

..... + $6O_2$ → ..... + $6CO_2$

(ii) Aerobic respiration produces ATP.

Describe how energy is released from ATP during exercise.

(2)

.....

.....

.....

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2. Question 5 · part b.iii · 5 marks

January 2023 Paper 2

Further practice · Attempt part b.iii only.

(b) The table shows the percentage change in mass of haemoglobin in the blood after different periods of time training at altitude.

Period of time in hours	Percentage change in mass of haemoglobin
0	0
100	0.7
200	2.4
300	4.1
400	5.9
500	6.5
600	7.2

(i) Plot the results on the grid.

(4)

(ii) Draw the curve of best fit.

(1)

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2. Question 5 · part b.iii · 5 marks

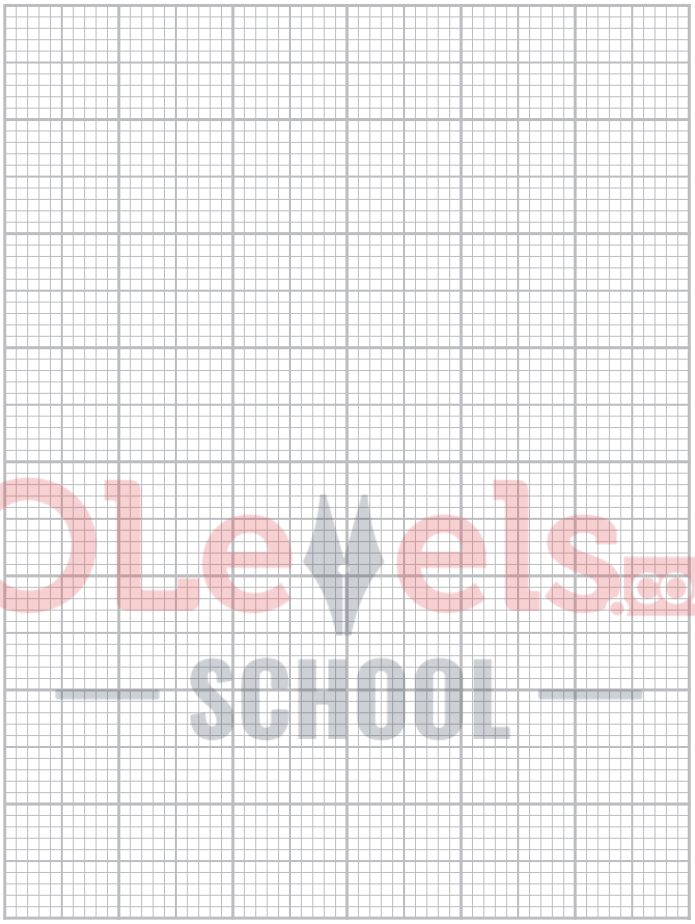
January 2023 Paper 2

Further practice · Attempt part b.iii only.

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2. Question 5 · part b.iii · 5 marks

January 2023 Paper 2

Further practice · Attempt part b.iii only.

(iii) Discuss how the percentage change in mass of haemoglobin, after training at altitude, could result in an increase in performance during exercise.

(5)

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(Total for Question 5 = 14 marks)

2. Question 5 · part b.iii · 5 marks · Working and answer

Working and answer

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3. Question 6 · part a.ii · 5 marks

June 2023 Paper 2

Further practice · Attempt part a.ii only.

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6 Sickle cell anaemia is an inherited condition. It is caused by a mutation of the gene that controls the production of haemoglobin.

This condition is caused by a recessive allele.

(a) (i) Explain how a mutation can cause a condition such as sickle cell anaemia.

(4)

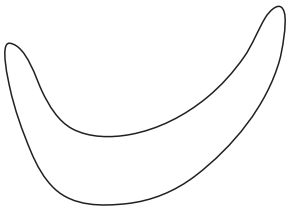
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3. Question 6 · part a.ii · 5 marks

June 2023 Paper 2

Further practice · Attempt part a.ii only.

(ii) In low oxygen concentrations the red blood cells of someone with sickle cell anaemia become like the cell shown in the diagram.



Explain why a person who has the sickle cell condition often feels tired.

(5)

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3. Question 6 · part a.ii · 5 marks

June 2023 Paper 2

Further practice · Attempt part a.ii only.

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(b) A woman, heterozygous for sickle cell anaemia, and a man of unknown genotype for sickle cell anaemia have a child. The child is born with sickle cell anaemia.

Explain the possible genotypes of the man.

Use H to represent the allele for unaffected haemoglobin and h to represent the allele for sickle haemoglobin.

(4)

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(Total for Question 6 = 13 marks)

3. Question 6 · part a.ii · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 5 · part b.i · 4 marks

November 2024 Paper 1

Further practice · Attempt part b.i only.

5 (a) The table gives information about some blood components in three people.

Blood component	Healthy person number per cm^3	Person A number per cm^3	Person B number per cm^3
white blood cells	7500	13 750	7250
red blood cells	5.0×10^6	5.0×10^6	5.0×10^6
platelets	3.0×10^5	3.0×10^5	0.8×10^5

(i) Calculate the ratio of the number of white blood cells in the healthy person compared with the number of white blood cells in person A.
Write your answer in the form of 1 : n

(2)

1 : .....

(ii) Explain a possible reason for the difference in the number of white blood cells in the healthy person and person A.

(2)

.....

.....

.....

.....

(iii) Describe the effect that the lower number of platelets per cm^3 would have on the health of person B.

(2)

.....

.....

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P 7 5 9 5 8 A 0 1 2 2 4

4. Question 5 · part b.i · 4 marks

November 2024 Paper 1

Further practice · Attempt part b.i only.

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(b) (i) Haemolytic anaemia (HA) is an inherited condition.
People with this condition have a low number of red blood cells in their blood.

Explain one symptom that people with HA will have.

(4)

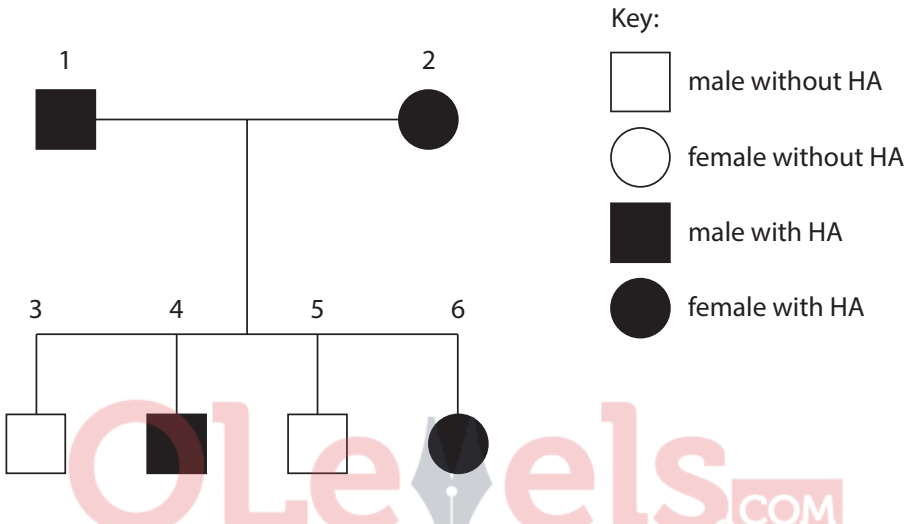
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4. Question 5 · part b.i · 4 marks

November 2024 Paper 1

Further practice · Attempt part b.i only.

(ii) The diagram shows the inheritance of HA in one family.



Explain how HA is inherited.

(4)

(Total for Question 5 = 14 marks)

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4. Question 5 · part b.i · 4 marks · Working and answer

Working and answer

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5. Question 4 · part b · 4 marks

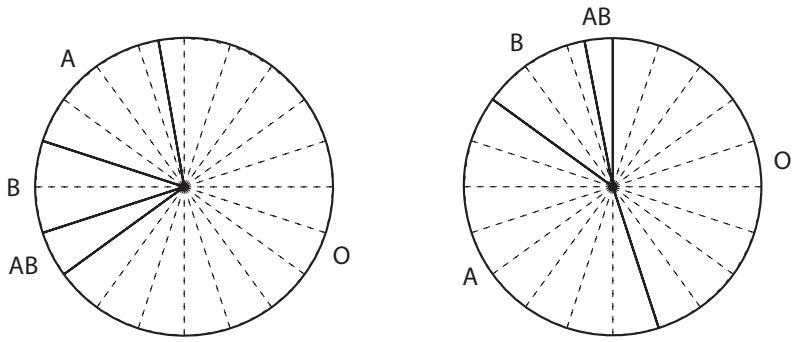
June 2022 Paper 1

Further practice · Attempt part b only.

4 (a) Humans belong to one of four blood groups.

These blood groups are A, B, AB and O.

The pie charts show estimates of the percentages of people from Africa and the percentages of people from Europe in each blood group.



(i) Complete the tables to give the percentage of each blood group.

Some information has been given.

(2)

Africa

Blood group	Percentage of blood group
O	68
A	
B	
AB	5

Europe

Blood group	Percentage of blood group
O	45
A	
B	12
AB	

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5. Question 4 · part b · 4 marks

June 2022 Paper 1

Further practice · Attempt part b only.

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(ii) Some people need a blood transfusion.

Explain why people with blood group A cannot receive a blood transfusion from people with blood group B.

(3)

(iii) A male with the genotype I^AI^O and a female with the genotype I^BI^O have a child with the genotype I^OI^O.

Draw a genetic diagram to show how the child inherits the genotype I^OI^O.

(3)

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(iv) Determine the probability that a child from these parents will inherit the blood group AB.

(1)

probability = ...

5. Question 4 · part b · 4 marks

June 2022 Paper 1

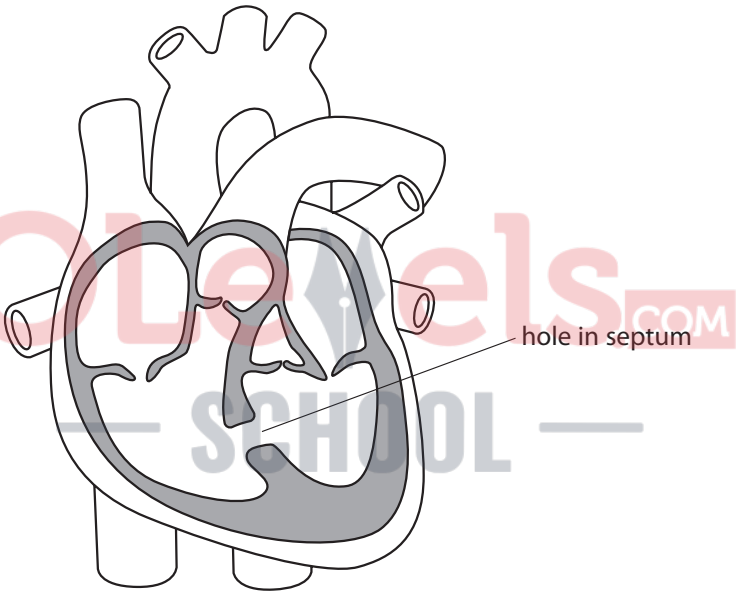
Further practice · Attempt part b only.

(b) The heart pumps the blood around the body.

The diagram shows a human heart.

This heart has a hole in the septum.

This means that blood from the left side of the heart mixes with blood from the right side.



Explain why a person with a hole in their heart breathes at a greater rate than a person with a healthy heart.

(4)

(Total for Question 4 = 13 marks)

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5. Question 4 · part b · 4 marks · Working and answer

Working and answer

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Practice workbook · Explain systolic, diastolic and changing pressure around the circulation

Meaning of systolic and diastolic pressure and reading or explaining pressure changes from heart to arteries, capillaries and veins.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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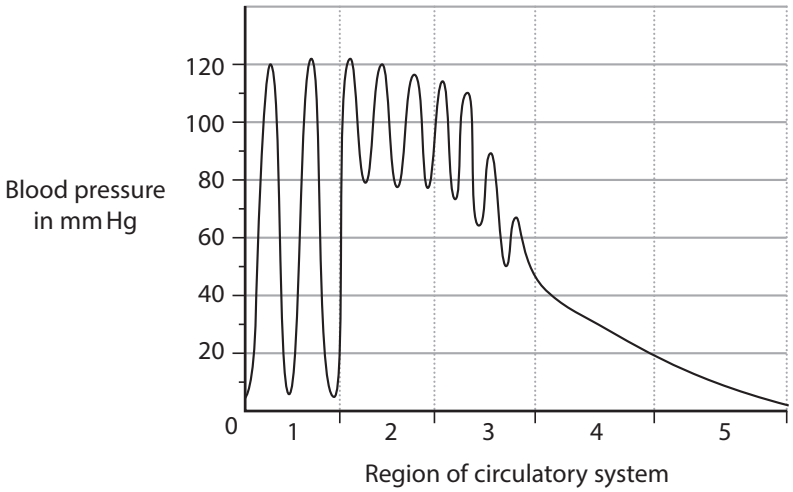
Type ID: HBI-EXPLAIN-BLOOD-PRESSURE-CHANGES

1. Question 3 · part b · 5 marks

June 2023 Paper 1R

Worked-example retry · Attempt part b only.

3 The graph shows how blood pressure varies in different regions of the circulatory system.



(a) (i) Estimate the difference between the maximum blood pressure and minimum blood pressure in region 1.

(2)

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difference in blood pressure = ... mm Hg

(ii) Which part of the circulatory system is represented by region 2?

(1)

- ☒ **A** arteries
- ☒ **B** capillaries
- ☒ **C** heart
- ☒ **D** veins

(iii) Which part of the circulatory system is represented by region 5?

(1)

- ☒ **A** arteries
- ☒ **B** capillaries
- ☒ **C** heart
- ☒ **D** veins

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P 7 2 5 8 6 A 0 6 2 0

1. Question 3 · part b · 5 marks

June 2023 Paper 1R

Worked-example retry · Attempt part b only.

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(b) Explain why blood pressure varies in region 1.

(5)

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(c) Explain why the blood pressure changes in region 4.

(4)

(Total for Question 3 = 13 marks)

1. Question 3 · part b · 5 marks · Working and answer

Working and answer

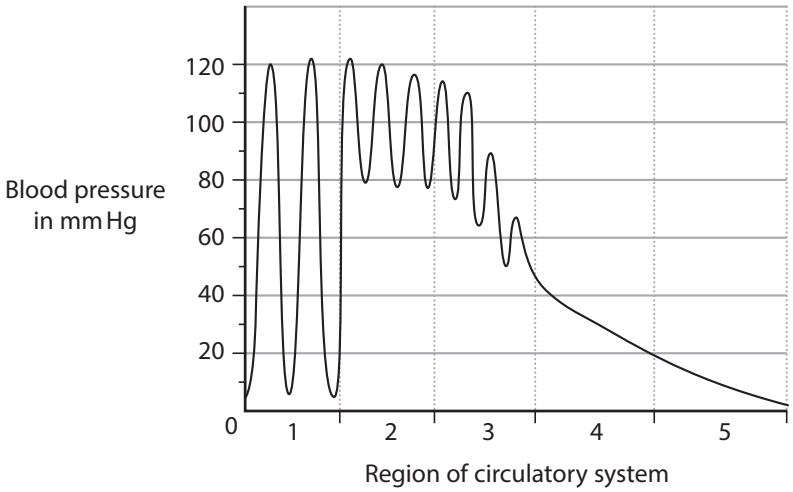
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2. Question 3 · part a.ii · 1 mark

June 2023 Paper 1R

Further practice · Attempt part a.ii only.

3 The graph shows how blood pressure varies in different regions of the circulatory system.



(a) (i) Estimate the difference between the maximum blood pressure and minimum blood pressure in region 1. **(2)**

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difference in blood pressure = ... mm Hg

(ii) Which part of the circulatory system is represented by region 2? **(1)**

- ☒ **A** arteries
- ☒ **B** capillaries
- ☒ **C** heart
- ☒ **D** veins

(iii) Which part of the circulatory system is represented by region 5? **(1)**

- ☒ **A** arteries
- ☒ **B** capillaries
- ☒ **C** heart
- ☒ **D** veins

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2. Question 3 · part a.ii · 1 mark · Working and answer

Working and answer

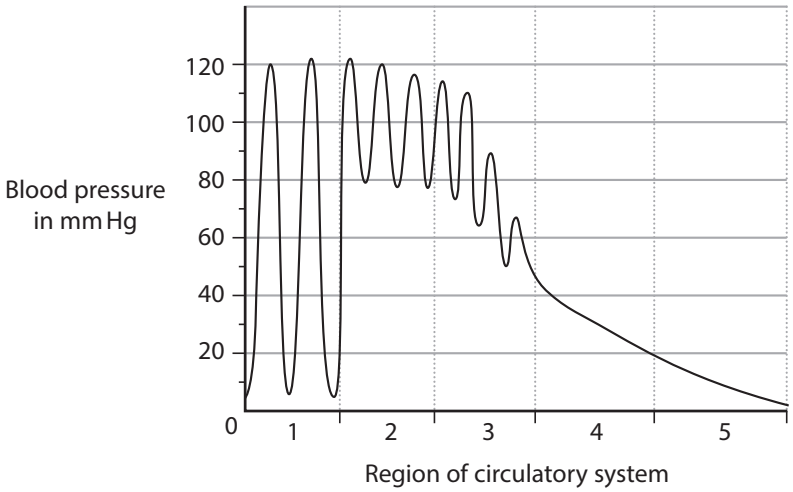
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3. Question 3 · part a.iii · 1 mark

June 2023 Paper 1R

Further practice · Attempt part a.iii only.

3 The graph shows how blood pressure varies in different regions of the circulatory system.



(a) (i) Estimate the difference between the maximum blood pressure and minimum blood pressure in region 1.

(2)

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difference in blood pressure = ... mm Hg

(ii) Which part of the circulatory system is represented by region 2?

(1)

- ☒ **A** arteries
- ☒ **B** capillaries
- ☒ **C** heart
- ☒ **D** veins

(iii) Which part of the circulatory system is represented by region 5?

(1)

- ☒ **A** arteries
- ☒ **B** capillaries
- ☒ **C** heart
- ☒ **D** veins

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3. Question 3 · part a.iii · 1 mark

June 2023 Paper 1R

Further practice · Attempt part a.iii only.

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(b) Explain why blood pressure varies in region 1.

(5)

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(c) Explain why the blood pressure changes in region 4.

(4)

(Total for Question 3 = 13 marks)

3. Question 3 · part a.iii · 1 mark · Working and answer

Working and answer

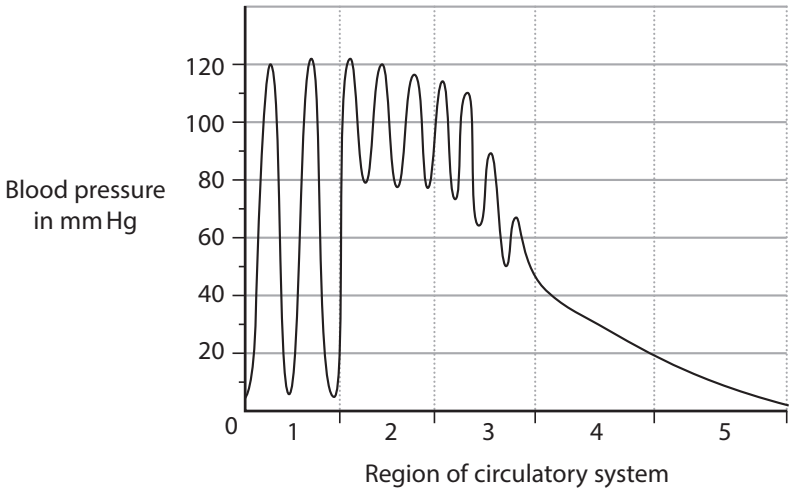
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4. Question 3 · part c · 4 marks

June 2023 Paper 1R

Further practice · Attempt part c only.

3 The graph shows how blood pressure varies in different regions of the circulatory system.



(a) (i) Estimate the difference between the maximum blood pressure and minimum blood pressure in region 1.

(2)

OLevels.COM SCHOOL

difference in blood pressure = ... mm Hg

(ii) Which part of the circulatory system is represented by region 2?

(1)

- ☒ **A** arteries
- ☒ **B** capillaries
- ☒ **C** heart
- ☒ **D** veins

(iii) Which part of the circulatory system is represented by region 5?

(1)

- ☒ **A** arteries
- ☒ **B** capillaries
- ☒ **C** heart
- ☒ **D** veins

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4. Question 3 · part c · 4 marks

June 2023 Paper 1R

Further practice · Attempt part c only.

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(b) Explain why blood pressure varies in region 1.

(5)

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(c) Explain why the blood pressure changes in region 4.

(4)

(Total for Question 3 = 13 marks)

4. Question 3 · part c · 4 marks · Working and answer

Working and answer

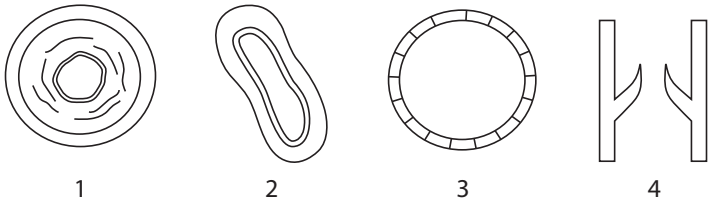
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5. Question 6 · part b · 4 marks

November 2024 Paper 2

Further practice · Attempt part b only.

6 (a) The diagram shows different sections through various blood vessels.



Not to scale

(i) Identify each type of blood vessel.

(4)

blood vessel 1

blood vessel 2

blood vessel 3

blood vessel 4

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5. Question 6 · part b · 4 marks

November 2024 Paper 2

Further practice · Attempt part b only.

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(ii) State one reason for your choice for each blood vessel.

(4)

blood vessel 1

blood vessel 2

blood vessel 3

blood vessel 4

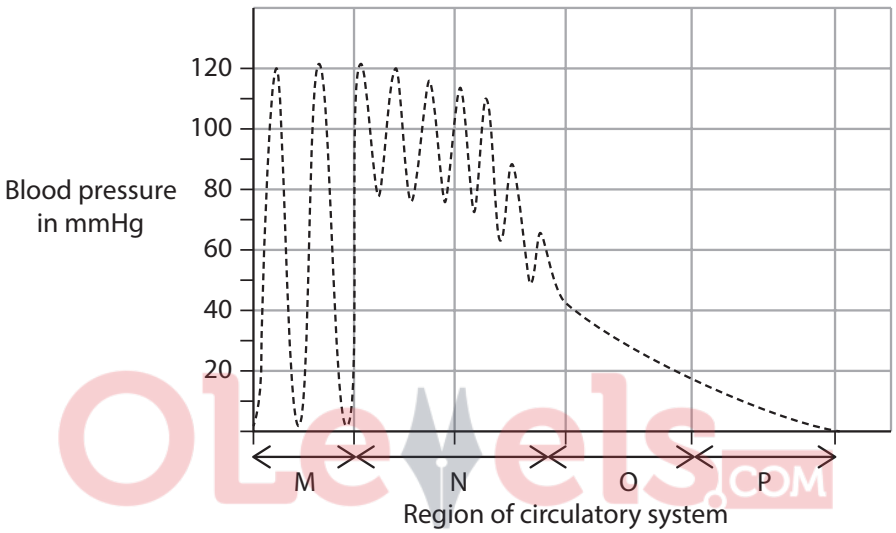
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5. Question 6 · part b · 4 marks

November 2024 Paper 2

Further practice · Attempt part b only.

(b) The graph shows changes in blood pressure in four regions of the circulatory system labelled M, N, O and P.



Identify each region, M, N, O and P, of the circulatory system.

(4)

M .....

N .....

O .....

P .....

(Total for Question 6 = 12 marks)

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5. Question 6 · part b · 4 marks · Working and answer

Working and answer

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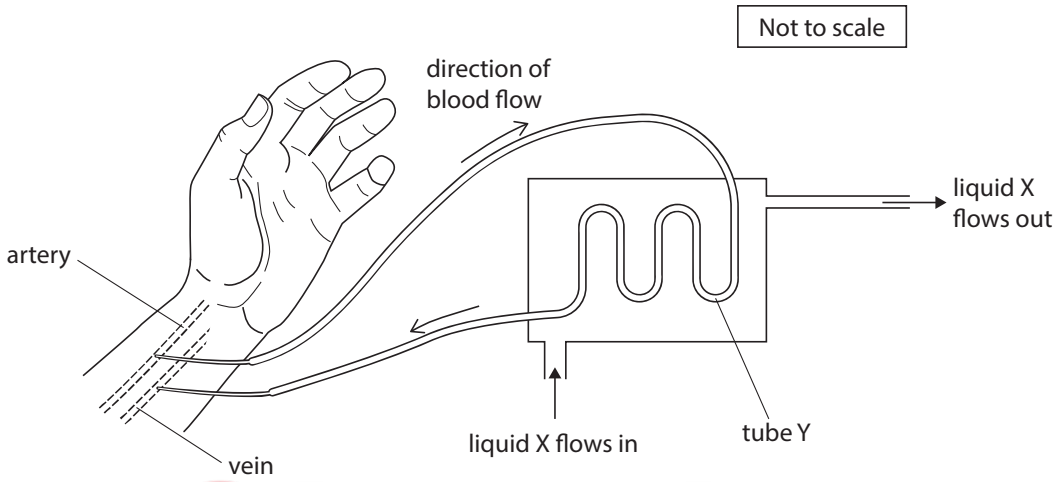
6. Question 5 · part a.i · 4 marks

June 2025 Paper 1R

Further practice · Attempt part a.i only.

5 The diagram shows a kidney dialysis machine.

The wall of tube Y is partially permeable so it allows the passage of small molecules.



(a) (i) Explain why the blood flow is in the direction shown in the diagram.

(4)

(ii) State which part of the kidney nephron has the same role as tube Y.

(1)

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6. Question 5 · part a.i · 4 marks · Working and answer

Working and answer

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7. Question 9 · part b.i · 2 marks

June 2022 Paper 1

Further practice · Attempt part b.i only.

9 In England, 31% of men and 26% of women have high blood pressure.

(a) Explain how medication can be used to treat people with high blood pressure.

(4)

(b) (i) Describe what is meant by systolic blood pressure.

(2)

(ii) Describe what is meant by diastolic blood pressure.

(2)

(Total for Question 9 = 8 marks)

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7. Question 9 · part b.i · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

8. Question 9 · part b.ii · 2 marks

June 2022 Paper 1

Further practice · Attempt part b.ii only.

9 In England, 31% of men and 26% of women have high blood pressure.

(a) Explain how medication can be used to treat people with high blood pressure.

(4)

(b) (i) Describe what is meant by systolic blood pressure.

(2)

(ii) Describe what is meant by diastolic blood pressure.

(2)

(Total for Question 9 = 8 marks)

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8. Question 9 · part b.ii · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

Practice workbook · Use ABO antigens and antibodies to choose donors and explain clumping

Antigens and antibodies of each blood group, which donors suit which recipients, counting compatible people, and why mismatches cause agglutination.

10 question parts: 1 worked-example retry and 9 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-ABO-ANTIGENS-AND-TRANSFUSION

1. Question 4 · part c · 4 marks

November 2020 Paper 1

Worked-example retry · Attempt part c only.

4 There are four different blood groups, A, B, AB and O.

The antigens on the surface of red blood cells determine a person's blood group. There are two types of antigen, antigen A and antigen B.

(a) The table lists the four different blood groups.

Complete the table by giving the correct type of antigens for each blood group.

(2)

Blood group	A	B	AB	O
Antigen				

(b) If a person needs a blood transfusion, it is important that they receive a suitable blood group.

Explain the effects on a person who receives a blood group that is not suitable.

(3)

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1. Question 4 · part c · 4 marks

November 2020 Paper 1

Worked-example retry · Attempt part c only.

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(c) Explain which blood group or groups can be given to people with each of the other blood groups during a transfusion.

(4)

(d) The table shows the percentage of students with each type of blood group in a school.

Blood group	Percentage (%)
A	42
B	10
AB	4
O	44

There are 750 students in the school.

Calculate the number of students who can safely receive blood group B.

(3)

number of students = ...

(Total for Question 4 = 12 marks)

1. Question 4 · part c · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 4 · part a · 2 marks

November 2020 Paper 1

Further practice · Attempt part a only.

4 There are four different blood groups, A, B, AB and O.

The antigens on the surface of red blood cells determine a person's blood group. There are two types of antigen, antigen A and antigen B.

(a) The table lists the four different blood groups.

Complete the table by giving the correct type of antigens for each blood group.

(2)

Blood group	A	B	AB	O
Antigen				

(b) If a person needs a blood transfusion, it is important that they receive a suitable blood group.

Explain the effects on a person who receives a blood group that is not suitable.

(3)

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2. Question 4 · part a · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 4 · part d · 3 marks

November 2020 Paper 1

Further practice · Attempt part d only.

4 There are four different blood groups, A, B, AB and O.

The antigens on the surface of red blood cells determine a person's blood group. There are two types of antigen, antigen A and antigen B.

(a) The table lists the four different blood groups.

Complete the table by giving the correct type of antigens for each blood group.

(2)

Blood group	A	B	AB	O
Antigen				

(b) If a person needs a blood transfusion, it is important that they receive a suitable blood group.

Explain the effects on a person who receives a blood group that is not suitable.

(3)

OLevels.com SCHOOL

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3. Question 4 · part d · 3 marks

November 2020 Paper 1

Further practice · Attempt part d only.

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(c) Explain which blood group or groups can be given to people with each of the other blood groups during a transfusion.

(4)

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(d) The table shows the percentage of students with each type of blood group in a school.

Blood group	Percentage (%)
A	42
B	10
AB	4
O	44

There are 750 students in the school.

Calculate the number of students who can safely receive blood group B.

(3)

number of students = ...

(Total for Question 4 = 12 marks)

3. Question 4 · part d · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 9 · part a.i · 2 marks

June 2024 Paper 1

Further practice · Attempt part a.i only.

9 (a) There are four main blood groups.

One of these groups is known as type O.

(i) Name the other three blood groups.

(2)

(ii) A person with blood type O needs a blood transfusion.

Explain which blood type this person should receive in a transfusion.

(5)

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4. Question 9 · part a.i · 2 marks · Working and answer

Working and answer

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5. Question 4 · part b · 3 marks

November 2020 Paper 1

Further practice · Attempt part b only.

4 There are four different blood groups, A, B, AB and O.

The antigens on the surface of red blood cells determine a person's blood group. There are two types of antigen, antigen A and antigen B.

(a) The table lists the four different blood groups.

Complete the table by giving the correct type of antigens for each blood group.

(2)

Blood group	A	B	AB	O
Antigen				

(b) If a person needs a blood transfusion, it is important that they receive a suitable blood group.

Explain the effects on a person who receives a blood group that is not suitable.

(3)

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5. Question 4 · part b · 3 marks

November 2020 Paper 1

Further practice · Attempt part b only.

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(c) Explain which blood group or groups can be given to people with each of the other blood groups during a transfusion.

(4)

(d) The table shows the percentage of students with each type of blood group in a school.

Blood group	Percentage (%)
A	42
B	10
AB	4
O	44

There are 750 students in the school.

Calculate the number of students who can safely receive blood group B.

(3)

number of students = ...

(Total for Question 4 = 12 marks)

5. Question 4 · part b · 3 marks · Working and answer

Working and answer

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6. Question 2 · part a.iii · 3 marks

November 2020 Paper 2R

Further practice · Attempt part a.iii only.

2 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Treatment of Ebola

Ebola is caused by an RNA virus. There is no proven treatment available for Ebola. However, whole blood collected from someone who has recovered from infection has been used as a treatment. The results have been promising in a small group of Ebola patients.

Guidance for national health authorities and blood transfusion services describes the stages needed to collect whole blood (CWB) or plasma (CP) from Ebola-recovered patients. This can be used for transfusion to patients as a treatment for early Ebola.

The guidance states how to

- identify people who have recovered from Ebola as potential blood donors
- screen donors' blood for A, B, O and Rhesus groupings
- screen donors' blood for transfusion-transmissible infections
- collect blood and care for donors
- obtain agreement of Ebola patients for the treatment
- identify patient's blood grouping
- store and transport CWB and CP to the places where transfusions are to be given
- select Ebola patients for transfusion
- transfuse the blood into the patient

Donated CWB should never be frozen and should be stored between 2 °C and 6 °C.

CWB and CP donations should be transfused to the Ebola patients using standard transfusion procedures.

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6. Question 2 · part a.iii · 3 marks

November 2020 Paper 2R

Further practice · Attempt part a.iii only.

(a) (i) Explain what is meant by an RNA virus (line 1).

(2)

(ii) The table lists five diseases. Put a tick (✓) next to the diseases that should be tested for in the donors' blood.

(2)

Disease	Blood tested (✓)
anaemia	
cystic fibrosis	
gonorrhoea	
HIV	
scurvy	

(iii) Explain why it is necessary to identify the blood group of the blood collected from donors who have recovered from Ebola (line 9).

(3)

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6. Question 2 · part a.iii · 3 marks

November 2020 Paper 2R

Further practice · Attempt part a.iii only.

(iv) Suggest why donated CWB should never be frozen (line 17).

(2)

(v) Explain how the blood of people who have recovered from Ebola is likely to be effective in treating the disease.

(2)

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(b) The nitrogenous base composition of another virus was found to be adenine 13%, guanine 26%, uracil 20% and cytosine 41%.

Explain how these figures prove that this is a virus with a single strand of RNA.

(3)

(Total for Question 2 = 14 marks)

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6. Question 2 · part a.iii · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

7. Question 4 · part a.ii · 3 marks

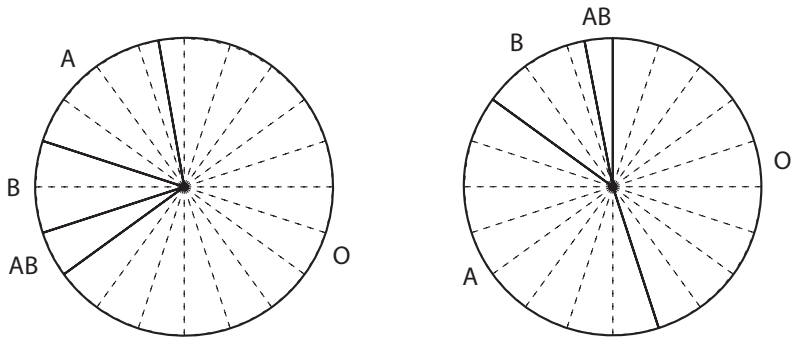
June 2022 Paper 1

Further practice · Attempt part a.ii only.

4 (a) Humans belong to one of four blood groups.

These blood groups are A, B, AB and O.

The pie charts show estimates of the percentages of people from Africa and the percentages of people from Europe in each blood group.



(i) Complete the tables to give the percentage of each blood group.

Some information has been given.

(2)

Africa

Blood group	Percentage of blood group
O	68
A	
B	
AB	5

Europe

Blood group	Percentage of blood group
O	45
A	
B	12
AB	

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P 7 0 9 7 3 A 0 1 2 2 4

7. Question 4 · part a.ii · 3 marks

June 2022 Paper 1

Further practice · Attempt part a.ii only.

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(ii) Some people need a blood transfusion.

Explain why people with blood group A cannot receive a blood transfusion from people with blood group B.

(3)

(iii) A male with the genotype I^AI^O and a female with the genotype I^BI^O have a child with the genotype I^OI^O.

Draw a genetic diagram to show how the child inherits the genotype I^OI^O.

(3)

OLevels.com SCHOOL

(iv) Determine the probability that a child from these parents will inherit the blood group AB.

(1)

probability = ...



7. Question 4 · part a.ii · 3 marks · Working and answer

Working and answer

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8. Question 9 · part a.ii · 5 marks

June 2024 Paper 1

Further practice · Attempt part a.ii only.

9 (a) There are four main blood groups.

One of these groups is known as type O.

(i) Name the other three blood groups.

(2)

(ii) A person with blood type O needs a blood transfusion.

Explain which blood type this person should receive in a transfusion.

(5)

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8. Question 9 · part a.ii · 5 marks

June 2024 Paper 1

Further practice · Attempt part a.ii only.

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(b) Haemophilia is a sex-linked genetic condition that affects the blood.

(i) Describe what is meant by a sex-linked genetic condition.

(2)

(ii) State one effect that haemophilia will have on an individual.

(1)

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(Total for Question 9 = 10 marks)

8. Question 9 · part a.ii · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

9. Question 9 · part b.i · 3 marks

June 2019 Paper 1

Further practice · Attempt part b.i only.

9 Inheritance of blood group involves codominant inheritance.

(a) (i) Which statement describes codominant inheritance in ABO blood groups?

(1)

- ☒ **A** the inheritance of two different alleles, both of which are expressed
- ☒ **B** the inheritance of two different alleles, only one of which is expressed
- ☒ **C** the inheritance of multiple alleles, only two of which are expressed
- ☒ **D** the inheritance of multiple alleles, only one of which is expressed

(ii) State the possible genotypes of a person with blood group A.

(1)

(b) (i) A person's blood group is determined by antigens.

These antigens are carbohydrate and protein molecules on the surface of red blood cells.

In 2007, a team of scientists used enzymes to convert blood groups A, B and AB into blood group O for transfusions.

Suggest how enzymes can convert blood groups A, B and AB into blood group O.

(3)

(ii) Suggest an advantage of producing blood group O using enzymes, compared with other methods of obtaining blood group O.

(1)

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P 5 8 5 5 6 8 A 0 2 4 2 8

9. Question 9 · part b.i · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

10. Question 9 · part b.ii · 1 mark

June 2019 Paper 1

Further practice · Attempt part b.ii only.

9 Inheritance of blood group involves codominant inheritance.

(a) (i) Which statement describes codominant inheritance in ABO blood groups?

(1)

- ☒ **A** the inheritance of two different alleles, both of which are expressed
- ☒ **B** the inheritance of two different alleles, only one of which is expressed
- ☒ **C** the inheritance of multiple alleles, only two of which are expressed
- ☒ **D** the inheritance of multiple alleles, only one of which is expressed

(ii) State the possible genotypes of a person with blood group A.

(1)

(b) (i) A person's blood group is determined by antigens.

These antigens are carbohydrate and protein molecules on the surface of red blood cells.

In 2007, a team of scientists used enzymes to convert blood groups A, B and AB into blood group O for transfusions.

Suggest how enzymes can convert blood groups A, B and AB into blood group O.

(3)

(ii) Suggest an advantage of producing blood group O using enzymes, compared with other methods of obtaining blood group O.

(1)

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10. Question 9 · part b.ii · 1 mark

June 2019 Paper 1

Further practice · Attempt part b.ii only.

(c) Haemophilia is a sex-linked blood disorder that reduces the ability of the blood to clot.

These are the genotypes of four offspring, P, Q, R and S.

P	Q	R	S
X^HX^h	X^hY	X^HX^H	X^HY

(i) Draw a genetic diagram to show how these offspring are produced from one set of parents.

(2)

(ii) These parents are expecting another baby.

Determine the probability that this baby will have haemophilia.

(1)

probability = ...

(Total for Question 9 = 9 marks)

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10. Question 9 · part b.ii · 1 mark · Working and answer

Working and answer

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Practice workbook · Explain how tissue fluid forms, what it does and why it builds up

Pressure at the capillary bed pushing fluid out and drawing it back, its composition compared with blood, exchange with cells, lymph drainage and swelling.

10 question parts: 1 worked-example retry and 9 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-TISSUE-FLUID-FORMATION-AND-FUNCTION

1. Question 8 · part b · 4 marks

November 2024 Paper 1

Worked-example retry · Attempt part b only.

8 Body cells are surrounded by tissue fluid.

(a) Describe the function of tissue fluid.

(2)

(b) Explain how tissue fluid is formed.

(4)

(c) Some people have a condition called cellular oedema.

In these people too much water enters body cells.

Explain how water can enter body cells from the surrounding tissue fluid.

(3)

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P 7 5 9 5 8 A 0 2 0 2 4

1. Question 8 · part b · 4 marks · Working and answer

Working and answer

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2. Question 4 · part a.ii · 4 marks

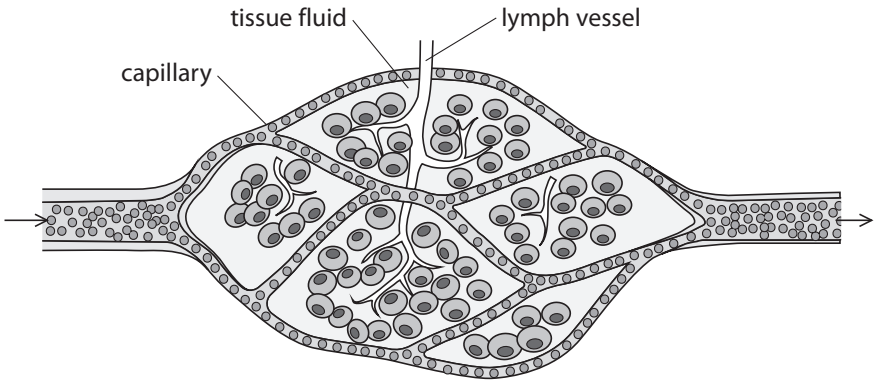
January 2023 Paper 2

Further practice · Attempt part a.ii only.

4 (a) When blood flows into a network of capillaries, tissue fluid is produced.

The tissue fluid supplies the surrounding cells with molecules.

The diagram shows a network of capillaries and surrounding cells.



(i) Give two differences between the composition of tissue fluid and the composition of blood entering the capillary network.

(2)

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(ii) Describe how tissue fluid is formed.

(4)

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2. Question 4 · part a.ii · 4 marks

January 2023 Paper 2

Further practice · Attempt part a.ii only.

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(iii) Lymphoedema occurs when tissue fluid builds up causing swelling.

This is more common in the legs and arms.

One treatment for this is to increase physical activity, resulting in more muscle contraction.

Explain why increased muscle contraction could reduce the swelling caused by the build up of tissue fluid.

(2)

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(b) Lymphocytes, found in lymph nodes, produce antibodies in response to specific antigens.

Explain why the reaction between an antigen and an antibody is specific.

(3)

(Total for Question 4 = 11 marks)

2. Question 4 · part a.ii · 4 marks · Working and answer

Working and answer

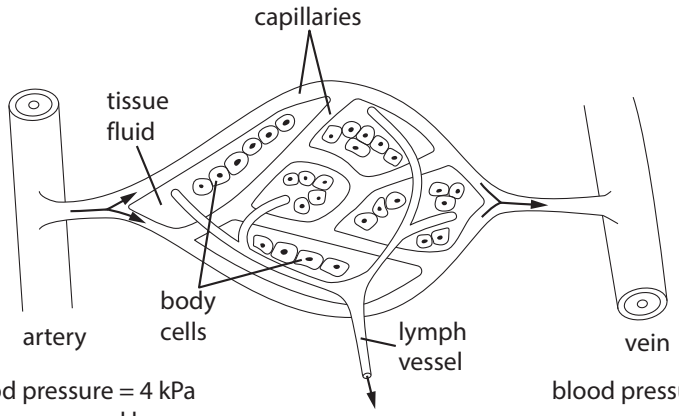
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3. Question 7 · part a.i · 2 marks

June 2025 Paper 1

Further practice · Attempt part a.i only.

7 The diagram shows a capillary network.



(a) (i) Explain the change in blood pressure from the artery to the vein.

(2)

(ii) Explain why the pressure caused by the plasma proteins does not change.

(2)

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3. Question 7 · part a.i · 2 marks · Working and answer

Working and answer

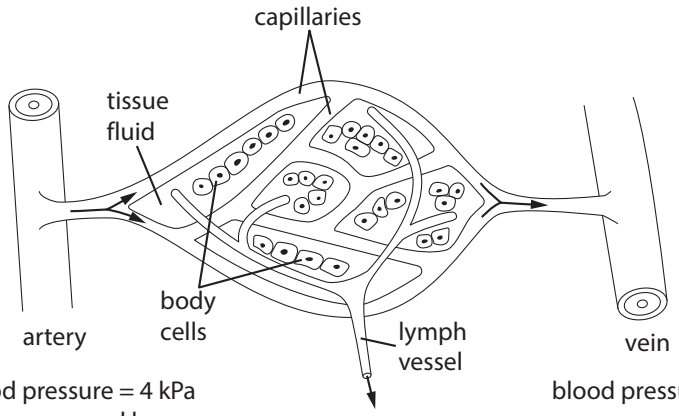
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4. Question 7 · part a.ii · 2 marks

June 2025 Paper 1

Further practice · Attempt part a.ii only.

7 The diagram shows a capillary network.



blood pressure = 4 kPa
pressure caused by plasma proteins = 3.3 kPa

blood pressure = 1.6 kPa
pressure caused by plasma proteins = 3.3 kPa

(a) (i) Explain the change in blood pressure from the artery to the vein.

(2)

(ii) Explain why the pressure caused by the plasma proteins does not change.

(2)

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4. Question 7 · part a.ii · 2 marks

June 2025 Paper 1

Further practice · Attempt part a.ii only.

(b) Describe how a sample of plasma could be tested for the presence of proteins.

(4)

(c) Tissue fluid is formed by the movement of molecules from the blood. This occurs in the capillary network.

Use data from the diagram to determine the direction of movement of fluid between the capillaries and tissue fluid at the artery end and at the vein end of the capillary network.

(4)

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4. Question 7 · part a.ii · 2 marks · Working and answer

Working and answer

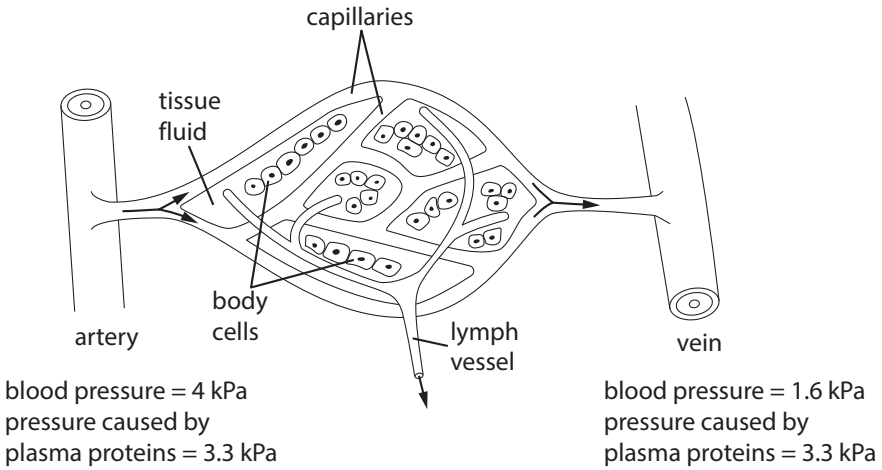
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5. Question 7 · part c · 4 marks

June 2025 Paper 1

Further practice · Attempt part c only.

7 The diagram shows a capillary network.



(a) (i) Explain the change in blood pressure from the artery to the vein.

(2)

(ii) Explain why the pressure caused by the plasma proteins does not change.

(2)

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5. Question 7 · part c · 4 marks

June 2025 Paper 1

Further practice · Attempt part c only.

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(b) Describe how a sample of plasma could be tested for the presence of proteins.

(4)

(c) Tissue fluid is formed by the movement of molecules from the blood. This occurs in the capillary network.

Use data from the diagram to determine the direction of movement of fluid between the capillaries and tissue fluid at the artery end and at the vein end of the capillary network.

(4)

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5. Question 7 · part c · 4 marks

June 2025 Paper 1

Further practice · Attempt part c only.

(d) A condition known as leaky capillary syndrome occurs in some people. When this occurs, much more of the fluid from the capillary passes out into the tissue fluid.

Suggest what effect leaky capillary syndrome would have on the circulatory system.

(2)

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(Total for Question 7 = 14 marks)

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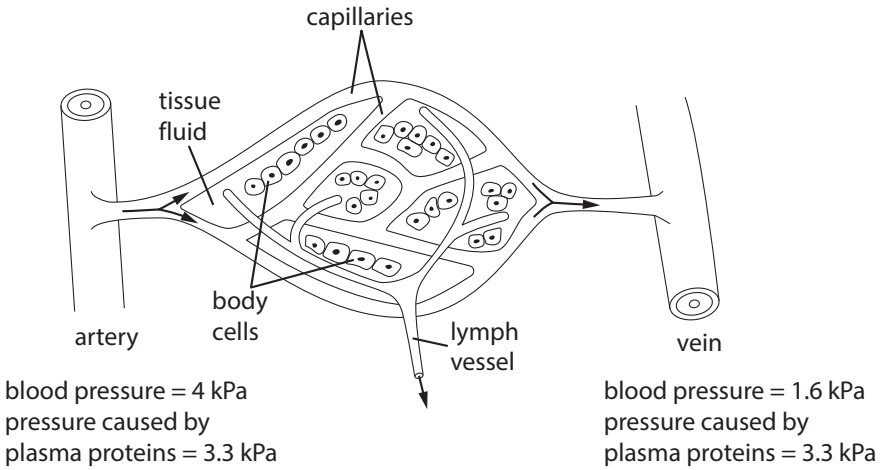
Working and answer

6. Question 7 · part d · 2 marks

June 2025 Paper 1

Further practice · Attempt part d only.

7 The diagram shows a capillary network.



(a) (i) Explain the change in blood pressure from the artery to the vein.

(2)

(ii) Explain why the pressure caused by the plasma proteins does not change.

(2)

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6. Question 7 · part d · 2 marks

June 2025 Paper 1

Further practice · Attempt part d only.

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(b) Describe how a sample of plasma could be tested for the presence of proteins.

(4)

(c) Tissue fluid is formed by the movement of molecules from the blood. This occurs in the capillary network.

Use data from the diagram to determine the direction of movement of fluid between the capillaries and tissue fluid at the artery end and at the vein end of the capillary network.

(4)

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6. Question 7 · part d · 2 marks

June 2025 Paper 1

Further practice · Attempt part d only.

(d) A condition known as leaky capillary syndrome occurs in some people. When this occurs, much more of the fluid from the capillary passes out into the tissue fluid.

Suggest what effect leaky capillary syndrome would have on the circulatory system.

(2)

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(Total for Question 7 = 14 marks)

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Working and answer

7. Question 4 · part a.iii · 2 marks

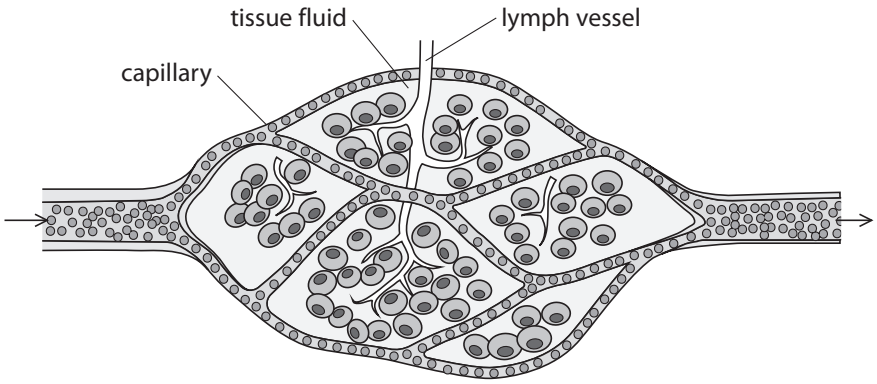
January 2023 Paper 2

Further practice · Attempt part a.iii only.

4 (a) When blood flows into a network of capillaries, tissue fluid is produced.

The tissue fluid supplies the surrounding cells with molecules.

The diagram shows a network of capillaries and surrounding cells.



(i) Give two differences between the composition of tissue fluid and the composition of blood entering the capillary network.

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(ii) Describe how tissue fluid is formed.

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7. Question 4 · part a.iii · 2 marks

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Further practice · Attempt part a.iii only.

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(iii) Lymphoedema occurs when tissue fluid builds up causing swelling.

This is more common in the legs and arms.

One treatment for this is to increase physical activity, resulting in more muscle contraction.

Explain why increased muscle contraction could reduce the swelling caused by the build up of tissue fluid.

(2)

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(b) Lymphocytes, found in lymph nodes, produce antibodies in response to specific antigens.

Explain why the reaction between an antigen and an antibody is specific.

(3)

(Total for Question 4 = 11 marks)

7. Question 4 · part a.iii · 2 marks · Working and answer

Working and answer

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8. Question 4 · part a.i · 2 marks

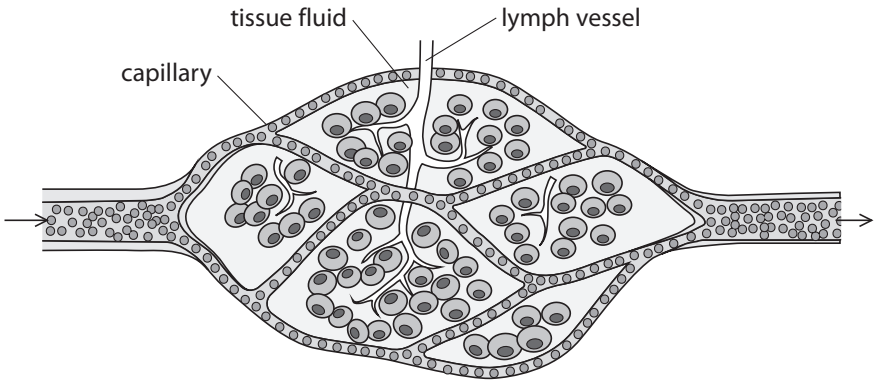
January 2023 Paper 2

Further practice · Attempt part a.i only.

4 (a) When blood flows into a network of capillaries, tissue fluid is produced.

The tissue fluid supplies the surrounding cells with molecules.

The diagram shows a network of capillaries and surrounding cells.



(i) Give two differences between the composition of tissue fluid and the composition of blood entering the capillary network.

(2)

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(ii) Describe how tissue fluid is formed.

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8. Question 4 · part a.i · 2 marks · Working and answer

Working and answer

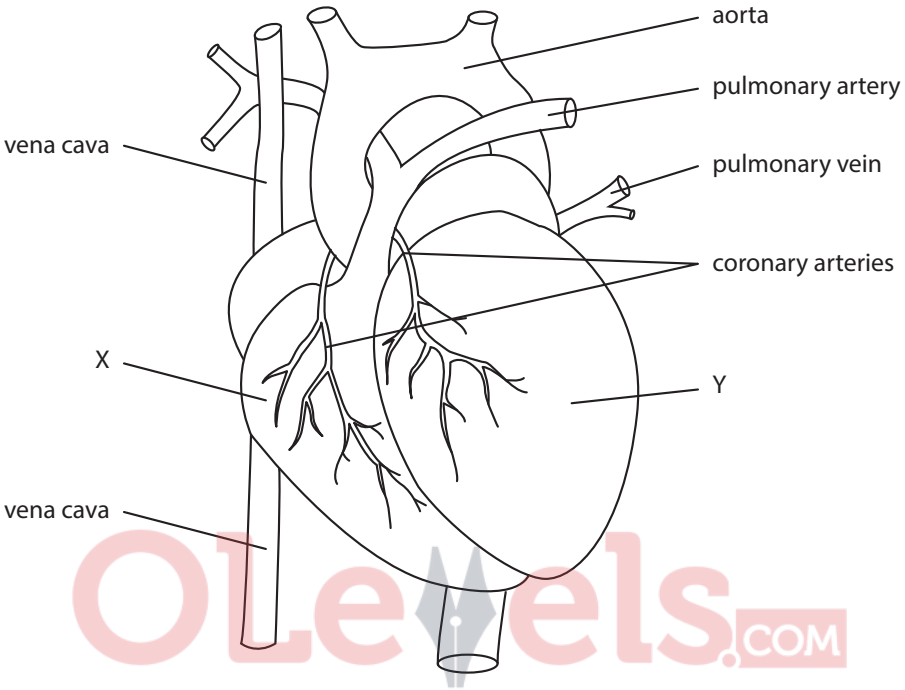
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9. Question 7 · part d · 3 marks

November 2023 Paper 1

Further practice · Attempt part d only.

7 The diagram shows the heart as seen from the front.



(a) The vena cava carries deoxygenated blood.

Name a different blood vessel, shown on the diagram, that also carries deoxygenated blood.

(1)

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9. Question 7 · part d · 3 marks

November 2023 Paper 1

Further practice · Attempt part d only.

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(b) Blood chambers X and Y have walls that are different in thickness.

Explain this difference.

(5)

(c) The coronary arteries can become blocked.

Explain the effect of this blockage on the functioning of the heart.

(3)

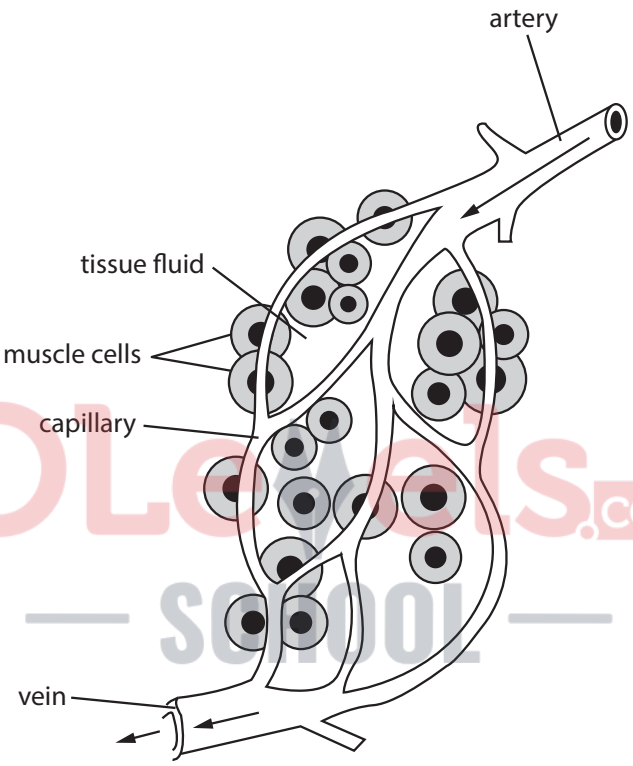
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9. Question 7 · part d · 3 marks

November 2023 Paper 1

Further practice · Attempt part d only.

(d) The diagram shows a capillary bed.



Use the diagram to explain the function of the tissue fluid.

(3)

(Total for Question 7 = 12 marks)

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9. Question 7 · part d · 3 marks · Working and answer

Working and answer

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10. Question 8 · part a · 2 marks

November 2024 Paper 1

Further practice · Attempt part a only.

8 Body cells are surrounded by tissue fluid.

(a) Describe the function of tissue fluid.

(2)

(b) Explain how tissue fluid is formed.

(4)

(c) Some people have a condition called cellular oedema.

In these people too much water enters body cells.

Explain how water can enter body cells from the surrounding tissue fluid.

(3)

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10. Question 8 · part a · 2 marks · Working and answer

Working and answer

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